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The equation $2x^4 + x + y^3 + y^2 + 2 = 0$ has no integer solutions

Size $h = 48$, length $l = 12$

Theorem 1. *The equation*

$$2x^4 + x + y^3 + y^2 + 2 = 0 \tag{1}$$

has no integer solutions.

The proof uses the cubic Hilbert reciprocity law over the field $K = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. It uses 3 graphs. For the g -th graph, we attach to a hypothetical integer solution the values of auxiliary polynomials $H_i^{(g)}$ with coefficients in $\mathbb{Z}[\omega]$, constants $c_i^{(g)} \in \mathbb{Z}[\omega]$, and a sum $B^{(g)}$ of cubic Hilbert symbols of these values, recorded by weighted edges. By the reciprocity law, the local contributions $B_p^{(g)}$ of all primes p add up to 0 modulo 3, for every g . We show that $B_p^{(g)} = 0$ for $p \notin S$, compute the vectors $(B_p^{(g)})_g$ for the primes $p \in S$, and obtain a contradiction. Two standard facts from class field theory are quoted without proof: the explicit formula for the cubic Hilbert symbol at the places not above 3, and the Hilbert reciprocity law. Apart from these, the proof uses polynomial identities, which can be checked by expanding both sides, and finite computations with residues.

1 Cubic Hilbert symbols over $\mathbb{Q}(\omega)$

The field. Let $K = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. Its ring of integers is $\mathbb{Z}[\omega]$, and every element of K can be written uniquely as $A + B\omega$ with $A, B \in \mathbb{Q}$. We write

$$\langle A, B \rangle = A + B\omega,$$

also when A and B are polynomials in x and y . Then $\langle A, B \rangle \langle C, D \rangle = \langle AC - BD, AD + BC - BD \rangle$, and the norm is $N(\langle A, B \rangle) = A^2 - AB + B^2$. In particular, a value $\langle A, B \rangle$ with $A, B \in \mathbb{Z}$ is zero if and only if $A = B = 0$.

Places. The finite places of K correspond to the non-zero prime ideals of $\mathbb{Z}[\omega]$. A prime $p \equiv 2 \pmod{3}$ is inert: there is one place above p , with uniformizer p and residue field $\mathbb{Z}[\omega]/p\mathbb{Z}[\omega]$ with p^2 elements. If $p \equiv 1 \pmod{3}$, the polynomial $t^2 + t + 1$ has two roots r modulo p , and for each of them there is a place v_r above p , at which ω is the root of $t^2 + t + 1$ in $\mathbb{Z}_p$ congruent to r modulo p ; the completion is $\mathbb{Q}_p$, the uniformizer is p , the residue field is $\mathbb{F}_p$, and ω reduces to r . The prime 3 is ramified: $\pi = 1 - \omega$ generates the unique prime ideal above 3, $\pi^2 = -3\omega$, and $\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega] = \mathbb{F}_3$, where $\langle A, B \rangle$ reduces to $A + B$. The only infinite place of K is complex. For a finite place v we write K_v for the completion and $v(a)$ for the normalized valuation, and we call a a *unit* at v if $v(a) = 0$.

Hilbert symbols. For a place v of K and $a, b \in K_v^\times$, the cubic Hilbert symbol $(a, b)_v$ is a cube root of unity, see [1, Chapter V, §3]. We write $(a, b)_v = \omega^{\beta_v(a, b)}$ with $\beta_v(a, b) \in \mathbb{Z}/3\mathbb{Z}$, and we use the following three facts [1, Chapter V, §3 and Chapter VI, §8].

(H1) $\beta_v(a, bc) = \beta_v(a, b) + \beta_v(a, c)$, $\beta_v(a, b) = -\beta_v(b, a)$, and $\beta_v(a, c^3) = 0$. In particular, $\beta_v(a, b)$ depends only on the classes of a and b in $K_v^\times/K_v^{\times 3}$.

(H2) Let v be a finite place not above 3, let q be the number of elements of its residue field, and let ϖ be a uniformizer at v . Write $a = \varpi^e u$ and $b = \varpi^f w$ with units u and w , and define $\chi(u) \in \mathbb{Z}/3\mathbb{Z}$ by $\bar{u}^{(q-1)/3} = \bar{\omega}^{\chi(u)}$, where the bar denotes the reduction to the residue field. Then $\beta_v(a, b) = e\chi(w) - f\chi(u)$. In particular, $\beta_v(a, b) = 0$ if a and b are both units.

(H3) (Hilbert reciprocity.) For $a, b \in K^\times$, $\beta_v(a, b) = 0$ for all but finitely many places v , and $\sum_v \beta_v(a, b) = 0$, where the sum is over all places of K . At the complex place, $\beta_v(a, b) = 0$.

The general formula in (H2) contains an additional term $ef\chi(-1)$, which vanishes because $-1 = (-1)^3$ is a cube. Formula (H2) fixes the normalization of the symbols; with the opposite sign convention all values β_v change sign, and the proof below is the same.

Classes at the places not above 3. Let $p \neq 3$ be a prime and v a place above p . For $a = p^e u \in K_v^\times$ with a unit u , put $c_v(a) = (e \bmod 3, \chi(u)) \in \mathbb{F}_3^2$. By (H2) and (H1), c_v is additive, $c_v(a)$ determines the class of a modulo cubes, and $\beta_v(a, b) = e\chi(w) - f\chi(u)$ for $c_v(a) = (e, \chi(u))$ and $c_v(b) = (f, \chi(w))$. We put $c_p(a) = c_v(a) \in \mathbb{F}_3^2$ if $p \equiv 2 \pmod{3}$, and $c_p(a) = (c_{v_r}(a), c_{v_{r'}}(a)) \in \mathbb{F}_3^4$ if $p \equiv 1 \pmod{3}$, where $r < r'$ are the roots of $t^2 + t + 1$ in $\{1, \dots, p-1\}$; accordingly, β_p is the sum of the forms β_v over the places v above p . For $p \equiv 2 \pmod{3}$ and a rational integer u prime to p , we have $\bar{u} \in \mathbb{F}_p$, so that $\bar{u}^{(p^2-1)/3} = (\bar{u}^{p-1})^{(p+1)/3} = 1$ and $\chi(u) = 0$.

The place above 3.

Lemma 2. *Let v be the place above 3. Every non-zero element of K_v is equal to $\pi^a \omega^b 2^c (1 + 3\omega)^d$ times a cube, for unique $a, b, c, d \in \mathbb{Z}/3\mathbb{Z}$, and every unit congruent to 1 modulo 9 is a cube. Put $c_3(A) = (a, b, c, d) \in \mathbb{F}_3^4$. Then c_3 is additive, and $\beta_v(A, B) = c_3(A)^\top M_3 c_3(B)$, where*

$$M_3 = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \pmod{3}.$$

Proof. For $\alpha \in \mathbb{Z}[\omega]$, the polynomial $g(s) = s + 3s^2 + 3s^3 - \alpha$ satisfies $g'(s) \equiv 1 \pmod{\pi}$, so by Hensel's lemma it has a root s in the completion of $\mathbb{Z}[\omega]$, and then $(1 + 3s)^3 = 1 + 9(s + 3s^2 + 3s^3) = 1 + 9\alpha$. Hence every unit congruent to 1 modulo 9 is a cube in K_v , and the class of a unit modulo cubes depends only on its residue modulo 9. There are 54 units in $\mathbb{Z}[\omega]/(9)$, and a direct enumeration shows that the 27 products $\omega^b 2^c (1 + 3\omega)^d$ with $b, c, d \in \{0, 1, 2\}$ are pairwise distinct modulo 9, also after multiplication by $-1 = (-1)^3$. Hence they represent all 27 classes of units modulo cubes. Since π^3 is a cube times a unit, the exponent of π is the valuation modulo 3, and the first statement follows.

By (H1), it remains to compute β_v on the 16 ordered pairs of the elements $\pi, \omega, 2$ and $1 + 3\omega$. Their norms are 3, 1, 4 and 7, hence, for each such pair, all the symbols at the places not above 2, 3 and 7 vanish by (H2), and, by (H3), β_v at the place above 3 is minus the sum of the values at the place above 2 and at the two places above 7. These places correspond to $\bar{\omega} = 2$ and $\bar{\omega} = 4$ in $\mathbb{F}_7$. The classes (e, χ) of the four elements at these places are

	2	$\bar{\omega} = 2$	$\bar{\omega} = 4$
π	(0, 2)	(0, 0)	(0, 2)
ω	(0, 1)	(0, 2)	(0, 2)
2	(1, 0)	(0, 2)	(0, 1)
$1 + 3\omega$	(0, 2)	(1, 0)	(0, 0)

where, at the places above 7, $\chi(u)$ is defined by $\bar{u}^2 = \bar{\omega}^{\chi(u)}$ in $\mathbb{F}_7$, and 7 is a uniformizer. Formula (H2) and the sum over these three places give M_3 . $\square$

We write β_3 for the form of Lemma 2. Since $3 = -\omega^2\pi^2$ and $4 = 2^2$, we have $c_3(3) = (2, 2, 0, 0)$ and $c_3(4) = (0, 0, 2, 0)$.

Residue classes. The local computations use the following description of the classes of the values of a polynomial on a residue class.

Lemma 3. *Let p be a prime, $k \geq 1$, and let $z \in \mathbb{Z}[\omega]$ be non-zero with $z \equiv \zeta \pmod{p^k}$, where $\zeta = \langle A, B \rangle$ with $0 \leq A, B < p^k$. Then $c_p(z) \in b + D$ for the vector b and the subspace D given as follows.*

- (i) $p \equiv 2 \pmod{3}$. If $\zeta \neq 0$, then $b = c_p(\zeta)$ and $D = 0$. If $\zeta = 0$, then $b = 0$, and D is $\mathbb{F}_3^2$, or the span of $(1, 0)$ if $z \in \mathbb{Z}$.
- (ii) $p \equiv 1 \pmod{3}$. For each root r , let ρ be the root of $t^2 + t + 1$ modulo p^k congruent to r . If $A + B\rho \not\equiv 0 \pmod{p^k}$, the two coordinates of $c_{v_r}(z)$ are those of $c_{v_r}(\zeta)$; otherwise these two coordinates are arbitrary.
- (iii) $p = 3$, $z \notin \mathbb{Z}$. If $\zeta \neq 0$, let $a = v(\zeta) < 2k$ be its valuation at π and $m = 2k - a$. Then $b = c_3(\zeta)$, and $D = 0$ if $m \geq 4$, while $D = D_m$ if $m \leq 3$, where D_1 is spanned by $(0, 1, 0, 0)$, $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, D_2 by $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, and D_3 by $(0, 0, 1, 1)$. If $\zeta = 0$, then $b = 0$ and $D = \mathbb{F}_3^4$.
- (iv) $p = 3$, $z \in \mathbb{Z}$, so that $B = 0$. If $3^k \nmid A$, let $j = v_3(A)$. Then $b = c_3(A)$ and $D = 0$ if $k - j \geq 2$, while $b = j c_3(3)$ and D is spanned by $c_3(4)$ if $k - j = 1$. If $3^k \mid A$, then $b = 0$ and D is spanned by $c_3(3)$ and $c_3(4)$.

Proof. (i) If $\zeta \neq 0$, then $e = v(z) = v(\zeta) < k$ and $z/p^e \equiv \zeta/p^e \pmod{p}$, so that χ is determined. If $z \in \mathbb{Z}$, its unit part is a rational integer, and $\chi = 0$. (ii) At v_r , z and ζ are the p -adic numbers $A' + B'\omega$ and $A + B\omega$ with $\omega \equiv \rho \pmod{p^k}$, and the argument of (i) applies. (iii) If $\zeta \neq 0$, then $v(z) = a$, and $z/\pi^a \equiv \zeta/\pi^a \pmod{\pi^m}$. By Lemma 2, the class of a unit depends only on its residue modulo $9 = \omega\pi^4$; the units congruent to 1 modulo π^m form a subgroup, and a direct enumeration of their residues modulo 9 shows that their classes form D_m . (iv) If $3^k \nmid A$, then $z = 3^j u$ with a rational unit $u \equiv A/3^j \pmod{3^{k-j}}$. The class of a rational unit depends only on its residue modulo 9, the units ± 1 are cubes, and the rational units congruent to 1 modulo 3 are 4^s times a unit congruent to 1 modulo 9; hence the classes are as stated, and if $3^k \mid A$ the valuation j is also unknown. $\square$

Lemma 4. *Let p be a prime, and consider vertices i with vectors b_i , subspaces D_i and vectors γ_i , and a set of edges (i, j) with $i < j$ and weights $w_{ij} \in \{1, 2\}$. Put $\rho_{ij} = w_{ij}$ and $\rho_{ji} = -w_{ij}$ for the edges, and $\sigma_i = \gamma_i + \sum_j \rho_{ij} b_j$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge (i, j) and all $d \in D_i, d' \in D_j$. Then*

$$\sum_{(i,j)} w_{ij} \beta_p(c_i, c_j) + \sum_i \beta_p(c_i, \gamma_i) = \sum_{(i,j)} w_{ij} \beta_p(b_i, b_j) + \sum_i \beta_p(b_i, \gamma_i)$$

whenever $c_i \in b_i + D_i$ for every i . By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and $\beta_p(u, v) = -\beta_p(v, u)$, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{(i,j)} w_{ij} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

2 The graphs

Put $F(x, y) = 2x^4 + x + y^3 + y^2 + 2$, so that (1) is the equation $F(x, y) = 0$. There are 3 certificates. The g -th certificate consists of polynomials $H_i^{(g)}$ with coefficients in $\mathbb{Z}[\omega]$ and constants $c_i^{(g)} \in \mathbb{Z}[\omega]$, listed in Table 1 and Table 2 and Table 3, and of *edges* (i, j) with $i < j$ and weights $w_{ij}^{(g)} \in \{1, 2\}$. For an edge we put $\rho_{ij}^{(g)} = w_{ij}^{(g)}$ and $\rho_{ji}^{(g)} = -w_{ij}^{(g)} \bmod 3$, and $\rho_{ij}^{(g)} = 0$ otherwise. In the rest of the proof we usually suppress the index g . The edges $(i, j; w_{ij})$ of the first certificate are (0, 6; 1), (0, 15; 2), (0, 27; 2), (0, 47; 2), (0, 48; 2), (0, 57; 2), (0, 63; 1), (0, 66; 2), (1, 3; 2), (1, 6; 1), (1, 16; 2), (1, 17; 2), (1, 18; 1), (1, 22; 2), (1, 37; 1), (1, 49; 2), (1, 50; 2), (1, 51; 1), (1, 52; 2), (1, 55; 1), (1, 65; 1), (2, 15; 1), (2, 17; 2), (2, 18; 1), (2, 26; 2), (2, 28; 1), (2, 35; 1), (2, 52; 2), (2, 63; 2), (3, 19; 2), (3, 25; 2), (3, 44; 1), (3, 59; 1), (3, 61; 1), (3, 66; 2), (4, 21; 2), (4, 30; 1), (4, 36; 1), (4, 47; 1), (4, 51; 1), (4, 52; 2), (4, 62; 2), (4, 64; 1), (5, 17; 1), (5, 18; 2), (5, 30; 2), (5, 33; 1), (5, 37; 1), (5, 39; 2), (5, 40; 1), (5, 41; 1), (5, 42; 1), (5, 45; 2), (5, 53; 1), (5, 56; 2), (5, 58; 2), (5, 60; 1), (5, 63; 2), (5, 66; 2), (6, 18; 1), (6, 23; 2), (6, 25; 1), (6, 26; 2), (6, 32; 2), (6, 38; 2), (6, 39; 1), (6, 46; 1), (6, 48; 1), (6, 50; 1), (6, 53; 1), (6, 60; 1), (6, 62; 1), (6, 64; 1), (6, 66; 2), (7, 41; 1), (7, 42; 1), (7, 45; 1), (7, 47; 1), (7, 51; 1), (7, 52; 2), (7, 56; 2), (8, 27; 1), (8, 28; 1), (8, 30; 2), (8, 37; 2), (8, 38; 2), (8, 39; 2), (8, 40; 2), (8, 42; 1), (8, 47; 2), (8, 58; 1), (8, 59; 2), (8, 63; 1), (8, 64; 2), (8, 65; 2), (8, 66; 2), (9, 22; 1), (9, 25; 2), (9, 43; 2), (9, 46; 2), (9, 48; 2), (9, 49; 1), (9, 58; 1), (10, 20; 2), (10, 26; 1), (10, 33; 1), (10, 42; 2), (10, 56; 1), (11, 20; 1), (11, 22; 1), (11, 25; 1), (11, 26; 1), (11, 29; 2), (11, 39; 1), (11, 42; 2), (11, 53; 2), (11, 59; 2), (12, 21; 1), (12, 28; 2), (12, 46; 2), (12, 52; 2), (12, 58; 2), (13, 23; 1), (13, 35; 1), (13, 36; 1), (14, 25; 1), (14, 26; 2), (14, 28; 1), (14, 31; 1), (14, 32; 2), (14, 33; 2), (14, 35; 2), (14, 37; 2), (14, 38; 1), (14, 42; 2), (14, 55; 1), (15, 22; 1), (15, 31; 1), (15, 32; 2), (15, 36; 1), (15, 37; 1), (15, 55; 2), (15, 64; 1), (16, 20; 2), (16, 26; 1), (16, 27; 2), (16, 44; 1), (16, 64; 1), (17, 30; 2), (17, 50; 2), (17, 54; 2), (17, 55; 2), (18, 20; 2), (18, 26; 2), (18, 32; 2), (18, 34; 2), (18, 50; 1), (18, 58; 1), (19, 22; 1), (19, 26; 2), (19, 45; 1), (19, 55; 2), (19, 56; 2), (19, 60; 2), (20, 33; 1), (20, 36; 1), (20, 38; 1), (20, 41; 2), (20, 42; 1), (20, 49; 1), (20, 50; 1), (20, 62; 2), (21, 27; 1), (21, 30; 1), (21, 50; 1), (21, 56; 1), (21, 57; 2), (21, 60; 1), (21, 61; 2), (21, 66; 1), (22, 24; 1), (22, 25; 2), (22, 26; 1), (22, 28; 1), (22, 43; 2), (22, 44; 2), (22, 49; 2), (22, 52; 1), (22, 56; 2), (23, 31; 1), (23, 32; 1), (23, 50; 1), (23, 56; 1), (24, 37; 2), (24, 40; 2), (24, 54; 2), (24, 57; 2), (24, 58; 2), (24, 61; 2), (25, 52; 2), (26, 41; 2), (26, 43; 1), (27, 34; 2), (27, 37; 1), (27, 43; 1), (27, 53; 2), (27, 55; 2), (27, 56; 2), (27, 57; 2), (28, 38; 1), (28, 41; 1), (28, 48; 1), (28, 51; 2), (28, 53; 1), (28, 56; 1), (28, 62; 2), (29, 57; 1), (29, 64; 1), (30, 50; 1), (30, 51; 2), (31, 33; 1), (31, 37; 2), (31, 43; 1), (31, 46; 1), (31, 47; 1), (31, 50; 2), (31, 52; 1), (31, 56; 2), (32, 44; 1), (32, 48; 2), (32, 55; 2), (32, 58; 1), (32, 59; 1), (32, 63; 2), (33, 42; 2), (33, 55; 1), (33, 58; 1), (33, 64; 1), (34, 41; 1), (34, 65; 1), (35, 60; 2), (36, 42; 1), (36, 49; 2), (36, 51; 1), (36, 57; 2), (36, 62; 2), (37, 38; 1), (37, 42; 2), (37, 48; 2), (37, 51; 1), (37, 52; 1), (37, 64; 1), (38, 50; 1), (38, 63; 2), (39, 56; 1), (40, 43; 2), (40, 59; 2), (40, 64; 2), (40, 66; 1), (41, 43; 2), (41, 64; 1), (41, 65; 1), (42, 45; 2), (42, 58; 1), (42, 60; 2), (42, 61; 2), (43, 53; 2), (43, 54; 2), (43, 59; 2), (44, 45; 2), (44, 48; 1), (44, 50; 2), (44, 56; 2), (45, 53; 2), (45, 55; 1), (45, 60; 1), (45, 62; 1), (45, 63; 1), (46, 60; 2), (46, 62; 1), (47, 52; 1), (48, 51; 1), (48, 56; 1), (51, 59; 2), (51, 61; 1), (51, 65; 2), (52, 61; 2), (53, 58; 1), (54, 55; 1), (54, 57; 2), (54, 63; 2), (58, 64; 2), (62, 63; 1). The edges $(i, j; w_{ij})$ of the second certificate are (0, 4; 2), (0, 24; 1), (0, 38; 1), (0, 43; 1), (0, 48; 2), (1, 3; 1), (1, 5; 2), (1, 8; 1), (1, 10; 1), (1, 11; 2), (1, 23; 1), (1, 31; 1), (1, 33; 1), (1, 36; 2), (1, 41; 2), (1, 43; 2), (1, 47; 1), (1, 48; 2), (1, 58; 1), (1, 59; 2), (2, 14; 2), (2, 16; 2), (2, 19; 2), (2, 35; 1), (2, 38; 2), (2, 39; 2), (2, 50; 1), (2, 52; 2), (2, 60; 2), (3, 14; 2), (3, 20; 2), (3, 27; 1), (3, 37; 2), (3, 41; 2), (3, 43; 2), (3, 45; 2), (3, 50; 2), (3, 62; 1), (3, 63; 2), (3, 65; 2), (3, 72; 2), (3, 73; 2), (4, 13; 2), (4, 14; 2), (4, 29; 2), (4, 39; 2), (4, 47; 2), (4, 48; 1), (4, 59; 2), (4, 66; 2), (4, 67; 2), (4, 72; 2), (5, 12; 1), (5, 14; 1), (5, 24; 2), (5, 28; 1), (5, 33; 1), (5, 36; 2), (5, 37; 1), (5, 48; 2), (5, 55; 2), (5, 56; 2), (5, 58; 1), (5, 59; 2), (6, 13; 2), (6, 14; 2), (6, 20; 1), (6, 25; 1), (6, 38; 2), (6, 39; 2), (6, 43; 2), (6, 53; 2), (7, 15; 2), (7, 21; 1), (7, 36; 1), (7, 42; 1), (7, 45; 1), (7, 56; 2),

$(7, 63; 2), (7, 64; 1), (7, 73; 2), (8, 14; 2), (8, 32; 2), (8, 45; 1), (8, 50; 2), (8, 65; 1), (9, 34; 2),$
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 $(25, 49; 2), (25, 69; 1), (26, 30; 2), (26, 33; 1), (26, 37; 2), (26, 42; 1), (26, 44; 2), (26, 45; 1),$
 $(26, 55; 2), (26, 56; 2), (26, 57; 2), (26, 62; 1), (27, 38; 1), (27, 45; 1), (27, 66; 2), (27, 70; 2),$
 $(28, 31; 1), (28, 41; 1), (29, 32; 1), (29, 50; 2), (29, 54; 1), (29, 61; 2), (29, 64; 1), (29, 65; 1),$
 $(29, 66; 2), (30, 33; 1), (30, 34; 1), (30, 57; 2), (30, 63; 2), (30, 70; 1), (31, 64; 2), (32, 56; 1),$
 $(32, 57; 1), (33, 36; 2), (33, 57; 1), (33, 59; 2), (33, 69; 1), (34, 35; 2), (34, 40; 2), (34, 45; 1),$
 $(34, 50; 1), (34, 60; 2), (34, 69; 2), (35, 38; 1), (35, 41; 1), (35, 43; 1), (35, 53; 1), (35, 55; 2),$
 $(35, 67; 1), (36, 43; 1), (36, 44; 2), (36, 54; 1), (36, 57; 2), (36, 66; 2), (37, 40; 2), (37, 44; 1),$
 $(37, 47; 1), (37, 56; 1), (37, 61; 1), (38, 46; 2), (38, 56; 1), (38, 63; 1), (39, 43; 2), (39, 49; 2),$
 $(39, 54; 2), (39, 56; 2), (39, 59; 2), (39, 70; 1), (40, 45; 1), (40, 56; 2), (40, 57; 2), (40, 59; 1),$
 $(40, 63; 1), (40, 69; 1), (41, 55; 2), (41, 56; 1), (42, 49; 2), (42, 56; 2), (42, 61; 1), (42, 64; 1),$
 $(42, 67; 1), (43, 44; 1), (43, 60; 2), (43, 62; 2), (44, 45; 1), (44, 48; 1), (44, 61; 2), (45, 59; 1),$
 $(45, 71; 2), (45, 72; 2), (47, 49; 2), (47, 50; 2), (47, 52; 1), (47, 54; 1), (47, 66; 2), (47, 68; 2),$
 $(48, 50; 1), (48, 51; 1), (48, 56; 2), (48, 71; 1), (49, 55; 1), (49, 57; 2), (49, 61; 1), (49, 64; 2),$
 $(50, 55; 1), (50, 56; 1), (50, 58; 2), (50, 59; 2), (51, 68; 2), (51, 72; 1), (51, 73; 1), (52, 53; 2),$
 $(52, 62; 2), (53, 54; 2), (53, 60; 2), (53, 63; 2), (53, 66; 1), (54, 57; 1), (54, 58; 1), (54, 64; 2),$
 $(56, 62; 2), (57, 59; 2), (57, 64; 1), (57, 67; 1), (58, 61; 2), (60, 63; 1), (60, 66; 2), (60, 68; 1),$
 $(61, 64; 2), (61, 71; 2), (62, 71; 1), (63, 64; 1), (63, 71; 2), (63, 73; 1), (65, 69; 1), (66, 68; 1).$ The
edges $(i, j; w_{ij})$ of the third certificate are $(0, 42; 2), (0, 43; 2), (0, 46; 1), (0, 47; 1), (1, 13; 1),$
 $(1, 20; 2), (1, 43; 2), (1, 46; 2), (1, 48; 2), (1, 50; 1), (2, 4; 1), (2, 6; 1), (2, 13; 1), (2, 15; 1), (2, 16; 2),$
 $(2, 17; 2), (2, 42; 1), (2, 50; 2), (2, 51; 1), (3, 13; 1), (3, 15; 2), (3, 21; 1), (3, 29; 2), (3, 35; 1),$
 $(3, 36; 1), (3, 39; 2), (3, 40; 2), (3, 42; 1), (3, 49; 2), (3, 50; 1), (3, 58; 1), (3, 60; 2), (4, 29; 2),$
 $(4, 30; 1), (4, 31; 1), (4, 34; 1), (4, 39; 2), (4, 57; 2), (5, 18; 2), (5, 27; 2), (5, 33; 1), (5, 39; 2),$
 $(5, 44; 1), (5, 45; 2), (5, 51; 1), (5, 52; 2), (5, 56; 2), (5, 60; 1), (5, 63; 2), (6, 21; 1), (6, 28; 2),$
 $(6, 35; 1), (6, 38; 1), (6, 50; 1), (6, 54; 2), (6, 59; 2), (6, 60; 2), (6, 62; 2), (7, 24; 1), (7, 25; 1),$
 $(7, 26; 2), (7, 28; 2), (7, 31; 2), (7, 32; 2), (7, 33; 1), (7, 35; 1), (7, 36; 1), (7, 37; 1), (7, 38; 2),$
 $(7, 46; 2), (7, 54; 2), (7, 61; 2), (8, 22; 2), (8, 27; 2), (8, 31; 1), (8, 36; 1), (8, 44; 1), (8, 45; 2),$
 $(8, 49; 1), (8, 50; 2), (8, 53; 1), (8, 61; 2), (8, 63; 1), (9, 20; 2), (9, 24; 1), (9, 28; 2), (9, 32; 2),$
 $(9, 33; 1), (9, 36; 1), (9, 42; 1), (9, 55; 2), (10, 21; 1), (10, 23; 1), (10, 41; 2), (10, 44; 2), (10, 45; 1),$
 $(10, 46; 1), (11, 24; 2), (11, 29; 2), (11, 40; 1), (11, 41; 2), (11, 51; 2), (11, 58; 1), (12, 19; 1),$
 $(12, 22; 2), (12, 29; 1), (12, 31; 2), (12, 32; 1), (12, 44; 2), (12, 52; 1), (12, 54; 1), (13, 16; 1),$
 $(13, 22; 1), (13, 27; 2), (13, 29; 1), (13, 32; 1), (13, 35; 2), (13, 37; 1), (14, 16; 2), (14, 18; 1),$
 $(14, 19; 2), (14, 21; 2), (14, 23; 2), (14, 24; 1), (14, 26; 1), (14, 30; 1), (14, 32; 1), (14, 36; 2),$
 $(14, 37; 2), (14, 40; 2), (14, 45; 1), (14, 58; 1), (15, 20; 2), (15, 21; 1), (15, 33; 1), (15, 35; 1),$

(15, 36; 1), (15, 58; 2), (16, 19; 2), (16, 20; 2), (16, 27; 2), (16, 30; 1), (16, 34; 1), (16, 54; 1),
(17, 18; 1), (17, 26; 2), (17, 30; 2), (17, 33; 2), (17, 38; 2), (17, 57; 1), (18, 30; 1), (18, 36; 1),
(18, 37; 2), (18, 39; 2), (18, 40; 2), (18, 46; 2), (18, 48; 2), (18, 50; 2), (18, 55; 1), (18, 56; 2),
(18, 58; 1), (18, 60; 1), (18, 63; 2), (19, 28; 1), (19, 30; 2), (19, 33; 1), (19, 36; 2), (19, 46; 1),
(19, 48; 1), (19, 51; 2), (19, 52; 1), (19, 53; 1), (19, 62; 1), (19, 63; 2), (20, 53; 1), (20, 54; 1),
(20, 57; 1), (20, 58; 2), (20, 59; 2), (21, 26; 2), (21, 37; 1), (21, 49; 2), (21, 51; 2), (21, 54; 1),
(21, 58; 1), (21, 63; 1), (22, 23; 2), (22, 28; 2), (22, 48; 2), (22, 54; 1), (22, 56; 2), (23, 30; 2),
(23, 39; 2), (23, 41; 1), (23, 44; 2), (23, 47; 1), (23, 53; 1), (23, 54; 2), (23, 56; 2), (23, 59; 2),
(24, 31; 2), (24, 32; 1), (24, 36; 1), (24, 48; 1), (24, 49; 2), (24, 51; 2), (24, 55; 1), (25, 41; 2),
(25, 58; 1), (25, 63; 2), (26, 28; 1), (26, 30; 1), (26, 35; 1), (26, 37; 2), (26, 40; 2), (26, 41; 2),
(26, 47; 2), (26, 48; 2), (27, 36; 1), (27, 53; 1), (27, 55; 2), (28, 37; 2), (28, 45; 2), (28, 50; 2),
(28, 58; 2), (29, 32; 2), (29, 58; 1), (30, 31; 2), (30, 39; 1), (30, 46; 1), (30, 50; 2), (30, 52; 2),
(30, 57; 2), (30, 58; 2), (30, 63; 2), (31, 44; 1), (31, 55; 2), (31, 63; 1), (32, 47; 1), (32, 49; 1),
(32, 59; 1), (32, 60; 2), (33, 34; 1), (33, 35; 2), (33, 43; 2), (33, 52; 2), (33, 54; 1), (34, 39; 1),
(34, 46; 1), (34, 55; 1), (35, 40; 1), (35, 53; 1), (35, 55; 2), (35, 56; 1), (35, 58; 2), (35, 62; 2),
(36, 40; 1), (36, 51; 2), (37, 48; 1), (37, 50; 2), (37, 51; 2), (37, 55; 1), (37, 60; 1), (38, 52; 1),
(39, 58; 2), (39, 59; 2), (40, 52; 1), (40, 54; 2), (40, 57; 1), (40, 58; 1), (40, 59; 1), (40, 61; 1),
(41, 52; 1), (41, 59; 2), (42, 58; 1), (42, 63; 1), (43, 44; 1), (43, 46; 2), (43, 47; 2), (43, 56; 1),
(44, 46; 2), (44, 50; 1), (44, 56; 1), (44, 59; 1), (45, 49; 2), (46, 51; 2), (46, 54; 2), (47, 50; 1),
(47, 58; 2), (47, 61; 1), (47, 62; 2), (47, 63; 2), (49, 50; 1), (50, 54; 1), (50, 61; 1), (50, 62; 2),
(52, 56; 2), (52, 59; 2), (52, 60; 2), (53, 57; 2), (54, 63; 2), (55, 59; 1), (56, 58; 2), (58, 60; 2),
(58, 63; 1). Let (ξ, η) denote (x, y) or (y, x) . We call a vertex i *linear* if $H_i = \varepsilon_i(\lambda_i \eta - s_i(\xi))$
for a unit ε_i of $\mathbb{Z}[\omega]$, an element $\lambda_i \in \mathbb{Z}[\omega]$ whose norm is a power of 3, and a polynomial s_i
with coefficients in $\mathbb{Z}[\omega]$; the last column of Table 1 and its analogues gives $\lambda_i \eta = s_i(\xi)$ for the
linear vertices, where $\lambda_i = 1$ if no factor is shown. For a linear vertex i , we put $f_i(t) = \lambda_i^d F$ and
 $g_{ij}(t) = \lambda_i^{d_j} H_j$ evaluated at $\xi = t, \eta = s_i(t)/\lambda_i$, where d and d_j are the degrees of F and H_j in
 η ; these are polynomials in t with coefficients in $\mathbb{Z}[\omega]$. For every linear vertex i , Table 4 gives
an integer $D_i \geq 1$ and a polynomial $w_i(t)$ with coefficients in $\mathbb{Z}[\omega]$ such that

$$D_i^3 \lambda_i^{e_i} c_i \prod_j g_{ij}^{\rho_{ij}} - w_i^3 = f_i Q_i \quad (2)$$

for a polynomial $Q_i(t)$ with coefficients in $\mathbb{Z}[\omega]$, where $\rho_{ij} \in \{0, 1, 2\}$; the table also gives an
integer $e_i \geq 0$ with $e_i + \sum_j \rho_{ij} d_j \equiv 0 \pmod{3}$, and $e_i = 0$ if $\lambda_i = 1$; Q_i is the quotient of the
division of the left side by f_i , and the remainder is zero. For every other vertex i , Table 4 gives
an integer $D_i \geq 1$ and polynomials W_i, A_i and B_i in x and y with coefficients in $\mathbb{Z}[\omega]$ such that

$$D_i^3 c_i \prod_j H_j^{\rho_{ij}} - W_i^3 = A_i F + B_i H_i. \quad (3)$$

For the edges (i, j) listed in Table 5, the table gives a linear endpoint $k \in \{i, j\}$ and the
factorization of the norm of the resultant $R_{ij} = \text{Res}_t(f_k, g_{kl})$, where $\{k, l\} = \{i, j\}$; it is non-
zero. These identities can be checked by expanding both sides and by polynomial division, using
 $\omega^2 = -1 - \omega$. They hold for each certificate, with its own data.

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials
denote their values at (x, y) , which lie in $\mathbb{Z}[\omega]$.

Step 1: the values are non-zero. For $i \in \{1, 2, 4, 5, 6, 7, 8, 9, 10, 13, 14, 15, 16, 19, 25, 26,$
 $28, 30, 31, 32, 33, 34, 35, 37, 38, 41, 44, 45, 46, 47, 48, 49, 54, 55, 60, 61, 62, 65\}$ in the first certificate,

there is no residue pair (a, b) modulo 2 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{2}$, as one checks for the 2^2 residue pairs; hence $H_i \neq 0$. For $i \in \{3, 11, 17, 18, 20, 22, 23, 24, 27, 36, 39, 51, 52, 53, 56, 57, 59, 63, 66\}$ in the first certificate, there is no residue pair (a, b) modulo 3 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{3}$, as one checks for the 3^2 residue pairs; hence $H_i \neq 0$. For $i \in \{50, 64\}$ in the first certificate, there is no residue pair (a, b) modulo 4 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{4}$, as one checks for the 4^2 residue pairs; hence $H_i \neq 0$. For $i \in \{12, 21, 29, 42, 43, 58\}$ in the first certificate, there is no residue pair (a, b) modulo 5 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{5}$, as one checks for the 5^2 residue pairs; hence $H_i \neq 0$. For $i \in \{40\}$ in the first certificate, there is no residue pair (a, b) modulo 7 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{7}$, as one checks for the 7^2 residue pairs; hence $H_i \neq 0$. For $i \in \{0\}$ in the first certificate, there is no residue pair (a, b) modulo 11 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{11}$, as one checks for the 11^2 residue pairs; hence $H_i \neq 0$. For $i \in \{0, 2, 3, 4, 5, 6, 7, 10, 11, 12, 15, 16, 17, 19, 20, 21, 26, 27, 28, 30, 35, 37, 38, 39, 46, 52, 53, 54, 55, 58, 59, 63, 64, 65, 67, 68, 70, 71, 72, 73\}$ in the second certificate, there is no residue pair (a, b) modulo 2 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{2}$, as one checks for the 2^2 residue pairs; hence $H_i \neq 0$. For $i \in \{8, 13, 14, 22, 31, 33, 34, 41, 43, 44, 47, 49, 56, 57, 61, 62, 66, 69\}$ in the second certificate, there is no residue pair (a, b) modulo 3 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{3}$, as one checks for the 3^2 residue pairs; hence $H_i \neq 0$. For $i \in \{18, 32, 40, 42\}$ in the second certificate, there is no residue pair (a, b) modulo 4 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{4}$, as one checks for the 4^2 residue pairs; hence $H_i \neq 0$. For $i \in \{9, 23, 25, 36, 45, 50, 51, 60\}$ in the second certificate, there is no residue pair (a, b) modulo 5 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{5}$, as one checks for the 5^2 residue pairs; hence $H_i \neq 0$. For $i \in \{24\}$ in the second certificate, there is no residue pair (a, b) modulo 9 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{9}$, as one checks for the 9^2 residue pairs; hence $H_i \neq 0$. For $i \in \{1\}$ in the second certificate, there is no residue pair (a, b) modulo 11 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{11}$, as one checks for the 11^2 residue pairs; hence $H_i \neq 0$. For $i \in \{48\}$ in the second certificate, there is no residue pair (a, b) modulo 13 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{13}$, as one checks for the 13^2 residue pairs; hence $H_i \neq 0$. For $i \in \{29\}$ in the second certificate, there is no residue pair (a, b) modulo 17 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{17}$, as one checks for the 17^2 residue pairs; hence $H_i \neq 0$. For $i \in \{1, 2, 5, 6, 7, 8, 9, 10, 13, 14, 15, 24, 25, 26, 28, 30, 31, 33, 34, 36, 37, 43, 44, 45, 46, 47, 55, 56, 57, 60, 61\}$ in the third certificate, there is no residue pair (a, b) modulo 2 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{2}$, as one checks for the 2^2 residue pairs; hence $H_i \neq 0$. For $i \in \{4, 11, 16, 17, 18, 20, 21, 22, 23, 27, 35, 38, 39, 49, 50, 52, 53, 59, 63\}$ in the third certificate, there is no residue pair (a, b) modulo 3 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{3}$, as one checks for the 3^2 residue pairs; hence $H_i \neq 0$. For $i \in \{32, 48, 51, 58\}$ in the third certificate, there is no residue pair (a, b) modulo 4 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{4}$, as one checks for the 4^2 residue pairs; hence $H_i \neq 0$. For $i \in \{12, 19, 29, 40, 41, 54, 62\}$ in the third certificate, there is no residue pair (a, b) modulo 5 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{5}$, as one checks for the 5^2 residue pairs; hence $H_i \neq 0$. For $i \in \{42\}$ in the third certificate, there is no residue pair (a, b) modulo 9 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{9}$, as one checks for the 9^2 residue pairs; hence $H_i \neq 0$. For $i \in \{0, 3\}$ in the third certificate, there is no residue pair (a, b) modulo 11 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{11}$, as one checks for the 11^2 residue pairs; hence $H_i \neq 0$. Also, all c_i are non-zero.

Step 2: reciprocity. For every place v of K , put

$$B_v^{(g)} = \sum_{(i,j)} w_{ij} \beta_v(H_i, H_j) + \sum_i \beta_v(H_i, c_i) \in \mathbb{Z}/3\mathbb{Z}, \quad (4)$$

the first sum over the edges of the g -th certificate, and, for a rational prime p , put $B_p^{(g)} = \sum_{v|p} B_v^{(g)}$. Applying (H3) to every pair (H_i, H_j) with an edge, multiplied by w_{ij} , and to every pair (H_i, c_i) , and adding, we find that $B_v^{(g)} = 0$ for all but finitely many v and $\sum_p B_p^{(g)} = 0$

for every g , as the complex place contributes 0. By (H1) and the definition of c_p , $B_p^{(g)} = \sum_{(i,j)} w_{ij} \beta_p(c_p(H_i), c_p(H_j)) + \sum_i \beta_p(c_p(H_i), c_p(c_i))$, where β_p is the form of Lemma 2 if $p = 3$.

Step 3: the primes outside S . Let $S = \{2, 3, 5, 7, 13, 19\}$. It contains 3 and the primes dividing the norms of the constants c_i , of the D_i , of the R_{ij} . Let $p \notin S$, and let v be a place above p , with residue field k_v ; then all these numbers are units at v . First, no edge (i, j) has both values divisible by v . Indeed, suppose that v divides H_i and H_j . If R_{ij} is defined, let k be the linear endpoint used for it and $\{k, l\} = \{i, j\}$, and put $t = \xi$. As $H_k = \varepsilon_k(\lambda_k \eta - s_k(\xi))$ and λ_k is a unit at v , we have $\eta \equiv s_k(t)/\lambda_k$ modulo v , hence $f_k(t) \equiv \lambda_k^d F(x, y) = 0$ and $g_{kl}(t) \equiv \lambda_k^{d_l} H_l \equiv 0$ modulo v . The resultant can be written as $R_{ij} = Af_k + Cg_{kl}$ with polynomials A and C with coefficients in $\mathbb{Z}[\omega]$, so v divides R_{ij} , a contradiction. By (H2), a term of (4) in which all arguments are units vanishes. Now let v divide H_i . Collecting the terms of (4) which contain H_i , and using (H1) and $\rho_{ji} = -w_{ij}$ for the edges (j, i) with $j < i$, we obtain $\beta_v(H_i, \Pi_i)$ with $\Pi_i = c_i \prod_j H_j^{\rho_{ij}}$, which is a unit. We claim that $D_i^3 \Pi_i \equiv W^3$ modulo v for some $W \in K$ integral at v . If i is linear, put $t = \xi$; then $g_{ij}(t) \equiv \lambda_i^{d_j} H_j$ and $f_i(t) \equiv 0$ modulo v as above, and (2) gives $D_i^3 \lambda_i^{E_i} \Pi_i \equiv w_i(t)^3$ modulo v , where $E_i = e_i + \sum_j \rho_{ij} d_j$ is divisible by 3; this gives the claim with $W = w_i(t)/\lambda_i^{E_i/3}$. Otherwise, (3) gives the claim with $W = W_i(x, y)$. Hence the reduction of Π_i is a non-zero cube in k_v , $\chi(\Pi_i) = 0$, and $\beta_v(H_i, \Pi_i) = v(H_i)\chi(\Pi_i) = 0$ by (H2). Hence $B_v^{(g)} = 0$, and $B_p^{(g)} = 0$.

Step 4: the primes in S . Let $p \in S$. Here 2 is inert, 5 is inert, the places above 7 are v_2 and v_4 , the places above 13 are v_3 and v_9 , and the places above 19 are v_7 and v_{11} . Suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Then $H_i \equiv H_i(a, b) \pmod{p^k}$, and Lemma 3, applied with ζ the residue of $H_i(a, b)$, gives a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$; when H_i has rational coefficients, $H_i \in \mathbb{Z}$, and we use this in (i) and (iv). With $\gamma_i = c_p(c_i)$, if the hypotheses of Lemma 4 hold, then $B_p^{(g)}$ is determined by a , b and k . Starting from the solutions of (1) modulo p , we refine a residue class $x \equiv a, y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until these hypotheses hold for every certificate, or until no solution remains. Table 6 lists the resulting terminal classes and the value of the vector $(B_p^{(g)})_g$ in each; every integer solution lies in one of them. The result is $(B_2^{(g)})_g = (1, 1, 2)$, $(B_3^{(g)})_g \in \{(0, 0, 1), (0, 0, 2), (0, 1, 0), (0, 2, 1), (0, 2, 2)\}$, $(B_5^{(g)})_g = (0, 0, 0)$, $(B_7^{(g)})_g = (0, 0, 1)$, $(B_{13}^{(g)})_g \in \{(0, 0, 0), (0, 1, 0), (0, 2, 0), (1, 0, 0), (1, 1, 0), (1, 2, 0), (2, 0, 0), (2, 2, 0)\}$, $(B_{19}^{(g)})_g = (0, 0, 0)$.

Step 5: contradiction. By Steps 2 and 3, $\sum_{p \in S} B_p^{(g)} = 0$ for every g . But by Step 4, the vector $(\sum_{p \in S} B_p^{(g)})_g$ lies in $\{(0, 0, 1), (0, 0, 2), (0, 1, 0), (0, 1, 1), (0, 1, 2), (0, 2, 0), (0, 2, 1), (0, 2, 2), (1, 0, 0), (1, 0, 1), (1, 0, 2), (1, 1, 0), (1, 1, 1), (1, 1, 2), (1, 2, 0), (1, 2, 1), (1, 2, 2), (2, 0, 0), (2, 0, 1), (2, 0, 2), (2, 1, 0), (2, 1, 1), (2, 1, 2), (2, 2, 0), (2, 2, 1), (2, 2, 2)\}$, which does not contain $(0, 0, 0)$. This contradiction proves the theorem. $\square$

References

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Table 1: The polynomials H_i , the constants c_i and, for the linear vertices, $\eta = r_i(\xi)$ of the first certificate. Here $\langle A, B \rangle = A + B\omega$.

i	H_i	c_i	chart
0	$\langle -y, 0 \rangle$	$\langle 0, -2 \rangle$	$y = \langle 0, 0 \rangle$
1	$\langle -y + 1, 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 1, 1 \rangle$
2	$\langle -y, -1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, -1 \rangle$
3	$\langle x, 0 \rangle$	$\langle 4, 0 \rangle$	$x = \langle 0, 0 \rangle$
4	$\langle x + 1, 1 \rangle$	$\langle 1, 3 \rangle$	$x = \langle -1, -1 \rangle$
5	$\langle x + 1, 0 \rangle$	$\langle 0, -3 \rangle$	$x = \langle -1, 0 \rangle$
6	$\langle x, 1 \rangle$	$\langle 4, 5 \rangle$	$x = \langle 0, -1 \rangle$
7	$\langle x, -1 \rangle$	$\langle -1, -1 \rangle$	$x = \langle 0, 1 \rangle$
8	$\langle x - 1, -1 \rangle$	$\langle -1, 2 \rangle$	$x = \langle 1, 1 \rangle$
9	$\langle x - 1, 1 \rangle$	$\langle 1, 0 \rangle$	$x = \langle 1, -1 \rangle$
10	$\langle x - 2, -1 \rangle$	$\langle 5, 8 \rangle$	$x = \langle 2, 1 \rangle$
11	$\langle x + 2, 0 \rangle$	$\langle 0, -2 \rangle$	$x = \langle -2, 0 \rangle$
12	$\langle x - 2, 0 \rangle$	$\langle 1, -1 \rangle$	$x = \langle 2, 0 \rangle$
13	$\langle 2x + 1, 0 \rangle$	$\langle 4, 0 \rangle$	–
14	$\langle 2x - 1, 0 \rangle$	$\langle -21, 0 \rangle$	–
15	$\langle 2x + 2, 1 \rangle$	$\langle -3, -8 \rangle$	–
16	$\langle 2x + 1, -1 \rangle$	$\langle 49, 0 \rangle$	–
17	$\langle -x - y, x \rangle$	$\langle 16, 6 \rangle$	$y = \langle -x, x \rangle$
18	$\langle -2x - y, -x \rangle$	$\langle 1, 0 \rangle$	$y = \langle -2x, -x \rangle$
19	$\langle x - y, -x + 1 \rangle$	$\langle 80, -19 \rangle$	$y = \langle x, -x + 1 \rangle$
20	$\langle x - y, -y \rangle$	$\langle 1, -1 \rangle$	$y = \langle 0, -x \rangle$
21	$\langle x - y, 0 \rangle$	$\langle 5, 8 \rangle$	$y = \langle x, 0 \rangle$
22	$\langle x, y \rangle$	$\langle 9, 3 \rangle$	$y = \langle x, x \rangle$
23	$\langle x + y, 0 \rangle$	$\langle -28, 32 \rangle$	$y = \langle -x, 0 \rangle$
24	$\langle x + y, y \rangle$	$\langle -10, -2 \rangle$	$y = \langle 0, x \rangle$
25	$\langle x + 1, -y + 1 \rangle$	$\langle -3, -2 \rangle$	$y = \langle -x, -x - 1 \rangle$
26	$\langle x + y + 1, 1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle -x - 1, -1 \rangle$
27	$\langle x + y + 1, y + 1 \rangle$	$\langle -6, 0 \rangle$	$y = \langle -1, x \rangle$
28	$\langle x - y + 1, -y \rangle$	$\langle 4, 0 \rangle$	$y = \langle 0, -x - 1 \rangle$
29	$\langle x - y + 1, 0 \rangle$	$\langle 4, 0 \rangle$	$y = \langle x + 1, 0 \rangle$
30	$\langle x + 1, -y \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x - 1, -x - 1 \rangle$
31	$\langle x + 1, y \rangle$	$\langle 0, -1 \rangle$	$y = \langle x + 1, x + 1 \rangle$
32	$\langle x + y + 1, y \rangle$	$\langle 2, 0 \rangle$	$y = \langle 0, x + 1 \rangle$
33	$\langle x - y, 1 \rangle$	$\langle -1, -1 \rangle$	$y = \langle x, 1 \rangle$
34	$\langle x + y, 1 \rangle$	$\langle -3, -8 \rangle$	$y = \langle -x, -1 \rangle$
35	$\langle x - y, -y - 1 \rangle$	$\langle -1, -1 \rangle$	$y = \langle -1, -x - 1 \rangle$
36	$\langle x, -y - 1 \rangle$	$\langle -4, -4 \rangle$	$y = \langle -x - 1, -x \rangle$
37	$\langle x + y, -1 \rangle$	$\langle -3, -2 \rangle$	$y = \langle -x, 1 \rangle$
38	$\langle x + y, y - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 1, x + 1 \rangle$
39	$\langle x - y - 1, 0 \rangle$	$\langle 0, -12 \rangle$	$y = \langle x - 1, 0 \rangle$
40	$\langle x + y - 1, 0 \rangle$	$\langle 2, 0 \rangle$	$y = \langle -x + 1, 0 \rangle$
41	$\langle x - y - 1, -1 \rangle$	$\langle -12, -32 \rangle$	$y = \langle x - 1, -1 \rangle$
42	$\langle x + y - 1, y + 1 \rangle$	$\langle -4, -8 \rangle$	$y = \langle -1, x - 2 \rangle$
43	$\langle x - 2, -y - 1 \rangle$	$\langle 2, 10 \rangle$	$y = \langle -x + 1, -x + 2 \rangle$
44	$\langle 2x + 1, y \rangle$	$\langle 222, 36 \rangle$	$y = \langle 2x + 1, 2x + 1 \rangle$
45	$\langle 2x + y + 1, y \rangle$	$\langle 0, -1 \rangle$	$y = \langle 0, 2x + 1 \rangle$
46	$\langle 2x - y - 1, -y \rangle$	$\langle 10, 16 \rangle$	$y = \langle 0, -2x + 1 \rangle$
47	$\langle 2x - 1, y \rangle$	$\langle -1, -1 \rangle$	$y = \langle 2x - 1, 2x - 1 \rangle$
48	$\langle -y + 1, -x^2 + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 1, -x^2 + 1 \rangle$
49	$\langle x^2 - y, x^2 + 1 \rangle$	$\langle 0, -1 \rangle$	$y = \langle x^2, x^2 + 1 \rangle$
50	$\langle -y, x^2 + x \rangle$	$\langle 5, 8 \rangle$	$y = \langle 0, x^2 + x \rangle$
51	$\langle -x^2 - y, x \rangle$	$\langle 0, -1 \rangle$	$y = \langle -x^2, x \rangle$
52	$\langle x^2 - x - y, x^2 \rangle$	$\langle 0, -1 \rangle$	$y = \langle x^2 - x, x^2 \rangle$
53	$\langle -y - 1, -x^2 \rangle$	$\langle 3, 4 \rangle$	$y = \langle -1, -x^2 \rangle$

i	H_i	c_i	chart
54	$\langle x^2 - y + 1, x^2 + x + 1 \rangle$	$\langle 3, -5 \rangle$	$y = \langle x^2 + 1, x^2 + x + 1 \rangle$
55	$\langle -y + 1, -x^2 + x + 1 \rangle$	$\langle -1, -2 \rangle$	$y = \langle 1, -x^2 + x + 1 \rangle$
56	$\langle -x^2 + x - y + 1, -x^2 + x \rangle$	$\langle -1, -1 \rangle$	$y = \langle -x^2 + x + 1, -x^2 + x \rangle$
57	$\langle x^2 + x - y + 1, 0 \rangle$	$\langle -4, -8 \rangle$	$y = \langle x^2 + x + 1, 0 \rangle$
58	$\langle x^2 - x - y + 1, 0 \rangle$	$\langle 10, 16 \rangle$	$y = \langle x^2 - x + 1, 0 \rangle$
59	$\langle -x^2 - x - y + 1, -x^2 \rangle$	$\langle -3, -8 \rangle$	$y = \langle -x^2 - x + 1, -x^2 \rangle$
60	$\langle x - y, -x^2 + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x, -x^2 + 1 \rangle$
61	$\langle -x - y, x^2 - x + 1 \rangle$	$\langle -2, -4 \rangle$	$y = \langle -x, x^2 - x + 1 \rangle$
62	$\langle -x - y, -x^2 - x + 1 \rangle$	$\langle -4, -4 \rangle$	$y = \langle -x, -x^2 - x + 1 \rangle$
63	$\langle -y - 1, -x^2 + x \rangle$	$\langle 0, -3 \rangle$	$y = \langle -1, -x^2 + x \rangle$
64	$\langle -x^2 - x - y - 1, 0 \rangle$	$\langle 4, 0 \rangle$	$y = \langle -x^2 - x - 1, 0 \rangle$
65	$\langle x - y + 1, x^2 + x - 1 \rangle$	$\langle 2, 3 \rangle$	$y = \langle x + 1, x^2 + x - 1 \rangle$
66	$\langle x + y^2 + y, 0 \rangle$	$\langle 2, 0 \rangle$	$x = \langle -y^2 - y, 0 \rangle$

Table 2: The polynomials H_i , the constants c_i and, for the linear vertices, $\eta = r_i(\xi)$ of the second certificate. Here $\langle A, B \rangle = A + B\omega$.

i	H_i	c_i	chart
0	$\langle -y, -1 \rangle$	$\langle 0, -2 \rangle$	$y = \langle 0, -1 \rangle$
1	$\langle -y - 1, 0 \rangle$	$\langle -4, -4 \rangle$	$y = \langle -1, 0 \rangle$
2	$\langle -y - 1, -1 \rangle$	$\langle -4, -8 \rangle$	$y = \langle -1, -1 \rangle$
3	$\langle x + 1, 1 \rangle$	$\langle 5, 8 \rangle$	$x = \langle -1, -1 \rangle$
4	$\langle x + 1, 0 \rangle$	$\langle 0, -3 \rangle$	$x = \langle -1, 0 \rangle$
5	$\langle x, 1 \rangle$	$\langle -5, -1 \rangle$	$x = \langle 0, -1 \rangle$
6	$\langle x, -1 \rangle$	$\langle 10, 1 \rangle$	$x = \langle 0, 1 \rangle$
7	$\langle x - 1, -1 \rangle$	$\langle 0, -3 \rangle$	$x = \langle 1, 1 \rangle$
8	$\langle x + 2, 0 \rangle$	$\langle 6, 2 \rangle$	$x = \langle -2, 0 \rangle$
9	$\langle x - 2, 0 \rangle$	$\langle -12, -12 \rangle$	$x = \langle 2, 0 \rangle$
10	$\langle 2x - 1, 0 \rangle$	$\langle 14, -14 \rangle$	–
11	$\langle 2x + 1, -1 \rangle$	$\langle 434, 518 \rangle$	–
12	$\langle -y - 1, -x + 1 \rangle$	$\langle -11, -13 \rangle$	$y = \langle -1, -x + 1 \rangle$
13	$\langle x - y, -y \rangle$	$\langle 7, 15 \rangle$	$y = \langle 0, -x \rangle$
14	$\langle x, y \rangle$	$\langle -4, 8 \rangle$	$y = \langle x, x \rangle$
15	$\langle x + y + 1, 1 \rangle$	$\langle 3, 1 \rangle$	$y = \langle -x - 1, -1 \rangle$
16	$\langle x - y + 1, -y \rangle$	$\langle 10, 1 \rangle$	$y = \langle 0, -x - 1 \rangle$
17	$\langle x + 1, -y \rangle$	$\langle -8, -5 \rangle$	$y = \langle -x - 1, -x - 1 \rangle$
18	$\langle x + y + 1, 0 \rangle$	$\langle 2, 0 \rangle$	$y = \langle -x - 1, 0 \rangle$
19	$\langle x + y + 1, y \rangle$	$\langle 52, 44 \rangle$	$y = \langle 0, x + 1 \rangle$
20	$\langle x + y, 1 \rangle$	$\langle 3, 1 \rangle$	$y = \langle -x, -1 \rangle$
21	$\langle x + y, -1 \rangle$	$\langle 0, -3 \rangle$	$y = \langle -x, 1 \rangle$
22	$\langle x + 2, y + 1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle x + 1, x + 2 \rangle$
23	$\langle x + y - 1, y + 1 \rangle$	$\langle -12, -8 \rangle$	$y = \langle -1, x - 2 \rangle$
24	$\langle 2x + y, 0 \rangle$	$\langle -12, -32 \rangle$	$y = \langle -2x, 0 \rangle$
25	$\langle 2x - y + 1, 0 \rangle$	$\langle -13, -13 \rangle$	$y = \langle 2x + 1, 0 \rangle$
26	$\langle 2x + y + 1, y \rangle$	$\langle -60, -28 \rangle$	$y = \langle 0, 2x + 1 \rangle$
27	$\langle 2x + y, y + 1 \rangle$	$\langle -52, -39 \rangle$	$y = \langle -1, 2x - 1 \rangle$
28	$\langle 2x - y - 1, -y \rangle$	$\langle -30, -14 \rangle$	$y = \langle 0, -2x + 1 \rangle$
29	$\langle 2x + y - 1, 0 \rangle$	$\langle 8, 3 \rangle$	$y = \langle -2x + 1, 0 \rangle$
30	$\langle -x^2 - y, -x^2 + 1 \rangle$	$\langle -3, 0 \rangle$	$y = \langle -x^2, -x^2 + 1 \rangle$
31	$\langle -x - y, -x^2 \rangle$	$\langle -1, -1 \rangle$	$y = \langle -x, -x^2 \rangle$
32	$\langle -x^2 - x - y, -x^2 - x \rangle$	$\langle -3, -8 \rangle$	$y = \langle -x^2 - x, -x^2 - x \rangle$
33	$\langle -y - 1, -x^2 \rangle$	$\langle -1, 3 \rangle$	$y = \langle -1, -x^2 \rangle$
34	$\langle -x^2 - y - 1, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 - 1, 0 \rangle$
35	$\langle -y - 1, x^2 - 1 \rangle$	$\langle -8, -5 \rangle$	$y = \langle -1, x^2 - 1 \rangle$
36	$\langle x^2 - x - y + 1, 0 \rangle$	$\langle 0, -2 \rangle$	$y = \langle x^2 - x + 1, 0 \rangle$
37	$\langle -x - y, -x^2 - x + 1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle -x, -x^2 - x + 1 \rangle$

i	H_i	c_i	chart
38	$\langle x - y, x^2 - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x, x^2 - 1 \rangle$
39	$\langle -x^2 - y, -x - 1 \rangle$	$\langle -1, -2 \rangle$	$y = \langle -x^2, -x - 1 \rangle$
40	$\langle -x^2 - x - y - 1, 0 \rangle$	$\langle -6, -6 \rangle$	$y = \langle -x^2 - x - 1, 0 \rangle$
41	$\langle x^2 - x - y - 1, -x \rangle$	$\langle 1, 0 \rangle$	$y = \langle x^2 - x - 1, -x \rangle$
42	$\langle -x^2 - x - y - 1, -x^2 - x \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 - x - 1, -x^2 - x \rangle$
43	$\langle x, -y^2 - 1 \rangle$	$\langle 4, 6 \rangle$	$x = \langle 0, y^2 + 1 \rangle$
44	$\langle x, 0 \rangle$	$\langle 2, 0 \rangle$	$x = \langle 0, 0 \rangle$
45	$\langle x + y + 2, 0 \rangle$	$\langle -1, -1 \rangle$	$y = \langle -x - 2, 0 \rangle$
46	$\langle x - y, -y - 1 \rangle$	$\langle 3, 1 \rangle$	$y = \langle -1, -x - 1 \rangle$
47	$\langle -y + 1, x + 2 \rangle$	$\langle -8, -6 \rangle$	$y = \langle 1, x + 2 \rangle$
48	$\langle -x^2 - y + 1, 0 \rangle$	$\langle -3, 0 \rangle$	$y = \langle -x^2 + 1, 0 \rangle$
49	$\langle x + 2, 2 \rangle$	$\langle -96, -11 \rangle$	$x = \langle -2, -2 \rangle$
50	$\langle x - 2, -y - 1 \rangle$	$\langle 9, 3 \rangle$	$y = \langle -x + 1, -x + 2 \rangle$
51	$\langle 3x - y, -2y \rangle$	$\langle 4, 6 \rangle$	$\langle 1, -1 \rangle y = \langle -3x, -3x \rangle$
52	$\langle 3x + 1, -1 \rangle$	$\langle 4, -23 \rangle$	$3x = \langle -1, 1 \rangle$
53	$\langle -y - 2, -1 \rangle$	$\langle -8, -4 \rangle$	$y = \langle -2, -1 \rangle$
54	$\langle x - 2, -1 \rangle$	$\langle 1, 3 \rangle$	$x = \langle 2, 1 \rangle$
55	$\langle 2x + 1, -y \rangle$	$\langle 156, 624 \rangle$	$y = \langle -2x - 1, -2x - 1 \rangle$
56	$\langle x + y, 0 \rangle$	$\langle -23, -22 \rangle$	$y = \langle -x, 0 \rangle$
57	$\langle -y + 1, -x \rangle$	$\langle -1, -10 \rangle$	$y = \langle 1, -x \rangle$
58	$\langle x + 1, y + 1 \rangle$	$\langle -1, -1 \rangle$	$y = \langle x, x + 1 \rangle$
59	$\langle x - y - 1, -1 \rangle$	$\langle -7, 8 \rangle$	$y = \langle x - 1, -1 \rangle$
60	$\langle x - y, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x, 0 \rangle$
61	$\langle x, -y - 1 \rangle$	$\langle 2, -2 \rangle$	$y = \langle -x - 1, -x \rangle$
62	$\langle x - y + 1, x \rangle$	$\langle 28, 60 \rangle$	$y = \langle x + 1, x \rangle$
63	$\langle x - y + 1, x + 1 \rangle$	$\langle -60, -28 \rangle$	$y = \langle x + 1, x + 1 \rangle$
64	$\langle x + y, y - 1 \rangle$	$\langle -3, -3 \rangle$	$y = \langle 1, x + 1 \rangle$
65	$\langle -x - y, -x - 1 \rangle$	$\langle 0, -3 \rangle$	$y = \langle -x, -x - 1 \rangle$
66	$\langle -x - y, x \rangle$	$\langle -8, -4 \rangle$	$y = \langle -x, x \rangle$
67	$\langle x - y + 1, 1 \rangle$	$\langle 1, 4 \rangle$	$y = \langle x + 1, 1 \rangle$
68	$\langle x - y, -y + 1 \rangle$	$\langle -3, -3 \rangle$	$y = \langle 1, -x + 1 \rangle$
69	$\langle x - y - 1, 0 \rangle$	$\langle 4, -4 \rangle$	$y = \langle x - 1, 0 \rangle$
70	$\langle 3x - 2y - 1, -y - 2 \rangle$	$\langle -30, -14 \rangle$	$\langle 1, -1 \rangle y = \langle -2, -3x - 1 \rangle$
71	$\langle 2x + 2, 1 \rangle$	$\langle 34, 58 \rangle$	—
72	$\langle 2x - 1, -y - 1 \rangle$	$\langle -30, -14 \rangle$	$y = \langle -2x, -2x + 1 \rangle$
73	$\langle -y + 1, x^2 + 1 \rangle$	$\langle -2, -1 \rangle$	$y = \langle 1, x^2 + 1 \rangle$

Table 3: The polynomials H_i , the constants c_i and, for the linear vertices, $\eta = r_i(\xi)$ of the third certificate. Here $\langle A, B \rangle = A + B\omega$.

i	H_i	c_i	chart
0	$\langle -y, 0 \rangle$	$\langle 4, 0 \rangle$	$y = \langle 0, 0 \rangle$
1	$\langle -y + 1, 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 1, 1 \rangle$
2	$\langle -y, -1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, -1 \rangle$
3	$\langle -y - 1, 0 \rangle$	$\langle 0, -1 \rangle$	$y = \langle -1, 0 \rangle$
4	$\langle x, 0 \rangle$	$\langle 1, 0 \rangle$	$x = \langle 0, 0 \rangle$
5	$\langle x + 1, 1 \rangle$	$\langle 5, 8 \rangle$	$x = \langle -1, -1 \rangle$
6	$\langle x + 1, 0 \rangle$	$\langle 1, 0 \rangle$	$x = \langle -1, 0 \rangle$
7	$\langle x, 1 \rangle$	$\langle 1, 3 \rangle$	$x = \langle 0, -1 \rangle$
8	$\langle x, -1 \rangle$	$\langle 6, 9 \rangle$	$x = \langle 0, 1 \rangle$
9	$\langle x - 1, -1 \rangle$	$\langle -1, -2 \rangle$	$x = \langle 1, 1 \rangle$
10	$\langle x - 1, 1 \rangle$	$\langle -1, -1 \rangle$	$x = \langle 1, -1 \rangle$
11	$\langle x + 2, 0 \rangle$	$\langle -42, -28 \rangle$	$x = \langle -2, 0 \rangle$
12	$\langle x - 2, 0 \rangle$	$\langle -4, -8 \rangle$	$x = \langle 2, 0 \rangle$
13	$\langle 2x + 1, 0 \rangle$	$\langle 0, -1 \rangle$	—
14	$\langle 2x - 1, 0 \rangle$	$\langle -8, 44 \rangle$	—

i	H_i	c_i	chart
15	$\langle 2x + 2, 1 \rangle$	$\langle -10, 6 \rangle$	–
16	$\langle -x - y, x \rangle$	$\langle 5, 8 \rangle$	$y = \langle -x, x \rangle$
17	$\langle -2x - y, -x \rangle$	$\langle -2, 4 \rangle$	$y = \langle -2x, -x \rangle$
18	$\langle x - y, -y \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, -x \rangle$
19	$\langle x - y, 0 \rangle$	$\langle 14, 0 \rangle$	$y = \langle x, 0 \rangle$
20	$\langle x, -y \rangle$	$\langle -22, -26 \rangle$	$y = \langle -x, -x \rangle$
21	$\langle x, y \rangle$	$\langle 2, 0 \rangle$	$y = \langle x, x \rangle$
22	$\langle x + y, 0 \rangle$	$\langle 0, -4 \rangle$	$y = \langle -x, 0 \rangle$
23	$\langle x - y + 1, -y + 1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle 1, -x \rangle$
24	$\langle x + 1, -y + 1 \rangle$	$\langle -5, -1 \rangle$	$y = \langle -x, -x - 1 \rangle$
25	$\langle x + 1, y + 1 \rangle$	$\langle -3, -2 \rangle$	$y = \langle x, x + 1 \rangle$
26	$\langle x + y + 1, 1 \rangle$	$\langle -1, -1 \rangle$	$y = \langle -x - 1, -1 \rangle$
27	$\langle x + y + 1, y + 1 \rangle$	$\langle 0, -3 \rangle$	$y = \langle -1, x \rangle$
28	$\langle x - y + 1, -y \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, -x - 1 \rangle$
29	$\langle x - y + 1, 0 \rangle$	$\langle 4, 0 \rangle$	$y = \langle x + 1, 0 \rangle$
30	$\langle x + 1, -y \rangle$	$\langle -1, 2 \rangle$	$y = \langle -x - 1, -x - 1 \rangle$
31	$\langle x + 1, y \rangle$	$\langle 2, 0 \rangle$	$y = \langle x + 1, x + 1 \rangle$
32	$\langle x + y + 1, 0 \rangle$	$\langle 0, -2 \rangle$	$y = \langle -x - 1, 0 \rangle$
33	$\langle x + y + 1, y \rangle$	$\langle -2, 1 \rangle$	$y = \langle 0, x + 1 \rangle$
34	$\langle x - y, -y + 1 \rangle$	$\langle -1, -1 \rangle$	$y = \langle 1, -x + 1 \rangle$
35	$\langle x, -y - 1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle -x - 1, -x \rangle$
36	$\langle x + y, -1 \rangle$	$\langle 5, 8 \rangle$	$y = \langle -x, 1 \rangle$
37	$\langle x + y, y - 1 \rangle$	$\langle 4, -1 \rangle$	$y = \langle 1, x + 1 \rangle$
38	$\langle x - y - 1, -y - 1 \rangle$	$\langle 4, 0 \rangle$	$y = \langle -1, -x \rangle$
39	$\langle x + y + 1, y - 1 \rangle$	$\langle 1, -1 \rangle$	$y = \langle 1, x + 2 \rangle$
40	$\langle x + y - 1, y + 1 \rangle$	$\langle 13, 11 \rangle$	$y = \langle -1, x - 2 \rangle$
41	$\langle x - 2, -y - 1 \rangle$	$\langle 70, 56 \rangle$	$y = \langle -x + 1, -x + 2 \rangle$
42	$\langle 2x + y, 0 \rangle$	$\langle 16, 6 \rangle$	$y = \langle -2x, 0 \rangle$
43	$\langle 2x + y + 1, y \rangle$	$\langle 0, -1 \rangle$	$y = \langle 0, 2x + 1 \rangle$
44	$\langle 2x - y - 1, -y \rangle$	$\langle -6, -4 \rangle$	$y = \langle 0, -2x + 1 \rangle$
45	$\langle 2x - 1, y \rangle$	$\langle 12, 4 \rangle$	$y = \langle 2x - 1, 2x - 1 \rangle$
46	$\langle -y + 1, -x^2 + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 1, -x^2 + 1 \rangle$
47	$\langle -x^2 - y, -x^2 + 1 \rangle$	$\langle 1, 3 \rangle$	$y = \langle -x^2, -x^2 + 1 \rangle$
48	$\langle -y, x^2 + x \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, x^2 + x \rangle$
49	$\langle x^2 - x - y, x^2 \rangle$	$\langle -3, 0 \rangle$	$y = \langle x^2 - x, x^2 \rangle$
50	$\langle -x - y, -x^2 \rangle$	$\langle -2, -1 \rangle$	$y = \langle -x, -x^2 \rangle$
51	$\langle -x^2 - x - y, -x^2 - x \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 - x, -x^2 - x \rangle$
52	$\langle -x^2 - y - 1, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 - 1, 0 \rangle$
53	$\langle x^2 + x - y + 1, 0 \rangle$	$\langle -6, -6 \rangle$	$y = \langle x^2 + x + 1, 0 \rangle$
54	$\langle x^2 - x - y + 1, 0 \rangle$	$\langle -1, 2 \rangle$	$y = \langle x^2 - x + 1, 0 \rangle$
55	$\langle x^2 - y, -x + 1 \rangle$	$\langle 0, -1 \rangle$	$y = \langle x^2, -x + 1 \rangle$
56	$\langle -x - y, -x^2 - x + 1 \rangle$	$\langle -2, -2 \rangle$	$y = \langle -x, -x^2 - x + 1 \rangle$
57	$\langle -x - y, -x^2 - x - 1 \rangle$	$\langle -1, 2 \rangle$	$y = \langle -x, -x^2 - x - 1 \rangle$
58	$\langle -x^2 - x - y - 1, 0 \rangle$	$\langle -1, -1 \rangle$	$y = \langle -x^2 - x - 1, 0 \rangle$
59	$\langle x^2 - x - y - 1, x^2 - x \rangle$	$\langle -1, -1 \rangle$	$y = \langle x^2 - x - 1, x^2 - x \rangle$
60	$\langle -x^2 - y - 1, -x^2 + 1 \rangle$	$\langle -1, -1 \rangle$	$y = \langle -x^2 - 1, -x^2 + 1 \rangle$
61	$\langle -x^2 - 3y + 1, -2x^2 - 1 \rangle$	$\langle -3, -3 \rangle$	$3y = \langle -x^2 + 1, -2x^2 - 1 \rangle$
62	$\langle -x^2 - x - 3y, -2x^2 - 2x \rangle$	$\langle -1, -1 \rangle$	$3y = \langle -x^2 - x, -2x^2 - 2x \rangle$
63	$\langle x + y^2 + 1, y^2 + 1 \rangle$	$\langle 20, 32 \rangle$	$x = \langle -y^2 - 1, -y^2 - 1 \rangle$

Table 4: The data of the identities (2) and (3); the entry $j:e$ means $\rho_{ij} = e \neq 0$.

The first certificate.

$i = 0$, $D_i = 2$, ρ_{ij} : 6:1, 15:2, 27:2, 47:2, 48:2, 57:2, 63:1, 66:2

w_i $\langle -4t^3 - 12t^2 - 15t - 10,$
 $14t^3 + 4t^2 - 9t - 14 \rangle$

$i = 1$, $D_i = 8$, ρ_{ij} : 3:2, 6:1, 16:2, 17:2, 18:1, 22:2, 37:1, 49:2, 50:2, 51:1, 52:2, 55:1, 65:1

w_i $\langle -34t^3 - 152t^2 - 130t + 16,$

	$138t^3 - 32t^2 - 160t - 60)$
$i = 2, D_i = 4, \rho_{ij}: 15:1, 17:2, 18:1, 26:2, 28:1, 35:1, 52:2, 63:2$	
w_i	$\langle -16t^3 - 2t^2 + 8t - 2,$ $12t^3 + 12t^2 - 10 \rangle$
$i = 3, D_i = 1, \rho_{ij}: 1:1, 19:2, 25:2, 44:1, 59:1, 61:1, 66:2$	
w_i	$\langle -2t^2 + 2t - 4,$ $-2t^2 - 2 \rangle$
$i = 4, D_i = 4, \rho_{ij}: 21:2, 30:1, 36:1, 47:1, 51:1, 52:2, 62:2, 64:1$	
w_i	$\langle 20t + 12,$ $32t + 36 \rangle$
$i = 5, D_i = 8, \rho_{ij}: 17:1, 18:2, 30:2, 33:1, 37:1, 39:2, 40:1, 41:1, 42:1, 45:2, 53:1, 56:2, 58:2, 60:1, 63:2, 66:2$	
w_i	$\langle 504t^2 - 72t - 1512,$ $864t^2 + 1152t \rangle$
$i = 6, D_i = 152, \rho_{ij}: 0:2, 1:2, 18:1, 23:2, 25:1, 26:2, 32:2, 38:2, 39:1, 46:1, 48:1, 50:1, 53:1, 60:1, 62:1, 64:1, 66:2$	
w_i	$\langle -4400t^2 - 17984t - 26568,$ $-16168t^2 - 25504t - 9648 \rangle$
$i = 7, D_i = 2, \rho_{ij}: 41:1, 42:1, 45:1, 47:1, 51:1, 52:2, 56:2$	
w_i	$\langle -8,$ $-2t^2 - 2t + 2 \rangle$
$i = 8, D_i = 152, \rho_{ij}: 27:1, 28:1, 30:2, 37:2, 38:2, 39:2, 40:2, 42:1, 47:2, 58:1, 59:2, 63:1, 64:2, 65:2, 66:2$	
w_i	$\langle 968t^2 - 10216t - 16800,$ $8200t^2 + 8560t - 9912 \rangle$
$i = 9, D_i = 2, \rho_{ij}: 22:1, 25:2, 43:2, 46:2, 48:2, 49:1, 58:1$	
w_i	$\langle -4t^2 + 24t + 28,$ $-2t^2 - 2t + 70 \rangle$
$i = 10, D_i = 2, \rho_{ij}: 20:2, 26:1, 33:1, 42:2, 56:1$	
w_i	$\langle 6t^2 - 2t - 18,$ $4t^2 + 8t - 12 \rangle$
$i = 11, D_i = 4, \rho_{ij}: 20:1, 22:1, 25:1, 26:1, 29:2, 39:1, 42:2, 53:2, 59:2$	
w_i	$\langle 32t^2 - 544t - 576,$ $140t^2 - 124t - 904 \rangle$
$i = 12, D_i = 2, \rho_{ij}: 21:1, 28:2, 46:2, 52:2, 58:2$	
w_i	$\langle 16t^2 - 24t + 144,$ $14t^2 - 30t + 180 \rangle$
$i = 13, D_i = 1, \rho_{ij}: 23:1, 35:1, 36:1$	
W	$\langle 2,$ $0 \rangle$
A	$\langle -4,$ $0 \rangle$
B	$\langle 4x^3 - 4y^2 - 4y,$ $-4xy - 4x - 4y^2 - 4y \rangle$
$i = 14, D_i = 32, \rho_{ij}: 25:1, 26:2, 28:1, 31:1, 32:2, 33:2, 35:2, 37:2, 38:1, 42:2, 55:1$	
W	$\langle 84y^2 - 126y - 84,$ $-504y^2 + 210y - 861 \rangle$
A	$\langle -688128x^{13}y - 688128x^{13} - 1032192x^{12}y^2 - 4472832x^{12}y - 4128768x^{12}$ $+ 688128x^{11}y^3 - 6193152x^{11}y^2 - 15482880x^{11}y - 9289728x^{11} + 1032192x^{10}y^4$ $- 5160960x^{10}y^3 - 18923520x^{10}y^2 - 18923520x^{10}y - 5505024x^{10} - 7225344x^9y^4$ $- 20643840x^9y^3 - 9117696x^9y^2 + 23568384x^9y + 23740416x^9 + 1032192x^8y^6$ $- 2236416x^8y^5 - 13418496x^8y^4 + 6021120x^8y^3 + 76382208x^8y^2 + 128507904x^8y$ $+ 78962688x^8 + 1376256x^7y^7 - 2064384x^7y^6 + 12730368x^7y^5 + 69844992x^7y^4$ $+ 135389184x^7y^3 + 214351872x^7y^2 + 237920256x^7y + 120766464x^7 - 12214272x^6y^7$ $+ 25116672x^6y^6 + 123518976x^6y^5 + 217964544x^6y^4 + 286089216x^6y^3$ $+ 273272832x^6y^2 + 209104896x^6y + 84467712x^6 - 1376256x^5y^9 - 4128768x^5y^8$ $+ 44556288x^5y^7 + 114573312x^5y^6 + 196976640x^5y^5 + 237232128x^5y^4$ $+ 175644672x^5y^3 + 33374208x^5y^2 - 64081920x^5y - 61157376x^5 - 1032192x^4y^{10}$ $+ 8773632x^4y^9 + 51179520x^4y^8 + 112422912x^4y^7 + 117239808x^4y^6$ $- 28815360x^4y^5 - 261918720x^4y^4 - 460271616x^4y^3 - 575016960x^4y^2$ $- 494850048x^4y - 255639552x^4 + 6193152x^3y^{10} + 36470784x^3y^9 + 70705152x^3y^8$ $- 36556800x^3y^7 - 324366336x^3y^6 - 782659584x^3y^5 - 1258930176x^3y^4$ $- 1402576896x^3y^3 - 1187235840x^3y^2 - 783218688x^3y - 357740544x^3 - 1032192x^2y^{12}$ $+ 5505024x^2y^{11} + 46018560x^2y^{10} + 12128256x^2y^9 - 185450496x^2y^8$ $- 551706624x^2y^7 - 1098252288x^2y^6 - 1626777600x^2y^5 - 1877299200x^2y^4$ $- 1716320256x^2y^3 - 1196826624x^2y^2 - 604735488x^2y - 247941120x^2 - 688128xy^{13}$ $+ 12386304xy^{12} + 33030144xy^{11} - 36900864xy^{10} - 195428352xy^9 - 499968000xy^8$ $- 923080704xy^7 - 1217771520xy^6 - 1260091392xy^5 - 1039159296xy^4 - 643958784xy^3$ $- 270606336xy^2 + 62060544xy + 56168448x + 688128y^{14} + 2408448y^{13} + 3010560y^{12} - 1419264y^{11}$ $+ 50147328y^{10} + 145281024y^9 + 408619008y^8 + 687224832y^7 + 1149012480y^6 + 1121933568y^5$ $+ 1543519488y^4 + 673319808y^3 + 1056018432y^2 - 6263040y + 254443560,$ $344064x^{14} + 688128x^{13}y + 688128x^{13} + 344064x^{12}y^2 - 344064x^{12}y - 1720320x^{12}$ $+ 1376256x^{11}y^3 - 2752512x^{11}y^2 - 12386304x^{11}y - 10493952x^{11} + 2408448x^{10}y^4 \rangle$

$$\begin{aligned}
& -1892352x^{10}y^3 - 21159936x^{10}y^2 - 37502976x^{10}y - 23396352x^{10} - 7569408x^9y^4 \\
& -23052288x^9y^3 - 47652864x^9y^2 - 54878208x^9y - 22880256x^9 - 1720320x^8y^6 \\
& -13590528x^8y^5 - 33030144x^8y^4 - 5023344x^8y^3 - 38363136x^8y^2 - 7569408x^8y \\
& +17633280x^8 - 1376256x^7y^7 - 10321920x^7y^6 - 25804800x^7y^5 - 32514048x^7y^4 \\
& +7569408x^7y^3 + 94273536x^7y^2 + 153624576x^7y + 103391232x^7 - 1720320x^6y^8 \\
& -14966784x^6y^7 - 12644352x^6y^6 + 27353088x^6y^5 + 105369600x^6y^4 \\
& +268541952x^6y^3 + 378384384x^6y^2 + 362901504x^6y + 186912768x^6 - 2752512x^5y^9 \\
& -11010048x^5y^8 + 9289728x^5y^7 + 47136768x^5y^6 + 191471616x^5y^5 \\
& +442810368x^5y^4 + 611057664x^5y^3 + 567189504x^5y^2 + 421994496x^5y + 179214336x^5 \\
& -2752512x^4y^{10} + 10493952x^4y^9 + 45158400x^4y^8 + 81457152x^4y^7 \\
& +233963520x^4y^6 + 425521152x^4y^5 + 538116096x^4y^4 + 507623424x^4y^3 \\
& +293099520x^4y^2 + 141324288x^4y + 25632768x^4 + 18235392x^3y^{10} + 39567360x^3y^9 \\
& +118013952x^3y^8 + 239640576x^3y^7 + 257230848x^3y^6 + 94359552x^3y^5 \\
& -165795840x^3y^4 - 413478912x^3y^3 - 514074624x^3y^2 - 416274432x^3y - 216803328x^3 \\
& +1376256x^2y^{12} + 11182080x^2y^{11} + 34836480x^2y^{10} + 93628416x^2y^9 \\
& +86231040x^2y^8 - 79779840x^2y^7 - 485517312x^2y^6 - 1090639872x^2y^5 \\
& -1569619968x^2y^4 - 1685053440x^2y^3 - 1381632000x^2y^2 - 880115712x^2y \\
& -385588224x^2 + 688128xy^{13} + 10321920xy^{12} + 39567360xy^{11} + 35954688xy^{10} - 68554752xy^9 \\
& -407156736xy^8 - 1019160576xy^7 - 1736663040xy^6 - 2298734592xy^5 - 2534203392xy^4 \\
& -2250995712xy^3 - 1533450240xy^2 - 838785024xy - 323463168x - 516096y^{13} - 5849088y^{12} \\
& -25761792y^{11} - 35180544y^{10} - 17074176y^9 + 133324800y^8 + 413930496y^7 + 867535872y^6 \\
& +1191671040y^5 + 1573778304y^4 + 1294134912y^3 + 1086465408y^2 + 559169520y + 188491296) \\
B \quad & (344064x^3y^{14} + 8085504x^3y^{13} - 7225344x^3y^{12} - 60555264x^3y^{11} \\
& -214867968x^3y^{10} - 426123264x^3y^9 - 717244416x^3y^8 - 812163072x^3y^7 \\
& -896426496x^3y^6 - 313512192x^3y^5 - 242441472x^3y^4 + 759921792x^3y^3 \\
& +95585280x^3y^2 + 831479040x^3y + 118134744x^3 + 688128x^2y^{15} + 1204224x^2y^{14} \\
& -9031680x^2y^{13} - 39911424x^2y^{12} - 100380672x^2y^{11} - 121712640x^2y^{10} \\
& +49072128x^2y^9 + 494226432x^2y^8 + 1352386560x^2y^7 + 2257334016x^2y^6 \\
& +3233435520x^2y^5 + 3415284096x^2y^4 + 3447893568x^2y^3 + 2078372352x^2y^2 \\
& +1530227328x^2y + 523231212x^2 - 172032xy^{15} - 2322432xy^{14} - 20987904xy^{13} \\
& -12214272xy^{12} + 118358016xy^{11} + 486592512xy^{10} + 1263682560xy^9 + 2485808640xy^8 \\
& +3963079680xy^7 + 5141141376xy^6 + 5788063680xy^5 + 5314615488xy^4 + 4389862176xy^3 \\
& +2570980608xy^2 + 1370128704xy + 479752182x + 1032192y^{16} - 2150400y^{15} - 10795008y^{14} \\
& +23955456y^{13} + 160849920y^{12} + 486377472y^{11} + 1125261312y^{10} + 2035966464y^9 \\
& +3006296832y^8 + 3953198592y^7 + 4260801216y^6 + 4225567584y^5 + 3140112864y^4 \\
& +2515368912y^3 + 1028216448y^2 + 637489440y + 56829927, \\
& 1376256x^3y^{14} + 12902400x^3y^{13} + 16084992x^3y^{12} - 16859136x^3y^{11} \\
& -147345408x^3y^{10} - 506720256x^3y^9 - 1203406848x^3y^8 - 2042170368x^3y^7 \\
& -2943101952x^3y^6 - 3305084160x^3y^5 - 3363384192x^3y^4 - 2527647360x^3y^3 \\
& -1682405760x^3y^2 + 740426736x^3y - 206533152x^3 + 1376256x^2y^{15} + 5160960x^2y^{14} \\
& -11268096x^2y^{13} - 59738112x^2y^{12} - 213233664x^2y^{11} - 526934016x^2y^{10} \\
& -969056256x^2y^9 - 1460573184x^2y^8 - 1820259840x^2y^7 - 1901716992x^2y^6 \\
& -1430405760x^2y^5 - 884194752x^2y^4 - 141314880x^2y^3 + 173785920x^2y^2 \\
& +358127112x^2y + 250592496x^2 + 344064xy^{16} - 1720320xy^{15} - 14020608xy^{14} - 54448128xy^{13} \\
& -156033024xy^{12} - 313528320xy^{11} - 442681344xy^{10} - 416747520xy^9 - 13751808xy^8 \\
& +793976064xy^7 + 1790858496xy^6 + 2820742848xy^5 + 3297727776xy^4 + 3206562912xy^3 \\
& +2309686176xy^2 + 1429026564xy + 649907832x - 1204224y^{16} - 8429568y^{15} - 36427776y^{14} \\
& -70232064y^{13} - 74274816y^{12} + 51652608y^{11} + 467260416y^{10} + 1236996096y^9 \\
& +2325856512y^8 + 3593998464y^7 + 4540690560y^6 + 4958700768y^5 + 4527589584y^4 \\
& +3750486768y^3 + 2274794256y^2 + 1260639114y + 547634220)
\end{aligned}$$

$i = 15$, $D_i = 16$, ρ_{ij} : 0:1, 2:2, 22:1, 31:1, 32:2, 36:1, 37:1, 55:2, 64:1

$$\begin{aligned}
W \quad & (44y^2 - 11y + 12, \\
& 94y^2 - 69y + 81)
\end{aligned}$$

$$\begin{aligned}
A \quad & (-10240x^8y^3 - 12288x^8y^2 + 16384x^8y - 49152x^7y^4 + 53248x^7y^3 - 71680x^7y^2 \\
& +53248x^7y - 40960x^6y^5 + 30720x^6y^4 - 30720x^6y^3 + 47104x^6y^2 + 14336x^6y \\
& +12288x^5y^6 - 71680x^5y^5 + 292864x^5y^4 - 218112x^5y^3 + 278528x^5y^2 \\
& -106496x^5y + 57344x^4y^7 - 134144x^4y^6 + 529408x^4y^5 - 399360x^4y^4 \\
& +401408x^4y^3 - 27648x^4y^2 - 114688x^4y + 49152x^3y^8 - 129024x^3y^7 \\
& +204800x^3y^6 + 254976x^3y^5 - 671744x^3y^4 + 697344x^3y^3 - 535552x^3y^2 \\
& +11264x^3y + 6144x^2y^9 - 81920x^2y^8 - 240640x^2y^7 + 626688x^2y^6 \\
& -1389568x^2y^5 + 543744x^2y^4 - 475648x^2y^3 - 368640x^2y^2 + 65536x^2y \\
& -384000xy^8 + 372736xy^7 - 904192xy^6 - 757760xy^5 + 709632xy^4 - 1518592xy^3 + 470528xy^2 \\
& -104448xy - 213760y^8 + 160384y^7 - 268864y^6 - 683936y^5 + 1232448y^4 - 1246792y^3 \\
& +731440y^2 - 7832y + 18792, \\
& 6144x^8y^3 - 32768x^8y^2 + 10240x^8y - 30720x^7y^4 + 8192x^7y^3 - 57344x^7y^2 \\
& +8192x^7y - 75776x^6y^5 + 81920x^6y^4 - 182272x^6y^3 + 92160x^6y^2 - 28672x^6y \\
& -67584x^5y^6 + 43008x^5y^5 - 55296x^5y^4 + 70656x^5y^3 + 73728x^5y^2 + 33792x^5y \\
& -14336x^4y^7 - 39936x^4y^6 + 424960x^4y^5 - 462848x^4y^4 + 852992x^4y^3 \\
& -274432x^4y^2 + 229376x^4y + 30720x^3y^8 - 93184x^3y^7 + 629760x^3y^6 \\
& -373760x^3y^5 + 533680x^3y^4 + 404480x^3y^3 - 240640x^3y^2 + 264192x^3y \\
& +16384x^2y^9 - 101376x^2y^8 + 345088x^2y^7 + 266240x^2y^6 - 611328x^2y^5 \\
& +1148928x^2y^4 - 858624x^2y^3 + 221184x^2y^2 - 84480x^2y - 50176xy^9 - 20480xy^8)
\end{aligned}$$

$$\begin{aligned}
& +508928xy^7 - 1374208xy^6 + 755712xy^5 - 641536xy^4 - 1206272xy^3 + 485376xy^2 - 629760xy \\
& - 50176y^9 + 27904y^8 + 105728y^7 - 771328y^6 - 170112y^5 + 1151696y^4 - 3069848y^3 \\
& + 2733592y^2 - 1678000y + 505224) \\
B \quad & \langle -2048x^3y^{10} - 213504x^3y^9 + 225024x^3y^8 + 41856x^3y^7 - 1291712x^3y^6 \\
& + 1804192x^3y^5 - 1341504x^3y^4 + 42056x^3y^3 + 522448x^3y^2 - 352104x^3y \\
& - 18792x^3 - 24576x^2y^{11} - 32256x^2y^{10} + 232704x^2y^9 + 567936x^2y^8 \\
& - 885248x^2y^7 + 1256512x^2y^6 + 818336x^2y^5 - 1233192x^2y^4 + 1768836x^2y^3 \\
& - 687324x^2y^2 + 387776x^2y - 233820x^2 - 3072xy^{12} + 58880xy^{11} + 111616xy^{10} \\
& + 369664xy^9 + 205056xy^8 + 502368xy^7 + 673520xy^6 + 997464xy^5 - 41492xy^4 + 283268xy^3 \\
& + 1044910xy^2 - 838674xy + 364824x - 1024y^{12} + 85504y^{11} + 71552y^{10} + 281920y^9 + 753952y^8 \\
& - 490832y^7 + 1662088y^6 + 349652y^5 - 1353764y^4 + 2912955y^3 - 2333944y^2 + 1196887y - 381267, \\
& 11264x^3y^{10} - 92672x^3y^9 + 186112x^3y^8 - 1004800x^3y^7 - 229120x^3y^6 \\
& + 832640x^3y^5 - 3499728x^3y^4 + 2708888x^3y^3 - 2592792x^3y^2 + 835248x^3y \\
& - 505224x^3 - 15360x^2y^{11} - 86016x^2y^{10} + 194048x^2y^9 - 354560x^2y^8 \\
& - 929600x^2y^7 + 963680x^2y^6 - 1347600x^2y^5 - 105080x^2y^4 - 229360x^2y^3 \\
& + 456612x^2y^2 - 662180x^2y + 262008x^2 - 8192xy^{12} + 39936xy^{11} + 101120xy^{10} \\
& + 30336xy^9 - 340672xy^8 + 356512xy^7 + 339120xy^6 - 225608xy^5 - 22576xy^4 + 1299510xy^3 \\
& - 827748xy^2 + 520178xy - 14094x + 30720y^{12} + 6400y^{11} - 45440y^{10} + 426688y^9 + 238176y^8 \\
& - 382720y^7 + 2365872y^6 - 999240y^5 + 536506y^4 + 2334319y^3 - 2097969y^2 + 1634152y - 427977) \\
i = 16, D_i = 8, \rho_{ij}: 1:1, 20:2, 26:1, 27:2, 44:1, 64:1 \\
W \quad & \langle 28y^2 + 28y + 7, \\
& -28 \rangle \\
A \quad & \langle 25088x^4y - 25088x^4 + 25088x^3y^2 + 100352x^3y - 37632x^3 + 37632x^2y^2 + 87808x^2y \\
& + 37632x^2 + 25088xy^4 + 75264xy^3 + 25088xy^2 - 62720xy + 137984x + 25088y^5 + 50176y^4 \\
& + 53312y^3 + 3136y^2 - 112896y + 76832, \\
& -25088x^4 + 12544x^3y^2 + 50176x^3y - 112896x^3 - 37632x^2y^3 + 12544x^2y^2 \\
& + 112896x^2y - 150528x^2 - 25088xy^4 - 100352xy^3 - 50176xy^2 - 87808x - 31360y^4 - 147392y^3 \\
& - 86240y^2 - 51744y - 40376) \\
B \quad & \langle -25088x^3y^5 - 12544x^3y^4 + 72128x^3y^3 - 40768x^3y^2 - 62720x^3y + 117600x^3 \\
& - 12544x^2y^6 - 25088x^2y^5 - 3136x^2y^4 - 47040x^2y^3 - 170128x^2y^2 \\
& - 201488x^2y - 25676x^2 + 12544xy^7 + 18816xy^6 + 3136xy^5 + 23520xy^4 + 47824xy^3 \\
& + 66248xy^2 + 101528xy - 162582x - 18816y^7 - 109760y^6 - 192864y^5 - 170912y^4 - 298312y^3 \\
& - 167776y^2 + 124264y - 110838, \\
& -31360x^3y^4 + 59584x^3y^3 + 4704x^3y^2 - 73696x^3y + 121912x^3 - 25088x^2y^6 \\
& - 94080x^2y^5 - 144256x^2y^4 - 123872x^2y^3 - 200704x^2y^2 - 214816x^2y + 162680x^2 \\
& + 6272xy^5 + 29792xy^4 + 169344xy^3 + 52920xy^2 - 111720xy + 31458x - 6272y^7 - 15680y^6 \\
& - 54880y^5 - 2352y^4 + 97608y^3 + 104860y^2 + 163268y - 4753) \\
i = 17, D_i = 8, \rho_{ij}: 1:1, 2:1, 5:2, 30:2, 50:2, 54:2, 55:2 \\
w_i \quad & \langle 578t^3 - 562t^2 - 138t - 420, \\
& -758t^3 + 36t^2 - 274t - 420 \rangle \\
i = 18, D_i = 8, \rho_{ij}: 1:2, 2:2, 5:1, 6:2, 20:2, 26:2, 32:2, 34:2, 50:1, 58:1 \\
w_i \quad & \langle -1192t^3 + 626t^2 - 72t + 144, \\
& -1074t^3 + 204t^2 - 262t - 316 \rangle \\
i = 19, D_i = 8, \rho_{ij}: 3:1, 22:1, 26:2, 45:1, 55:2, 56:2, 60:2 \\
w_i \quad & \langle -2192t^3 - 1020t^2 + 1866t - 1314, \\
& 830t^3 - 3830t^2 + 1552t - 946 \rangle \\
i = 20, D_i = 4, \rho_{ij}: 10:1, 11:2, 16:1, 18:1, 33:1, 36:1, 38:1, 41:2, 42:1, 49:1, 50:1, 62:2 \\
w_i \quad & \langle 7t^3 + 12t^2 - 31t - 6, \\
& -21t^3 - 19t^2 + 3t + 30 \rangle \\
i = 21, D_i = 160, \rho_{ij}: 4:1, 12:2, 27:1, 30:1, 50:1, 56:1, 57:2, 60:1, 61:2, 66:1 \\
w_i \quad & \langle 1764t^3 + 276t^2 - 1236t + 120, \\
& 2002t^3 + 1066t^2 - 1706t - 452 \rangle \\
i = 22, D_i = 8, \rho_{ij}: 1:1, 9:2, 11:2, 15:2, 19:2, 24:1, 25:2, 26:1, 28:1, 43:2, 44:2, 49:2, 52:1, 56:2 \\
w_i \quad & \langle 263t^3 - 24t^2 - 263t - 534, \\
& 328t^3 + 27t^2 + 218t - 108 \rangle \\
i = 23, D_i = 1, \rho_{ij}: 6:1, 13:2, 31:1, 32:1, 50:1, 56:1 \\
w_i \quad & \langle 8t^3 + 4t^2 + 4t + 2, \\
& 4t^3 + 10t^2 + 8t + 4 \rangle \\
i = 24, D_i = 16, \rho_{ij}: 22:2, 37:2, 40:2, 54:2, 57:2, 58:2, 61:2 \\
w_i \quad & \langle 27t^3 + 67t^2 - 3t + 82, \\
& 312t^3 - 121t^2 - 30t + 260 \rangle \\
i = 25, D_i = 2, \rho_{ij}: 3:1, 6:2, 9:1, 11:2, 14:2, 22:1, 52:2 \\
w_i \quad & \langle 8t^3 + 2t^2 - 10t, \\
& -4t^3 + 20t^2 - 2t \rangle \\
i = 26, D_i = 2, \rho_{ij}: 2:1, 6:1, 10:2, 11:2, 14:1, 16:2, 18:1, 19:1, 22:2, 41:2, 43:1 \\
w_i \quad & \langle -42t^3 - 8t^2 + 58t + 22, \\
& -28t^3 - 26t^2 + 10t - 2 \rangle \\
i = 27, D_i = 32, \rho_{ij}: 0:1, 8:2, 16:1, 21:2, 34:2, 37:1, 43:1, 53:2, 55:2, 56:2, 57:2 \\
w_i \quad & \langle 88t^3 + 96t^2 + 231t - 208, \\
& -319t^3 + 72t^2 + 660t + 418 \rangle
\end{aligned}$$

$i = 28, D_i = 8, \rho_{ij}: 2:2, 8:2, 12:1, 14:2, 22:2, 38:1, 41:1, 48:1, 51:2, 53:1, 56:1, 62:2$
 $w_i \quad \langle 89t^3 + 21t^2 - 192t - 7, \\ 123t^3 - 28t^2 - 120t - 56 \rangle$

$i = 29, D_i = 1, \rho_{ij}: 11:1, 57:1, 64:1$
 $w_i \quad \langle -2t^2, \\ -2t^2 \rangle$

$i = 30, D_i = 2, \rho_{ij}: 4:2, 5:1, 8:1, 17:1, 21:2, 50:1, 51:2$
 $w_i \quad \langle -4t^3 + 3t^2 + 6t + 5, \\ 3t^3 + 14t^2 + 14t + 6 \rangle$

$i = 31, D_i = 4, \rho_{ij}: 14:2, 15:2, 23:2, 33:1, 37:2, 43:1, 46:1, 47:1, 50:2, 52:1, 56:2$
 $w_i \quad \langle -136t^3 - 279t^2 - 220t - 11, \\ -177t^3 - 230t^2 - 108t + 50 \rangle$

$i = 32, D_i = 8, \rho_{ij}: 6:1, 14:1, 15:1, 18:1, 23:2, 44:1, 48:2, 55:2, 58:1, 59:1, 63:2$
 $w_i \quad \langle -83t^3 - 168t^2 - 150t - 20, \\ 4t^3 - 91t^2 - 176t - 99 \rangle$

$i = 33, D_i = 4, \rho_{ij}: 5:2, 10:2, 14:1, 20:2, 31:2, 42:2, 55:1, 58:1, 64:1$
 $w_i \quad \langle -38t^3 - 19t^2 + 33t - 25, \\ 15t^3 - 48t^2 + 15t + 9 \rangle$

$i = 34, D_i = 2, \rho_{ij}: 18:1, 27:1, 41:1, 65:1$
 $w_i \quad \langle -6t + 4, \\ -2t + 6 \rangle$

$i = 35, D_i = 1, \rho_{ij}: 2:2, 13:2, 14:1, 60:2$
 $w_i \quad \langle 2t - 1, \\ 0 \rangle$

$i = 36, D_i = 2, \rho_{ij}: 4:2, 13:2, 15:2, 20:2, 42:1, 49:2, 51:1, 57:2, 62:2$
 $w_i \quad \langle 20t^3 - 32t^2 - 58t - 56, \\ 46t^3 + 24t^2 - 30t - 52 \rangle$

$i = 37, D_i = 16, \rho_{ij}: 1:2, 5:2, 8:1, 14:1, 15:2, 24:1, 27:2, 31:1, 38:1, 42:2, 48:2, 51:1, 52:1, 64:1$
 $w_i \quad \langle -935t^3 + 803t^2 + 2322t + 1080, \\ 632t^3 + 2381t^2 + 1513t + 69 \rangle$

$i = 38, D_i = 4, \rho_{ij}: 6:1, 8:1, 14:2, 20:2, 28:2, 37:2, 50:1, 63:2$
 $w_i \quad \langle 40t^3 - 24t^2 - 44t + 4, \\ 8t^3 - 56t^2 - 4t + 4 \rangle$

$i = 39, D_i = 1, \rho_{ij}: 5:1, 6:2, 8:1, 11:2, 56:1$
 $w_i \quad \langle 4t^3 + 2t^2 - 4t + 2, \\ 2t^3 + 2 \rangle$

$i = 40, D_i = 4, \rho_{ij}: 5:2, 8:1, 24:1, 43:2, 59:2, 64:2, 66:1$
 $w_i \quad \langle 17t^3 - 12t^2 + 12t - 4, \\ 0 \rangle$

$i = 41, D_i = 2, \rho_{ij}: 5:2, 7:2, 20:1, 26:1, 28:2, 34:2, 43:2, 64:1, 65:1$
 $w_i \quad \langle -56t^3 + 18t^2 + 6t - 26, \\ -42t^3 + 34t^2 - 12t - 4 \rangle$

$i = 42, D_i = 16, \rho_{ij}: 5:2, 7:2, 8:2, 10:1, 11:1, 14:1, 20:2, 33:1, 36:2, 37:1, 45:2, 58:1, 60:2, 61:2$
 $w_i \quad \langle -1105t^3 + 2022t^2 - 3408t - 532, \\ 2491t^3 - 9282t^2 + 11883t - 7322 \rangle$

$i = 43, D_i = 8, \rho_{ij}: 9:1, 22:1, 26:2, 27:2, 31:2, 40:1, 41:1, 53:2, 54:2, 59:2$
 $w_i \quad \langle 5219t^3 - 9618t^2 + 9129t - 694, \\ 3709t^3 - 8862t^2 + 11982t - 4856 \rangle$

$i = 44, D_i = 8, \rho_{ij}: 3:2, 16:2, 22:1, 32:2, 45:2, 48:1, 50:2, 56:2$
 $w_i \quad \langle -44296t^3 - 79064t^2 - 44956t - 2568, \\ -68768t^3 - 94672t^2 - 41516t + 3804 \rangle$

$i = 45, D_i = 16, \rho_{ij}: 5:1, 7:2, 19:2, 42:1, 44:1, 53:2, 55:1, 60:1, 62:1, 63:1$
 $w_i \quad \langle 56t^3 - 64t^2 - 128t - 80, \\ -64t^3 - 224t^2 - 136t - 72 \rangle$

$i = 46, D_i = 8, \rho_{ij}: 6:2, 9:1, 12:1, 31:2, 60:2, 62:1$
 $w_i \quad \langle 64t^3 - 120t^2 + 40t - 40, \\ 24t^3 + 32t^2 - 48t - 8 \rangle$

$i = 47, D_i = 8, \rho_{ij}: 0:1, 4:2, 7:2, 8:1, 31:2, 52:1$
 $w_i \quad \langle -8t^3 + 8t^2 + 8t + 16, \\ 0 \rangle$

$i = 48, D_i = 4, \rho_{ij}: 0:1, 6:2, 9:1, 28:2, 32:1, 37:1, 44:2, 51:1, 56:1$
 $w_i \quad \langle -12t^5 + 16t^4 + 12t^3 + 8t^2 + 4t, \\ -4t^3 + 20t^2 + 12t \rangle$

$i = 49, D_i = 2, \rho_{ij}: 1:1, 9:2, 20:2, 22:1, 36:1$
 $w_i \quad \langle 2t^3 - 2t - 4, \\ 4t^2 - 2 \rangle$

$i = 50, D_i = 256, \rho_{ij}: 1:1, 6:2, 17:1, 18:2, 20:2, 21:2, 23:2, 30:2, 31:1, 38:2, 44:1$
 $w_i \quad \langle -29184t^5 - 87552t^4 - 8192t^3 + 14848t^2 - 9472t - 36352, \\$

$$-33792t^5 - 22528t^4 + 27392t^3 + 19456t^2 - 15872t - 12288)$$

$i = 51, D_i = 8192, \rho_{ij}: 1:2, 4:2, 7:2, 28:1, 30:1, 36:2, 37:2, 48:2, 59:2, 61:1, 65:2$
 $w_i \quad \langle -1343488t^5 + 253952t^4 + 1581056t^3 + 196608t^2 + 352256t + 868352, \\ -1204224t^5 + 2514944t^4 + 1114112t^3 - 524288t^2 + 204800t + 1269760 \rangle$

$i = 52, D_i = 256, \rho_{ij}: 1:1, 2:1, 4:1, 7:1, 12:1, 22:2, 25:1, 31:2, 37:2, 47:2, 61:2$
 $w_i \quad \langle -1024t^5 - 3072t^4 + 1792t^3 - 1024t^2 + 1024t - 4096, \\ 3840t^5 - 13056t^4 + 6144t^3 - 1792t^2 - 1536t - 5120 \rangle$

$i = 53, D_i = 4, \rho_{ij}: 5:2, 6:2, 11:1, 27:1, 28:2, 43:1, 45:1, 58:1$
 $w_i \quad \langle 4t^3 + 8t^2 + 8t + 16, \\ 4t^5 - 12t^3 + 8t^2 \rangle$

$i = 54, D_i = 32, \rho_{ij}: 17:1, 24:1, 43:1, 55:1, 57:2, 63:2$
 $w_i \quad \langle 128t^5 + 96t^4 + 288t^3 + 288t^2 + 160t, \\ 160t^5 + 288t^4 + 192t^3 + 192t^2 + 32t \rangle$

$i = 55, D_i = 64, \rho_{ij}: 1:2, 14:2, 15:1, 17:1, 19:1, 27:1, 32:1, 33:2, 45:2, 54:2$
 $w_i \quad \langle 4544t^5 + 4032t^4 - 22400t^3 + 10496t^2 + 5184t - 2432, \\ 10816t^5 + 384t^4 - 4096t^3 + 15232t^2 - 6720t + 1088 \rangle$

$i = 56, D_i = 1024, \rho_{ij}: 5:1, 7:1, 10:2, 19:1, 21:2, 22:1, 23:2, 27:1, 28:2, 31:1, 39:2, 44:1, 48:2$
 $w_i \quad \langle 212992t^5 - 138240t^4 - 575488t^3 + 332800t^2 + 136192t - 190464, \\ 147456t^5 - 444416t^4 - 52224t^3 + 359424t^2 - 417792t - 323584 \rangle$

$i = 57, D_i = 2, \rho_{ij}: 0:1, 21:1, 24:1, 27:1, 29:2, 36:1, 54:1$
 $w_i \quad \langle -4t^4 - 4t^3 - 4t^2, \\ 4t^4 + 4t^3 + 4t^2 \rangle$

$i = 58, D_i = 2048, \rho_{ij}: 5:1, 8:2, 9:2, 12:1, 18:2, 24:1, 32:2, 33:2, 42:2, 53:2, 64:2$
 $w_i \quad \langle 561152t^5 - 798720t^4 + 937984t^3 - 491520t^2 + 450560t - 114688, \\ 65536t^5 - 823296t^4 + 892928t^3 - 991232t^2 + 376832t - 524288 \rangle$

$i = 59, D_i = 8, \rho_{ij}: 3:2, 8:1, 11:1, 32:2, 40:1, 43:1, 51:1$
 $w_i \quad \langle 16t^4 + 16t^3 - 32t^2, \\ 24t^4 + 24t^3 + 8t^2 \rangle$

$i = 60, D_i = 32, \rho_{ij}: 5:2, 6:2, 19:1, 21:2, 35:1, 42:1, 45:2, 46:1$
 $w_i \quad \langle -32t^3 - 32t^2 + 32t + 32, \\ 32t^4 + 32t^3 - 128t^2 - 32t + 96 \rangle$

$i = 61, D_i = 32, \rho_{ij}: 3:2, 21:1, 24:1, 42:1, 51:2, 52:1$
 $w_i \quad \langle -32t^3 + 128t^2 - 64t + 96, \\ 32t^5 - 32t^4 + 96t^3 + 64 \rangle$

$i = 62, D_i = 64, \rho_{ij}: 4:1, 6:2, 20:1, 28:1, 36:1, 45:2, 46:2, 63:1$
 $w_i \quad \langle 160t^5 + 160t^4 - 320t^3 + 192t^2 + 128t + 160, \\ 192t^5 - 352t^3 - 192t^2 + 64t + 32 \rangle$

$i = 63, D_i = 128, \rho_{ij}: 0:2, 2:1, 5:1, 8:2, 32:1, 38:1, 45:2, 54:1, 62:2$
 $w_i \quad \langle 128t^5 - 128t^4 - 2688t^3 + 256t^2 + 384t, \\ -1664t^5 - 256t^4 - 1152t^3 - 256t^2 - 1152 \rangle$

$i = 64, D_i = 128, \rho_{ij}: 4:2, 6:2, 8:1, 15:2, 16:2, 29:2, 33:2, 37:2, 40:1, 41:2, 58:1$
 $w_i \quad \langle -5888t^5 - 8960t^4 - 11520t^3 - 7680t^2 - 3584t + 1024, \\ 512t^5 - 4352t^4 - 7424t^3 - 8960t^2 - 6144t - 3584 \rangle$

$i = 65, D_i = 2, \rho_{ij}: 1:2, 8:1, 34:2, 41:2, 51:1$
 $w_i \quad \langle 2t^3 - 10t^2 - 12t + 2, \\ -4t^3 - 8t^2 - 4t + 10 \rangle$

$i = 66, D_i = 2, \rho_{ij}: 0:1, 3:1, 5:1, 6:1, 8:1, 21:2, 40:2$
 $w_i \quad \langle -2t^4 - 4t^3 - 2t^2 - 4t, \\ -2t^4 - 4t^3 - 2t^2 - 4t \rangle$

The second certificate.

$i = 0, D_i = 14, \rho_{ij}: 4:2, 24:1, 38:1, 43:1, 48:2$
 $w_i \quad \langle -12t^3 - 26t^2 + 20, \\ -12t^3 + 26t + 20 \rangle$

$i = 1, D_i = 4, \rho_{ij}: 3:1, 5:2, 8:1, 10:1, 11:2, 23:1, 31:1, 33:1, 36:2, 41:2, 43:2, 47:1, 48:2, 58:1, 59:2$
 $w_i \quad \langle -38t^3 - 78t^2 + 73t - 70, \\ -126t^3 + 80t^2 + 105t - 62 \rangle$

$i = 2, D_i = 4, \rho_{ij}: 14:2, 16:2, 19:2, 35:1, 38:2, 39:2, 50:1, 52:2, 60:2$
 $w_i \quad \langle 32t^3 + 66t^2 + 60t + 28, \\ -56t^3 - 60t^2 - 36t + 14 \rangle$

$i = 3, D_i = 8, \rho_{ij}: 1:2, 14:2, 20:2, 27:1, 37:2, 41:2, 43:2, 45:2, 50:2, 62:1, 63:2, 65:2, 72:2, 73:2$
 $w_i \quad \langle 128t^2 - 472t + 552, \\ 48t^2 + 1440t - 136 \rangle$

$i = 4, D_i = 4, \rho_{ij}: 0:1, 13:2, 14:2, 29:2, 39:2, 47:2, 48:1, 59:2, 66:2, 67:2, 72:2$
 $w_i \quad \langle 12t^2 + 36t + 96, \\ -36t^2 + 192t - 96 \rangle$

$i = 5, D_i = 4, \rho_{ij}: 1:1, 12:1, 14:1, 24:2, 28:1, 33:1, 36:2, 37:1, 48:2, 55:2, 56:2, 58:1, 59:2$
 $w_i \quad \langle -52t^2 - 136t - 84, \\ -128t^2 - 128t + 84 \rangle$

$$\begin{aligned}
i &= 6, D_i = 4, \rho_{ij}: 13:2, 14:2, 20:1, 25:1, 38:2, 39:2, 43:2, 53:2 \\
w_i &\langle -4t + 44, \\
&\quad 36t - 32 \rangle \\
i &= 7, D_i = 4, \rho_{ij}: 15:2, 21:1, 36:1, 42:1, 45:1, 56:2, 63:2, 64:1, 73:2 \\
w_i &\langle -4t^2 - 28t - 12, \\
&\quad -8t^2 - 8t - 24 \rangle \\
i &= 8, D_i = 2, \rho_{ij}: 1:2, 14:2, 32:2, 45:1, 50:2, 65:1 \\
w_i &\langle 8t^2 + 8t + 64, \\
&\quad -2t^2 - 2t - 128 \rangle \\
i &= 9, D_i = 2, \rho_{ij}: 34:2, 36:2, 50:2, 51:1, 56:1, 60:2, 63:2 \\
w_i &\langle 48t^2 + 48t + 576, \\
&\quad 60t^2 + 132t - 72 \rangle \\
i &= 10, D_i = 128, \rho_{ij}: 1:2, 13:2, 14:2, 16:1, 17:1, 18:1, 21:1, 25:2, 40:2, 47:1, 53:1, 57:1, 59:2, 60:2, 63:2, 64:1, 66:2, 73:1 \\
W &\langle 4284y^2 + 630y - 3654, \\
&\quad 1260y^2 - 3276y - 5859 \rangle \\
A &\langle 352321536x^{23}y^3 + 1233125376x^{23}y^2 + 1409286144x^{23}y + 528482304x^{23} \\
&\quad - 1585446912x^{22}y^4 - 2466250752x^{22}y^3 + 6165626880x^{22}y^2 + 13388218368x^{22}y \\
&\quad + 6341787648x^{22} + 2965372928x^{21}y^5 - 2040528896x^{21}y^4 - 28831645696x^{21}y^3 \\
&\quad - 6253707264x^{21}y^2 + 50998542336x^{21}y + 33426505728x^{21} - 1952448512x^{20}y^6 \\
&\quad + 10731126784x^{20}y^5 + 38065405952x^{20}y^4 - 79125544960x^{20}y^3 \\
&\quad - 117102870528x^{20}y^2 + 90062192640x^{20}y + 102657687552x^{20} - 2994733056x^{19}y^7 \\
&\quad - 5563744256x^{19}y^6 - 13696499712x^{19}y^5 + 158456610816x^{19}y^4 - 30886854656x^{19}y^3 \\
&\quad - 388654694400x^{19}y^2 + 5240782848x^{19}y + 193424523264x^{19} + 8573157376x^{18}y^8 \\
&\quad - 28449964032x^{18}y^7 + 7707033600x^{18}y^6 - 99736354816x^{18}y^5 + 234022240256x^{18}y^4 \\
&\quad + 281446187008x^{18}y^3 - 631184031744x^{18}y^2 - 357958680576x^{18}y + 176182788096x^{18} \\
&\quad - 9189720064x^{17}y^9 + 66177728512x^{17}y^8 - 97894006784x^{17}y^7 - 18056478720x^{17}y^6 \\
&\quad - 73407660032x^{17}y^5 + 58712915968x^{17}y^4 + 670387142656x^{17}y^3 \\
&\quad - 332679610368x^{17}y^2 - 885582200832x^{17}y - 169841000448x^{17} + 4051697664x^{16}y^{10} \\
&\quad - 59828600832x^{16}y^9 + 215356538880x^{16}y^8 - 98268348416x^{16}y^7 \\
&\quad - 424664891392x^{16}y^6 + 256768999424x^{16}y^5 - 182443835392x^{16}y^4 \\
&\quad + 361423175680x^{16}y^3 + 851429031936x^{16}y^2 - 953316016128x^{16}y - 956949331968x^{16} \\
&\quad + 1967128576x^{15}y^{11} + 4829741056x^{15}y^{10} - 181599731712x^{15}y^9 \\
&\quad + 388111532032x^{15}y^8 + 427079761920x^{15}y^7 - 1458647859200x^{15}y^6 \\
&\quad + 586681417728x^{15}y^5 + 319647383552x^{15}y^4 - 1404918824960x^{15}y^3 \\
&\quad + 2325498298368x^{15}y^2 + 235438866432x^{15}y - 1914790477824x^{15} - 5270142976x^{14}y^{12} \\
&\quad + 47299166208x^{14}y^{11} - 18276679680x^{14}y^{10} - 371515719680x^{14}y^9 \\
&\quad + 280014880768x^{14}y^8 + 1817164382208x^{14}y^7 - 2622354882560x^{14}y^6 \\
&\quad + 127250464768x^{14}y^5 + 2357878849536x^{14}y^4 - 4146853838848x^{14}y^3 \\
&\quad + 255511284908x^{14}y^2 + 2709231501312x^{14}y - 2364528918528x^{14} + 7398752256x^{13}y^{13} \\
&\quad - 54140076032x^{13}y^{12} + 178186616832x^{13}y^{11} - 59979071488x^{13}y^{10} \\
&\quad - 611982508032x^{13}y^9 - 324322983936x^{13}y^8 + 3622804914176x^{13}y^7 \\
&\quad - 2875751137280x^{13}y^6 - 1675087052800x^{13}y^5 + 5402736984064x^{13}y^4 \\
&\quad - 5769411952640x^{13}y^3 + 393796386816x^{13}y^2 + 5075621117952x^{13}y - 1654810214400x^{13} \\
&\quad - 9395240960x^{12}y^{14} + 31738298368x^{12}y^{13} - 144205938688x^{12}y^{12} \\
&\quad + 401349279744x^{12}y^{11} + 25242370048x^{12}y^{10} - 934492504064x^{12}y^9 \\
&\quad - 1121035747328x^{12}y^8 + 4890461470720x^{12}y^7 - 1368398495744x^{12}y^6 \\
&\quad - 4074903175168x^{12}y^5 + 7140678565888x^{12}y^4 - 4116311965696x^{12}y^3 \\
&\quad - 3336947367936x^{12}y^2 + 5173775695872x^{12}y + 185678954496x^{12} + 8162115584x^{11}y^{15} \\
&\quad - 14224982016x^{11}y^{14} + 17799577600x^{11}y^{13} - 255356043264x^{11}y^{12} \\
&\quad + 627811287040x^{11}y^{11} + 429322141696x^{11}y^{10} - 1553191141376x^{11}y^9 \\
&\quad - 1828963483648x^{11}y^8 + 4435682263040x^{11}y^7 + 1452496912384x^{11}y^6 \\
&\quad - 4899770466304x^{11}y^5 + 5064370683904x^{11}y^4 + 644834656256x^{11}y^3 \\
&\quad - 5858171289600x^{11}y^2 + 1887755304960x^{11}y + 2179609657344x^{11} - 1937768448x^{10}y^{16} \\
&\quad + 5688524800x^{10}y^{15} + 39804993536x^{10}y^{14} - 66302509056x^{10}y^{13} \\
&\quad - 414120935424x^{10}y^{12} + 713663971328x^{10}y^{11} + 1397004435456x^{10}y^{10} \\
&\quad - 2044931080192x^{10}y^9 - 1863683670016x^{10}y^8 + 1823359369216x^{10}y^7 \\
&\quad + 3478182428672x^{10}y^6 - 2184110931968x^{10}y^5 - 844505546752x^{10}y^4 \\
&\quad + 5286812188672x^{10}y^3 - 4563642875904x^{10}y^2 - 3391540690944x^{10}y + 2915537780736x^{10} \\
&\quad - 5079302144x^9y^{17} + 3860856832x^9y^{16} - 25455230976x^9y^{15} + 129448804352x^9y^{14} \\
&\quad - 156579397632x^9y^{13} - 584435367936x^9y^{12} + 468374781952x^9y^{11} \\
&\quad + 2385148903424x^9y^{10} - 2171913633792x^9y^9 - 869769936896x^9y^8 \\
&\quad - 1697281474560x^9y^7 + 2341840879616x^9y^6 + 3569883283456x^9y^5 \\
&\quad - 6858959880192x^9y^4 + 5795522281472x^9y^3 + 448725516288x^9y^2 - 7189385183232x^9y \\
&\quad + 1592263372096x^8y^9 + 7237271552x^8y^{18} - 10173284352x^8y^{17} + 15443427328x^8y^{16} \\
&\quad + 3723231232x^8y^{15} + 249417957376x^8y^{14} - 246226878464x^8y^{13} \\
&\quad - 826333462528x^8y^{12} - 20023607296x^8y^{11} + 2850442706944x^8y^{10} \\
&\quad - 1797869273088x^8y^9 + 606717870080x^8y^8 - 3642408304640x^8y^7 \\
&\quad - 2353350967296x^8y^6 + 8507631992832x^8y^5 - 7702212116480x^8y^4 \\
&\quad + 1026938372096x^8y^3 + 5857909800960x^8y^2 - 6304736083968x^8y - 1188416323584x^8 \\
&\quad - 4404019200x^7y^{19} + 4653580288x^7y^{18} - 23436722176x^7y^{17} - 65733656576x^7y^{16} \\
&\quad - 12222988288x^7y^{15} + 340188463104x^7y^{14} - 151689101312x^7y^{13} \rangle
\end{aligned}$$

$$\begin{aligned}
& -966691389440x^7y^{12} - 628908621824x^7y^{11} + 2351845343232x^7y^{10} \\
& -798000939008x^7y^9 + 1280650248192x^7y^8 - 1929743958016x^7y^7 \\
& -7557943787520x^7y^6 + 7673694257152x^7y^5 - 1138139856896x^7y^4 \\
& -5605512708096x^7y^3 + 7391455346688x^7y^2 - 465606672384x^7y - 3586115764224x^7 \\
& + 822083584x^6y^{20} + 4961861632x^6y^{19} + 16900423680x^6y^{18} + 18181259264x^6y^{17} \\
& -73561800704x^6y^{16} + 55954898944x^6y^{15} + 336351461376x^6y^{14} \\
& -49198399488x^6y^{13} - 958054006784x^6y^{12} - 1264696688640x^6y^{11} \\
& + 811548803072x^6y^{10} + 859523252224x^6y^9 + 8613527552x^6y^8 + 2874306985984x^6y^7 \\
& -8156861038592x^6y^6 - 477949853696x^6y^5 + 8967391870976x^6y^4 \\
& -8329442623488x^6y^3 + 3267259269120x^6y^2 + 6834030379008x^6y - 3694739521536x^6 \\
& + 675282944x^5y^{21} - 6606028800x^5y^{20} + 7340032x^5y^{19} + 11479810048x^5y^{18} \\
& -14379122688x^5y^{17} - 184204525568x^5y^{16} + 26438795264x^5y^{15} \\
& + 275733807104x^5y^{14} + 176309403648x^5y^{13} - 353752842240x^5y^{12} \\
& -1365412937728x^5y^{11} - 1077746073600x^5y^{10} + 3011620634624x^5y^9 \\
& -2787574415360x^5y^8 + 6802698993664x^5y^7 - 1395474038784x^5y^6 \\
& -11256658853888x^5y^5 + 14355074449408x^5y^4 - 3796952219648x^5y^3 \\
& -3965879844864x^5y^2 + 10291855687680x^5y - 994847293440x^5 - 528482304x^4y^{22} \\
& + 1636827136x^4y^{21} - 5093982208x^4y^{20} - 1313865728x^4y^{19} + 26287407104x^4y^{18} \\
& -700055552x^4y^{17} - 151304667136x^4y^{16} + 4578344960x^4y^{15} + 86241705984x^4y^{14} \\
& + 114928386048x^4y^{13} + 361546121216x^4y^{12} - 833271627776x^4y^{11} \\
& -2444353601536x^4y^{10} + 4889374687232x^4y^9 - 4302530412544x^4y^8 \\
& + 5299130597376x^4y^7 + 9260871581696x^4y^6 - 15909129420800x^4y^5 \\
& + 8556160614400x^4y^4 + 5384430944256x^4y^3 - 8891255095296x^4y^2 + 6528114819072x^4y \\
& + 2981738446848x^4 + 146800640x^3y^{23} + 1805647872x^3y^{22} + 1284505600x^3y^{21} \\
& -1089994752x^3y^{20} + 22256812032x^3y^{19} + 65275822080x^3y^{18} - 6550061056x^3y^{17} \\
& -75514249216x^3y^{16} - 3748921344x^3y^{15} + 82079907840x^3y^{14} + 198251511808x^3y^{13} \\
& + 918959161344x^3y^{12} + 305365516288x^3y^{11} - 2783397937152x^3y^{10} \\
& + 4467496255488x^3y^9 - 1465586941952x^3y^8 - 2281833824256x^3y^7 \\
& + 15053752827904x^3y^6 - 8305861394432x^3y^5 - 6813618438144x^3y^4 \\
& + 11752510128128x^3y^3 - 7219035242496x^3y^2 - 2952201240576x^3y + 5433485524992x^3 \\
& -14680064x^2y^{24} - 1350565888x^2y^{23} - 249561088x^2y^{22} - 6306922496x^2y^{21} \\
& -26789281792x^2y^{20} - 25337790464x^2y^{19} - 16317808640x^2y^{18} - 82995576832x^2y^{17} \\
& + 29809246208x^2y^{16} - 42341892096x^2y^{15} + 116968914944x^2y^{14} \\
& + 414263148544x^2y^{13} + 643289579520x^2y^{12} + 1571526541312x^2y^{11} \\
& -1803111890944x^2y^{10} - 170169008128x^2y^9 + 5502240292864x^2y^8 \\
& -10453774041088x^2y^7 + 8569211420672x^2y^6 + 8694491086848x^2y^5 \\
& -21203310313472x^2y^4 + 8649993748480x^2y^3 + 564151418880x^2y^2 \\
& -11979958222848x^2y + 4192163168256x^2 + 88080384xy^{24} + 1600126976xy^{23} + 5501353984xy^{22} \\
& + 4404019200xy^{21} + 15270936576xy^{20} + 74926587904xy^{19} + 61291102208xy^{18} \\
& -10402201600xy^{17} + 170712170496xy^{16} - 120820596736xy^{15} + 97727938560xy^{14} \\
& + 763509211136xy^{13} - 761602637824xy^{12} + 1865520054272xy^{11} + 435931611136xy^{10} \\
& -8260097212416xy^9 + 11179159814144xy^8 - 11051457708032xy^7 - 8530053496832xy^6 \\
& + 23122058444800xy^5 - 21798080446464xy^4 - 3544472092672xy^3 + 8908931039232xy^2 \\
& -13555913097216xy - 496021929984x - 99090432y^{24} - 532152320y^{23} - 2359820288y^{22} \\
& -73859072y^{21} - 1862533120y^{20} - 37896585216y^{19} - 35626680320y^{18} + 35438133248y^{17} \\
& -67208544256y^{16} + 56216387584y^{15} - 103687127040y^{14} - 480740900864y^{13} \\
& + 119503519744y^{12} - 1311230197760y^{11} + 651085897728y^{10} + 3216293040128y^9 \\
& -6323093790720y^8 + 7962417333248y^7 - 209325932032y^6 - 11641787483648y^5 \\
& + 15950870334336y^4 - 1699173202944y^3 - 2947996442496y^2 + 9331475072304y - 1598912040600, \\
& 176160768x^{23}y^3 + 880803840x^{23}y^2 + 1233125376x^{23}y + 528482304x^{23} \\
& -352321536x^{22}y^4 - 1937768448x^{22}y^3 + 1409286144x^{22}y^2 + 7222591488x^{22}y \\
& + 4227858432x^{22} - 719323136x^{21}y^5 + 2657091584x^{21}y^4 - 8250195968x^{21}y^3 \\
& -15061745664x^{21}y^2 + 8852078592x^{21}y + 12287213568x^{21} + 4066377728x^{20}y^6 \\
& -10554966016x^{20}y^5 + 8000634880x^{20}y^4 + 15091105792x^{20}y^3 - 54653878272x^{20}y^2 \\
& -45405437952x^{20}y + 1717567488x^{20} - 7751073792x^{19}y^7 + 33220984832x^{19}y^6 \\
& -22812819456x^{19}y^5 - 67909976064x^{19}y^4 + 104962457600x^{19}y^3 + 924844032x^{19}y^2 \\
& -217030066176x^{19}y - 108867354624x^{19} + 7017070592x^{18}y^8 - 57472450560x^{18}y^7 \\
& + 89137348608x^{18}y^6 + 124949364736x^{18}y^5 - 355088728064x^{18}y^4 \\
& + 76600573952x^{18}y^3 + 436218101760x^{18}y^2 - 409199443968x^{18}y - 442405748736x^{18} \\
& -322961408x^{17}y^9 + 47335866368x^{17}y^8 - 173459636224x^{17}y^7 - 27525120000x^{17}y^6 \\
& + 726876028928x^{17}y^5 - 654951055360x^{17}y^4 - 603732312064x^{17}y^3 \\
& + 1399597301760x^{17}y^2 - 213132509184x^{17}y - 1028228382720x^{17} - 7398752256x^{16}y^{10} \\
& + 8081375232x^{16}y^9 + 146081316864x^{16}y^8 - 218424672256x^{16}y^7 \\
& -697831522304x^{16}y^6 + 1691811315712x^{16}y^5 - 224707739648x^{16}y^4 \\
& -2366264836096x^{16}y^3 + 2378963091456x^{16}y^2 + 862747361280x^{16}y - 1617155850240x^{16} \\
& + 10584326144x^{15}y^{11} - 63843598336x^{15}y^{10} + 40803237888x^{15}y^9 \\
& + 282598572032x^{15}y^8 + 79030124544x^{15}y^7 - 1891071164416x^{15}y^6 \\
& + 2286797979648x^{15}y^5 + 1646791229440x^{15}y^4 - 4640544391168x^{15}y^3 \\
& + 2198112043008x^{15}y^2 + 2710090285056x^{15}y - 1685825519616x^{15} - 9527361536x^{14}y^{12} \\
& + 70992789504x^{14}y^{11} - 207627485184x^{14}y^{10} + 76842795008x^{14}y^9 \\
& + 460675088384x^{14}y^8 + 606851825664x^{14}y^7 - 3029146796032x^{14}y^6 \\
& + 1676151357440x^{14}y^5 + 4501491154944x^{14}y^4 - 5601024278528x^{14}y^3
\end{aligned}$$

$$\begin{aligned}
& +58958807040x^{14}y^2 + 4145646403584x^{14}y - 799362514944x^{14} + 7751073792x^{13}y^{13} \\
& -35305553920x^{13}y^{12} + 178505908224x^{13}y^{11} - 441374474240x^{13}y^{10} \\
& -46297251840x^{13}y^9 + 1030705643520x^{13}y^8 + 1063963328512x^{13}y^7 \\
& -3431178698752x^{13}y^6 - 129790115840x^{13}y^5 + 6233166184448x^{13}y^4 \\
& -3504175316992x^{13}y^3 - 3162581237760x^{13}y^2 + 3494743375872x^{13}y + 871269138432x^{13} \\
& -4873781248x^{12}y^{14} + 2216689664x^{12}y^{13} - 14448852992x^{12}y^{12} \\
& +345065914368x^{12}y^{11} - 657659527168x^{12}y^{10} - 798400970752x^{12}y^9 \\
& +2078491541504x^{12}y^8 + 1299949027328x^{12}y^7 - 2806864936960x^{12}y^6 \\
& -1735710212096x^{12}y^5 + 4472246632448x^{12}y^4 + 1239980965888x^{12}y^3 \\
& -4973447872512x^{12}y^2 + 130650734592x^{12}y + 2390308945920x^{12} - 1291845632x^{11}y^{15} \\
& +8741978112x^{11}y^{14} - 82692800512x^{11}y^{13} + 47816638464x^{11}y^{12} \\
& +570995769344x^{11}y^{11} - 586325426176x^{11}y^{10} - 2333498933248x^{11}y^9 \\
& +2976656392192x^{11}y^8 + 1162426187776x^{11}y^7 - 1776203333632x^{11}y^6 \\
& -1301584019456x^{11}y^5 - 913643143168x^{11}y^4 + 5489573232640x^{11}y^3 \\
& -3214284423168x^{11}y^2 - 4377039077376x^{11}y + 2584129830912x^{11} + 8103395328x^{10}y^{16} \\
& -11413749760x^{10}y^{15} + 56158584832x^{10}y^{14} - 225871134720x^{10}y^{13} \\
& +89841991680x^{10}y^{12} + 841026371584x^{10}y^{11} - 20186923008x^{10}y^{10} \\
& -3902074781696x^{10}y^9 + 3094618046464x^{10}y^8 + 902381699072x^{10}y^7 \\
& -1404406857728x^{10}y^6 + 1601998684160x^{10}y^5 - 6585361104896x^{10}y^4 \\
& +5541977391104x^{10}y^3 + 1660590489600x^{10}y^2 - 6875334574080x^{10}y + 973332283392x^{10} \\
& -9850322944x^9y^{17} + 12081692672x^9y^{16} - 15612248064x^9y^{15} + 74011377664x^9y^{14} \\
& -385076428800x^9y^{13} + 11511005184x^9y^{12} + 1007868968960x^9y^{11} \\
& +659222953984x^9y^{10} - 4734981242880x^9y^9 + 1737791111168x^9y^8 \\
& +974851670016x^9y^7 - 2119289274368x^9y^6 + 4917506736128x^9y^5 \\
& -7599941615616x^9y^4 + 593426907136x^9y^3 + 6304061718528x^9y^2 - 4901092589568x^9y \\
& -1632407519232x^9 + 5446303744x^8y^{18} - 5328863232x^8y^{17} + 13821280256x^8y^{16} \\
& +56028299264x^8y^{15} + 60547923968x^8y^{14} - 548104044544x^8y^{13} \\
& -231522959360x^8y^{12} + 1148812263424x^8y^{11} + 1480508309504x^8y^{10} \\
& -3594758651904x^8y^9 - 974684684288x^8y^8 + 1573807456256x^8y^7 \\
& -3036640051200x^8y^6 + 5094682263552x^8y^5 - 1697638383616x^8y^4 \\
& -5793474412544x^8y^3 + 6851093200896x^8y^2 + 1069739016192x^8y - 3469527613440x^8 \\
& -5798625280x^7y^{18} - 16937123840x^7y^{17} - 32167690240x^7y^{16} + 155265531904x^7y^{15} \\
& +108487507968x^7y^{14} - 517424545792x^7y^{13} - 557792886784x^7y^{12} \\
& +776487305216x^7y^{11} + 1691121352704x^7y^{10} - 200103952384x^7y^9 \\
& -3677295476736x^7y^8 + 2435969449984x^7y^7 - 2158897004544x^7y^6 \\
& +271357313024x^7y^5 + 7626498768896x^7y^4 - 8070497304576x^7y^3 \\
& +2162497290240x^7y^2 + 7324671148032x^7y - 2997865414656x^7 - 2143289344x^6y^{20} \\
& +8558477312x^6y^{19} + 627572736x^6y^{18} - 18357420032x^6y^{17} - 52365623296x^6y^{16} \\
& +214011478016x^6y^{15} + 56437506048x^6y^{14} - 471064903680x^6y^{13} \\
& -639795724288x^6y^{12} + 154003046400x^6y^{11} + 844164235264x^6y^{10} \\
& +3906528346112x^6y^9 - 4701729062912x^6y^8 + 2311246708736x^6y^7 \\
& +1642444095488x^6y^6 - 6975933841408x^6y^5 + 12694918856704x^6y^4 \\
& -3370957406208x^6y^3 - 4785426530304x^6y^2 + 9158616219648x^6y - 140481331200x^6 \\
& +1438646272x^5y^{21} - 2003828736x^5y^{20} + 11773411328x^5y^{19} + 13986430976x^5y^{18} \\
& -40965636096x^5y^{17} - 38490210304x^5y^{16} + 344390631424x^5y^{15} \\
& +308740096000x^5y^{14} - 116475297792x^5y^{13} - 270935261184x^5y^{12} \\
& -154408583168x^5y^{11} - 630198632448x^5y^{10} + 6203120680960x^5y^9 \\
& -2676245856256x^5y^8 - 135304314880x^5y^7 + 6431423201280x^5y^6 \\
& -10544623452160x^5y^5 + 7527559725056x^5y^4 + 5385458548736x^5y^3 \\
& -8750059487232x^5y^2 + 4328788918272x^5y + 3369417375744x^5 - 352321536x^4y^{22} \\
& -3178233856x^4y^{21} - 7105150976x^4y^{20} + 1038614528x^4y^{19} + 8036417536x^4y^{18} \\
& -74349936640x^4y^{17} - 52507836416x^4y^{16} + 176013967360x^4y^{15} \\
& +311771529216x^4y^{14} + 75424333824x^4y^{13} - 68431118336x^4y^{12} \\
& +11357782016x^4y^{11} - 1830528745472x^4y^{10} + 4299743494144x^4y^9 \\
& +2230012739584x^4y^8 - 4433399513088x^4y^7 + 7591399653376x^4y^6 \\
& -5586782322688x^4y^5 - 6253582483456x^4y^4 + 11002707050496x^4y^3 \\
& -6165648900096x^4y^2 - 4736919011328x^4y + 4968592441344x^4 - 14680064x^3y^{23} \\
& +2598371328x^3y^{22} + 1798307840x^3y^{21} + 9342025728x^3y^{20} + 38447087616x^3y^{19} \\
& +39165493248x^3y^{18} - 50137923584x^3y^{17} + 46904639488x^3y^{16} + 83219447808x^3y^{15} \\
& +445127983104x^3y^{14} + 452379017216x^3y^{13} - 400014311424x^3y^{12} \\
& +388465688576x^3y^{11} - 1681443520512x^3y^{10} - 2052120182784x^3y^9 \\
& +7369363226624x^3y^8 - 7503013281792x^3y^7 + 1537551368192x^3y^6 \\
& +6281605578752x^3y^5 - 18832061595648x^3y^4 + 7600573775872x^3y^3 \\
& +1732319576064x^3y^2 - 11904243499008x^3y + 3068106080256x^3 + 14680064x^2y^{24} \\
& -146800640x^2y^{23} - 1181745152x^2y^{22} - 10582491136x^2y^{21} - 15544352768x^2y^{20} \\
& +4063625216x^2y^{19} - 16203120640x^2y^{18} - 25611665408x^2y^{17} + 114243469312x^2y^{16} \\
& -152739643392x^2y^{15} + 269318619136x^2y^{14} + 807381499904x^2y^{13} \\
& -1310548033536x^2y^{12} + 219098120192x^2y^{11} + 357293031424x^2y^{10} \\
& -9255399587840x^2y^9 + 8814891696128x^2y^8 - 5120717815808x^2y^7 \\
& -9487664119808x^2y^6 + 16404300595200x^2y^5 - 18691594485760x^2y^4 \\
& -3827051855872x^2y^3 + 9230785019904x^2y^2 - 11323010678784x^2y - 1544227356672x^2 \\
& -572522496xy^{24} + 767033344xy^{23} + 840433664xy^{22} - 3627810816xy^{21} + 12307857408xy^{20}
\end{aligned}$$

$$\begin{aligned}
& +25321275392xy^{19} - 94001954816xy^{18} - 130925527040xy^{17} - 70231719936xy^{16} \\
& - 518890717184xy^{15} - 368572366848xy^{14} + 766532386816xy^{13} - 2099158777856xy^{12} \\
& - 1341262168064xy^{11} + 2950004965376xy^{10} - 11634244091904xy^9 + 3581330292736xy^8 \\
& + 3806939021312xy^7 - 16676321492992xy^6 + 15055106834432xy^5 - 2200174362624xy^4 \\
& - 14735474819072xy^3 + 10148175249408xy^2 - 1960329412608xy - 5853713596416x + 253231104y^{24} \\
& - 205520896y^{23} + 47710208y^{22} + 3726442496y^{21} - 5624299520y^{20} - 19501547520y^{19} \\
& + 50718703616y^{18} + 88174428160y^{17} - 305070080y^{16} + 259141664768y^{15} + 56420990976y^{14} \\
& - 608169361408y^{13} + 1412857757696y^{12} + 501744287744y^{11} - 1243949113344y^{10} \\
& + 7979848425472y^9 - 4781507082240y^8 + 473409276928y^7 + 9672227289088y^6 \\
& - 11708530941184y^5 + 8251909002624y^4 + 6858753154176y^3 - 6928262357184y^2 + 5155751794704y \\
& + 2499355340496) \\
B \quad & \langle 205520896x^3y^{25} - 792723456x^3y^{24} + 249561088x^3y^{23} + 4014997504x^3y^{22} \\
& - 9289728000x^3y^{21} - 11105468416x^3y^{20} + 55977836544x^3y^{19} - 875298816x^3y^{18} \\
& - 35122970624x^3y^{17} + 290991439872x^3y^{16} - 258064515072x^3y^{15} \\
& + 49099309056x^3y^{14} + 1668284219392x^3y^{13} - 2330113572864x^3y^{12} \\
& + 1722971357184x^3y^{11} + 2798279106560x^3y^{10} - 17579002728448x^3y^9 \\
& + 14688580100096x^3y^8 - 8467047515136x^3y^7 - 23565188874752x^3y^6 \\
& + 33464041866752x^3y^5 - 19918827752320x^3y^4 - 14316439438336x^3y^3 \\
& + 13957899935616x^3y^2 - 14383489271088x^3y - 4099569168744x^3 - 73400320x^2y^{26} \\
& - 491782144x^2y^{25} - 1669857280x^2y^{24} - 280756224x^2y^{23} - 8891531264x^2y^{22} \\
& - 33799929856x^2y^{21} - 480772096x^2y^{20} + 58361511936x^2y^{19} - 48354754560x^2y^{18} \\
& - 19689177088x^2y^{17} + 117863022592x^2y^{16} - 603072626688x^2y^{15} \\
& - 738174173184x^2y^{14} + 1573602623488x^2y^{13} - 3378112905216x^2y^{12} \\
& - 1792430194688x^2y^{11} + 6976785141760x^2y^{10} - 19528262928384x^2y^9 \\
& + 2700093075456x^2y^8 + 10833219436032x^2y^7 - 33594979000576x^2y^6 \\
& + 18424515904256x^2y^5 + 12541477960256x^2y^4 - 24476368863744x^2y^3 \\
& + 11199166738368x^2y^2 + 1750435716456x^2y - 9579351693492x^2 + 7340032xy^{27} \\
& + 572522496xy^{26} + 524812288xy^{25} + 1870790656xy^{24} + 14022213632xy^{23} + 14815395840xy^{22} \\
& + 9471393792xy^{21} + 80079290368xy^{20} + 56698994688xy^{19} - 30491869184xy^{18} \\
& + 38587006976xy^{17} - 306288984064xy^{16} - 697363595264xy^{15} - 1622715465728xy^{14} \\
& + 95845105664xy^{13} - 1005171679232xy^{12} - 6102239223808xy^{11} + 7541745297408xy^{10} \\
& - 4515085231104xy^9 - 18368868575232xy^8 + 27651138789120xy^7 - 14825456291968xy^6 \\
& - 15650334940800xy^5 + 44080992918816xy^4 - 15222044606720xy^3 - 1515115153440xy^2 \\
& + 19633132629684xy - 8733828050010x - 40370176y^{27} - 455081984y^{26} - 2774532096y^{25} \\
& - 3101622272y^{24} - 4321902592y^{23} - 45418283008y^{22} - 72819539968y^{21} - 2063925248y^{20} \\
& - 106301095936y^{19} - 81922555904y^{18} + 23581687808y^{17} - 718639595520y^{16} \\
& - 249714769920y^{15} - 1401038700544y^{14} - 1623006027776y^{13} + 4106532777984y^{12} \\
& - 5728116101120y^{11} + 2942229341184y^{10} + 14185677481472y^9 - 24497389077504y^8 \\
& + 20233917517184y^7 + 13893264586432y^6 - 27808561646656y^5 + 38186632767696y^4 \\
& + 3809003904384y^3 - 7944479429712y^2 + 18706741875906y - 3071436074721, \\
& 190840832x^3y^{25} + 528482304x^3y^{24} + 4121427968x^3y^{23} + 3477340160x^3y^{22} \\
& - 6586761216x^3y^{21} + 19943784448x^3y^{20} + 34365112320x^3y^{19} - 107207589888x^3y^{18} \\
& - 109379780608x^3y^{17} - 69741772800x^3y^{16} - 574405214208x^3y^{15} \\
& - 727238443008x^3y^{14} + 1192638218240x^3y^{13} - 2244102389760x^3y^{12} \\
& - 3757957496832x^3y^{11} + 5155076448256x^3y^{10} - 12809484197888x^3y^9 \\
& - 2158837237760x^3y^8 + 13596975608832x^3y^7 - 20411368929280x^3y^6 \\
& + 11743908633856x^3y^5 + 12868107430528x^3y^4 - 21777583807616x^3y^3 \\
& + 9743455248576x^3y^2 + 5533288966128x^3y - 9002000821968x^3 + 7340032x^2y^{26} \\
& - 917504000x^2y^{25} - 2161639424x^2y^{24} - 3773693952x^2y^{23} - 18966642688x^2y^{22} \\
& - 28139847680x^2y^{21} + 11950489600x^2y^{20} - 46239449088x^2y^{19} \\
& - 172102189056x^2y^{18} - 228218568704x^2y^{17} - 524805406720x^2y^{16} \\
& - 321834713088x^2y^{15} - 1024391380992x^2y^{14} + 300302041088x^2y^{13} \\
& + 1781389983744x^2y^{12} - 4980829020160x^2y^{11} + 4225243516928x^2y^{10} \\
& + 3304036356096x^2y^9 - 16763119715328x^2y^8 + 23126782597632x^2y^7 \\
& - 1385641671680x^2y^6 - 13277543592320x^2y^5 + 39697874515264x^2y^4 \\
& - 12157771042368x^2y^3 - 3010037501856x^2y^2 + 20332393321464x^2y - 6796992812904x^2 \\
& - 7340032xy^{27} + 55050240xy^{26} + 58720256xy^{25} + 4206755840xy^{24} + 9207611392xy^{23} \\
& - 1542782976xy^{22} + 13262979072xy^{21} + 47532670976xy^{20} - 81003675648xy^{19} \\
& - 10234298368xy^{18} - 79753576448xy^{17} - 858711719936xy^{16} + 517530976256xy^{15} \\
& - 70719373312xy^{14} - 1368387141632xy^{13} + 7671792730112xy^{12} - 1821372772352xy^{11} \\
& - 1549863942144xy^{10} + 24046273456128xy^9 - 22441163039232xy^8 + 15922789182720xy^7 \\
& + 28392107000320xy^6 - 33272492390592xy^5 + 40686731754144xy^4 + 11549522761952xy^3 \\
& - 15037777717200xy^2 + 22469372045820xy + 1072587748428x + 282591232y^{27} + 47710208y^{26} \\
& - 539492352y^{25} + 3429171200y^{24} - 2925461504y^{23} - 2435143680y^{22} + 40725250048y^{21} \\
& + 129141899264y^{20} + 48660742144y^{19} + 359686799360y^{18} + 491647270912y^{17} \\
& - 553092513792y^{16} + 1321239478272y^{15} + 2026122821632y^{14} - 1957956345856y^{13} \\
& + 9563053473792y^{12} + 4202710722560y^{11} - 6795937311744y^{10} + 26104981095424y^9 \\
& - 11599493977344y^8 - 2509803384704y^7 + 34504085939456y^6 - 23543437774304y^5 \\
& + 14323662819984y^4 + 20987881127472y^3 - 11276076131208y^2 + 10617139538166y + 5140329800382) \\
i = 11, D_i = 8, p_{ij}: 1:1, 15:2, 16:2, 17:2, 48:1, 57:1, 64:2, 67:1 \\
W \quad \langle 182y^2 - 91y + 133, \\
\quad 182y^2 + 56 \rangle
\end{aligned}$$

A	$(132608x^8y + 132608x^8 + 64512x^7y^2 + 1082368x^7y + 1017856x^7 - 534016x^6y^3$ $+ 562688x^6y^2 + 4530176x^6y + 3433472x^6 - 1128960x^5y^4 - 2361856x^5y^3$ $+ 3304448x^5y^2 + 11079936x^5y + 6542592x^5 - 792064x^4y^5 - 4123392x^4y^4$ $- 3161088x^4y^3 + 9739520x^4y^2 + 16436224x^4y + 6866944x^4 + 537600x^3y^6$ $- 3677184x^3y^5 - 4895744x^3y^4 + 1643264x^3y^3 + 15000832x^3y^2 + 13647872x^3y$ $+ 971264x^3 + 1329664x^2y^7 - 249088x^2y^6 - 5605376x^2y^5 + 1003520x^2y^4$ $+ 7395584x^2y^3 + 12321792x^2y^2 + 323456x^2y - 9632896x^2 + 860160xy^8 + 2277632xy^7$ $- 1017856xy^6 - 2128896xy^5 + 4338432xy^4 + 5607168xy^3 + 1671936xy^2 - 17283840xy - 17380608x$ $- 265216y^9 + 1421056y^8 - 877184y^7 + 3891776y^6 - 7019040y^5 + 12982704y^4 + 32480y^3$ $+ 2557968y^2 - 7568960y - 11701760,$ $21504x^8y + 21504x^8 - 333312x^7y^2 + 60928x^7y + 394240x^7 - 774144x^6y^3$ $- 2200576x^6y^2 + 620032x^6y + 2046464x^6 - 526848x^5y^4 - 4049920x^5y^3$ $- 5766656x^5y^2 + 3171840x^5y + 5415424x^5 + 559104x^4y^5 - 3075072x^4y^4$ $- 9106944x^4y^3 - 6155520x^4y^2 + 7650048x^4y + 8332800x^4 + 1462272x^3y^6$ $- 80640x^3y^5 - 7211008x^3y^4 - 8843520x^3y^3 + 1286656x^3y^2 + 10750208x^3y$ $+ 6288128x^3 + 903168x^2y^7 + 3053568x^2y^6 - 3544576x^2y^5 - 6598144x^2y^4$ $- 404992x^2y^3 + 12655104x^2y^2 + 9105152x^2y - 3051776x^2 - 204288xy^8 + 2718464xy^7$ $+ 1725696xy^6 - 3840256xy^5 + 1964032xy^4 + 6008576xy^3 + 18920832xy^2 + 2382464xy - 15137024x$ $- 43008y^9 - 628992y^8 + 2235520y^7 - 1116416y^6 + 358624y^5 - 630560y^4 + 14119672y^3$ $+ 4935784y^2 + 8786792y - 10271688)$
B	$(261632x^3y^9 + 1057280x^3y^8 + 2051840x^3y^7 - 1858752x^3y^6 + 10496416x^3y^5$ $- 17900848x^3y^4 - 2857568x^3y^3 - 12751760x^3y^2 - 19253696x^3y - 2696960x^3$ $- 333312x^2y^{10} + 1467648x^2y^9 - 660352x^2y^8 + 2736832x^2y^7 + 599648x^2y^6$ $- 3145184x^2y^5 - 8713208x^2y^4 - 6758612x^2y^3 - 9225860x^2y^2 - 2102380x^2y$ $+ 6371484x^2 - 132608xy^{11} - 370944xy^{10} + 8064xy^9 + 1153152xy^8 - 416416xy^7$ $- 6557376xy^6 - 767144xy^5 - 1345960xy^4 - 9596538xy^3 + 13464178xy^2 + 7948402xy + 15577814x$ $- 686336y^{11} + 266112y^{10} - 2644992y^9 + 1819104y^8 - 2768416y^7 - 7110152y^6 + 13824692y^5$ $- 7005026y^4 + 23178022y^3 + 4773776y^2 + 17040870y + 8579466,$ $- 645120x^3y^9 + 1603840x^3y^8 - 1414784x^3y^7 + 3508736x^3y^6 + 3757600x^3y^5$ $+ 8198176x^3y^4 - 9593080x^3y^3 + 7276696x^3y^2 - 14712040x^3y - 10025400x^3$ $- 397824x^2y^{10} - 163072x^2y^9 - 651392x^2y^8 + 2907520x^2y^7 - 1199072x^2y^6$ $+ 10243632x^2y^5 - 10839864x^2y^4 + 4053448x^2y^3 - 17913056x^2y^2 - 5572728x^2y$ $- 3124296x^2 - 21504xy^{11} - 1263360xy^{10} + 488320xy^9 - 1718080xy^8 + 3520608xy^7$ $- 1761648xy^6 - 10820768xy^5 + 6799548xy^4 - 20466866xy^3 - 3203746xy^2 - 3821566xy + 13539862x$ $- 340480y^{11} + 215040y^{10} - 2763712y^9 + 1598464y^8 - 1419376y^7 - 11437552y^6 + 2267804y^5$ $- 14705152y^4 + 3357753y^3 - 1375577y^2 + 1274371y + 13590073)$
$i = 12, D_i = 2, \rho_{ij}: 5:2, 23:1, 41:1, 42:1, 51:1$	
w_i	$(-2t + 10,$ $-10t + 8)$
$i = 13, D_i = 64, \rho_{ij}: 4:1, 6:1, 10:1, 17:1, 23:1, 24:2, 30:1, 31:1, 36:1, 37:2, 39:2, 42:2, 43:1, 47:1, 52:2, 54:1, 71:1$	
w_i	$(1500t^3 - 1067t^2 + 1474t + 1380,$ $6117t^3 - 940t^2 - 3061t - 4386)$
$i = 14, D_i = 32, \rho_{ij}: 2:1, 3:1, 4:1, 5:2, 6:1, 8:1, 10:1, 15:2, 35:2, 40:1, 41:2, 43:1, 45:2, 48:1, 65:1, 67:2, 71:1$	
w_i	$(1317t^3 + 4196t^2 + 5163t + 2206,$ $-1780t^3 + 2661t^2 + 5298t + 3652)$
$i = 15, D_i = 2, \rho_{ij}: 7:1, 11:1, 14:1, 24:1, 30:2, 50:2, 60:2$	
w_i	$(6t^3 + 10t^2 + 12t - 4,$ $16t^3 + 8t^2 + 18t + 8)$
$i = 16, D_i = 64, \rho_{ij}: 2:1, 10:2, 11:1, 32:2, 33:1, 35:2, 38:2, 40:1, 41:2, 43:2, 59:2, 65:2$	
w_i	$(16227t^3 + 22367t^2 + 12464t - 1653,$ $8769t^3 + 15508t^2 + 12984t + 2480)$
$i = 17, D_i = 4, \rho_{ij}: 10:2, 11:1, 13:2, 34:1, 35:2, 38:1, 40:1, 45:2, 47:2, 53:1, 66:1$	
w_i	$(-141t^3 - 266t^2 - 210t - 34,$ $-103t^3 - 35t^2 + 70t + 89)$
$i = 18, D_i = 4, \rho_{ij}: 10:2, 19:2, 29:2, 31:2, 32:2, 35:2, 41:1, 65:2$	
w_i	$(2t^3 + 24t^2 - 4,$ $27t^3 + 26t^2 - 4t - 6)$
$i = 19, D_i = 32, \rho_{ij}: 2:1, 18:1, 30:1, 32:2, 34:1, 36:2, 40:2, 43:1, 52:2, 60:2, 66:2, 68:1, 71:1$	
w_i	$(1297t^3 - 16298t^2 - 25134t - 14598,$ $13838t^3 + 8303t^2 - 3036t - 8577)$
$i = 20, D_i = 4, \rho_{ij}: 3:1, 6:2, 29:2, 32:2, 59:2, 65:2, 67:1, 72:2, 73:2$	
w_i	$(-30t^3 + 58t^2 + 24t + 8,$ $4t^3 + 38t^2 + 50t - 2)$
$i = 21, D_i = 2, \rho_{ij}: 7:2, 10:2, 38:1, 57:1, 60:2, 63:1, 71:2, 73:1$	
w_i	$(-52t^3 - 44t^2 + 44t + 42,$ $-20t^3 + 38t^2 + 76t + 30)$
$i = 22, D_i = 2, \rho_{ij}: 27:2, 36:1, 37:1, 42:1, 61:1, 70:1$	
w_i	$(-4t^3 + 4t^2 + 16t + 24,$ $-8t^3 - 12t^2 - 8t + 8)$
$i = 23, D_i = 2, \rho_{ij}: 1:2, 12:2, 13:2, 26:2, 27:1, 34:2, 36:1, 40:1, 46:1, 49:1$	

$$\begin{aligned}
w_i & \langle -168t^3 + 292t^2 - 192t + 128, \\
& \quad 56t^3 + 64t^2 - 44t + 104 \rangle \\
i = 24, D_i = 8, \rho_{ij}: & 0:2, 5:1, 13:1, 15:2, 26:2, 30:2, 31:2, 39:1, 43:1, 45:2, 48:2, 72:2 \\
w_i & \langle -97328t^3 + 66220t^2 + 23744t + 29456, \\
& \quad -22568t^3 + 15512t^2 + 5572t + 6888 \rangle \\
i = 25, D_i = 8, \rho_{ij}: & 6:2, 10:1, 29:2, 30:2, 34:2, 35:2, 39:2, 42:2, 49:2, 69:1 \\
w_i & \langle -2080t^3 - 4940t^2 - 3640t - 1248, \\
& \quad 0 \rangle \\
i = 26, D_i = 8, \rho_{ij}: & 23:1, 24:1, 30:2, 33:1, 37:2, 42:1, 44:2, 45:1, 55:2, 56:2, 57:2, 62:1 \\
w_i & \langle -10560t^3 - 10212t^2 - 2748t - 2520, \\
& \quad 168t^3 + 10980t^2 + 8712t + 3192 \rangle \\
i = 27, D_i = 8, \rho_{ij}: & 3:2, 22:1, 23:2, 38:1, 45:1, 66:2, 70:2 \\
w_i & \langle 208t^3 + 312t^2 - 156t + 208, \\
& \quad -520t^3 + 936t^2 - 468t - 52 \rangle \\
i = 28, D_i = 2, \rho_{ij}: & 5:2, 31:1, 41:1 \\
w_i & \langle -8t + 2, \\
& \quad -2t - 6 \rangle \\
i = 29, D_i = 8, \rho_{ij}: & 4:1, 18:1, 20:1, 25:1, 32:1, 50:2, 54:1, 61:2, 64:1, 65:1, 66:2 \\
w_i & \langle 104t^3 - 120t^2 + 56t - 32, \\
& \quad 312t^3 - 360t^2 + 168t - 96 \rangle \\
i = 30, D_i = 4, \rho_{ij}: & 13:2, 15:1, 19:2, 24:1, 25:1, 26:1, 33:1, 34:1, 57:2, 63:2, 70:1 \\
w_i & \langle -8t^5 - 4t^4 + 32t^3 + 84t^2 + 60t + 28, \\
& \quad -4t^5 + 40t^4 + 76t^3 + 48t^2 - 24t - 28 \rangle \\
i = 31, D_i = 16, \rho_{ij}: & 1:2, 13:2, 18:1, 24:1, 28:2, 64:2 \\
w_i & \langle -32t^5 + 64t^4 - 32t^3 - 16t^2, \\
& \quad -32t^5 + 112t^4 - 80t^3 + 32t^2 - 32t + 32 \rangle \\
i = 32, D_i = 32, \rho_{ij}: & 8:1, 16:1, 18:1, 19:1, 20:1, 29:2, 56:1, 57:1 \\
w_i & \langle -32t^5 - 288t^4 - 224t^3 - 32t, \\
& \quad 64t^5 - 96t^4 - 224t^3 + 64t \rangle \\
i = 33, D_i = 4, \rho_{ij}: & 1:2, 5:2, 16:2, 26:2, 30:2, 36:2, 57:1, 59:2, 69:1 \\
w_i & \langle -28t^5 + 44t^4 - 8t^3 + 4t^2 + 16t + 32, \\
& \quad -20t^5 + 24t^4 + 24t^3 - 12t^2 + 4t + 8 \rangle \\
i = 34, D_i = 8, \rho_{ij}: & 9:1, 17:2, 19:2, 23:1, 25:1, 30:2, 35:2, 40:2, 45:1, 50:1, 60:2, 69:2 \\
w_i & \langle -48t^5 - 24t^4 + 48t^3 + 32t^2 - 136t - 112, \\
& \quad 0 \rangle \\
i = 35, D_i = 4, \rho_{ij}: & 2:2, 14:1, 16:1, 17:1, 18:1, 25:1, 34:1, 38:1, 41:1, 43:1, 53:1, 55:2, 67:1 \\
w_i & \langle -28t^5 + 16t^4 - 12t^3 - 24t^2 + 76t + 104, \\
& \quad 24t^4 - 4t^3 - 36t^2 + 44t + 16 \rangle \\
i = 36, D_i = 2048, \rho_{ij}: & 1:1, 5:1, 7:2, 9:1, 13:2, 19:1, 22:2, 23:2, 33:1, 43:1, 44:2, 54:1, 57:2, 66:2 \\
w_i & \langle -143360t^5 - 581632t^4 + 610304t^3 - 913408t^2 + 245760t - 507904, \\
& \quad -606208t^5 + 413696t^4 - 544768t^3 - 4096t^2 - 344064t - 147456 \rangle \\
i = 37, D_i = 64, \rho_{ij}: & 3:1, 5:2, 13:1, 22:2, 26:1, 40:2, 44:1, 47:1, 56:1, 61:1 \\
w_i & \langle -320t^5 - 320t^4 - 832t^3 - 384t^2 + 128t, \\
& \quad -64t^5 - 320t^4 - 640t^3 - 256t^2 - 384t - 448 \rangle \\
i = 38, D_i = 512, \rho_{ij}: & 0:2, 2:1, 6:1, 16:1, 17:2, 21:2, 27:2, 35:2, 46:2, 56:1, 63:1 \\
w_i & \langle -1024t^5 + 8192t^4 + 18944t^3 + 16896t^2 + 4608t + 2048, \\
& \quad -1024t^5 + 15360t^4 + 17920t^3 - 512t^2 - 10752t - 1024 \rangle \\
i = 39, D_i = 512, \rho_{ij}: & 2:1, 4:1, 6:1, 13:1, 24:2, 25:1, 43:2, 49:2, 54:2, 56:2, 59:2, 70:1 \\
w_i & \langle 44032t^5 + 60416t^4 - 36864t^3 - 57344t^2 - 39424t + 13824, \\
& \quad 8192t^5 + 59392t^4 + 81408t^3 - 8704t^2 - 57344t - 3072 \rangle \\
i = 40, D_i = 2048, \rho_{ij}: & 10:1, 14:2, 16:2, 17:2, 19:1, 23:2, 34:1, 37:1, 45:1, 56:2, 57:2, 59:1, 63:1, 69:1 \\
w_i & \langle -131072t^5 - 151552t^3 - 118784t^2 - 49152t + 131072, \\
& \quad 802816t^5 + 471040t^4 + 1228800t^3 + 815104t^2 + 249856t - 638976 \rangle \\
i = 41, D_i = 256, \rho_{ij}: & 1:1, 3:1, 12:2, 14:1, 16:1, 18:2, 28:2, 35:2, 55:2, 56:1 \\
w_i & \langle -14592t^5 + 16128t^4 + 9984t^3 - 24320t^2 - 17152t - 9728, \\
& \quad 7168t^5 + 768t^4 - 20736t^3 - 12032t^2 - 6144t - 18432 \rangle \\
i = 42, D_i = 256, \rho_{ij}: & 7:2, 12:2, 13:1, 22:2, 25:1, 26:2, 49:2, 56:2, 61:1, 64:1, 67:1 \\
w_i & \langle -512t^4 - 1536t^3 + 512t^2 + 768t, \\
& \quad -256t^5 - 1024t^4 + 512t^3 + 2048t^2 + 768t - 1536 \rangle \\
i = 43, D_i = 4, \rho_{ij}: & 0:2, 1:1, 3:1, 6:1, 13:2, 14:2, 16:1, 19:2, 24:2, 35:2, 36:2, 39:1, 44:1, 60:2, 62:2 \\
w_i & \langle 8t^7 + 125t^6 + 26t^5 + 290t^4 + 4t^3 + 255t^2 - 18t + 64, \\
& \quad -58t^7 + 72t^6 - 122t^5 + 169t^4 - 113t^3 + 120t^2 - 34t - 16 \rangle \\
i = 44, D_i = 2, \rho_{ij}: & 26:1, 36:1, 37:2, 43:2, 45:1, 48:1, 61:2 \\
w_i & \langle -4t + 4, \\
& \quad 2t^2 + 2t + 4 \rangle \\
i = 45, D_i = 32, \rho_{ij}: & 3:1, 7:2, 8:2, 14:1, 17:1, 24:1, 26:2, 27:2, 34:2, 40:2, 44:2, 59:1, 71:2, 72:2 \\
w_i & \langle 10974t^3 + 20310t^2 + 18606t + 4788, \\
& \quad 9585t^3 + 16437t^2 + 14217t + 3558 \rangle
\end{aligned}$$

$i = 46, D_i = 1, \rho_{ij}: 23:2, 38:1$
 $w_i \quad \langle 2t^2 - t, 2t^2 - t \rangle$

$i = 47, D_i = 4, \rho_{ij}: 1:2, 4:1, 10:2, 13:2, 17:1, 37:2, 49:2, 50:2, 52:1, 54:1, 66:2, 68:2$
 $w_i \quad \langle 128t^3 + 3152t^2 + 4876t + 1864, -788t^3 + 544t^2 + 56t - 760 \rangle$

$i = 48, D_i = 4, \rho_{ij}: 0:1, 1:1, 4:2, 5:1, 11:2, 14:2, 24:1, 44:2, 50:1, 51:1, 56:2, 71:1$
 $w_i \quad \langle -152t^5 + 96t^4 + 168t^3 - 64t^2 - 88t + 64, -76t^5 + 48t^4 + 84t^3 - 32t^2 - 44t + 32 \rangle$

$i = 49, D_i = 2, \rho_{ij}: 23:2, 25:1, 39:1, 42:1, 47:1, 55:1, 57:2, 61:1, 64:2$
 $w_i \quad \langle 40t^2 - 30t + 358, 88t^2 - 66t - 450 \rangle$

$i = 50, D_i = 8, \rho_{ij}: 2:2, 3:1, 8:1, 9:1, 15:1, 29:1, 34:2, 47:1, 48:2, 55:1, 56:1, 58:2, 59:2$
 $w_i \quad \langle -2257t^3 + 3206t^2 - 1868t - 1436, -3368t^3 + 7504t^2 - 9319t + 2606 \rangle$

$i = 51, D_i = 18, e_i = 1, \rho_{ij}: 9:2, 12:2, 48:2, 68:2, 72:1, 73:1$
 $w_i \quad \langle 729t^3 - 1107t^2 + 351t + 486, 1377t^3 + 135t^2 + 621t + 162 \rangle$

$i = 52, D_i = 54, e_i = 0, \rho_{ij}: 2:1, 13:1, 19:1, 47:2, 53:2, 62:2$
 $w_i \quad \langle -1782t + 2160, -810t + 2322 \rangle$

$i = 53, D_i = 2, \rho_{ij}: 6:1, 10:2, 17:2, 35:2, 52:1, 54:2, 60:2, 63:2, 66:1$
 $w_i \quad \langle -14t^3 - 4t^2 - 4t + 22, -16t^3 + 22t^2 + 46t - 10 \rangle$

$i = 54, D_i = 4, \rho_{ij}: 13:2, 29:2, 36:2, 39:1, 47:2, 53:1, 57:1, 58:1, 64:2$
 $w_i \quad \langle -16t^2 + 24t + 420, -20t^2 + 128t + 168 \rangle$

$i = 55, D_i = 4, \rho_{ij}: 5:1, 26:1, 35:1, 41:1, 49:2, 50:2$
 $w_i \quad \langle -88t^3 - 152t^2 - 268t - 120, 64t^3 + 224t^2 + 124t - 12 \rangle$

$i = 56, D_i = 512, \rho_{ij}: 5:1, 7:1, 9:2, 26:1, 32:2, 37:2, 38:2, 39:1, 40:1, 41:2, 42:1, 48:1, 50:2, 62:2$
 $w_i \quad \langle 7548t^3 + 9636t^2 + 10188t + 16248, 40614t^3 + 15786t^2 + 5550t - 9588 \rangle$

$i = 57, D_i = 16, \rho_{ij}: 10:2, 11:2, 21:2, 26:1, 30:1, 32:2, 33:2, 36:1, 40:1, 49:1, 54:2, 59:2, 64:1, 67:1$
 $w_i \quad \langle -2652t^3 + 4252t^2 + 10386t + 4880, -4134t^3 - 1888t^2 + 6672t + 5848 \rangle$

$i = 58, D_i = 2, \rho_{ij}: 1:2, 5:2, 50:1, 54:2, 61:2$
 $w_i \quad \langle 0, 4t - 2 \rangle$

$i = 59, D_i = 4, \rho_{ij}: 1:1, 4:1, 5:1, 10:1, 16:1, 20:1, 33:1, 39:1, 40:2, 45:2, 50:1, 57:1$
 $w_i \quad \langle -6t^3 - 2t^2 - 48t + 56, 70t^3 - 72t^2 - 38t + 14 \rangle$

$i = 60, D_i = 10, \rho_{ij}: 2:1, 9:1, 10:1, 15:1, 19:1, 21:1, 34:1, 43:1, 53:1, 63:1, 66:2, 68:1$
 $w_i \quad \langle -66t^3 - 66t^2 - 34t + 68, -2t^3 - 2t^2 - 18t + 36 \rangle$

$i = 61, D_i = 2, \rho_{ij}: 22:2, 29:1, 37:2, 42:2, 44:1, 49:2, 58:1, 64:2, 71:2$
 $w_i \quad \langle 50t^3 + 64t^2 + 50t + 4, 10t^3 + 56t^2 + 64t + 20 \rangle$

$i = 62, D_i = 1, \rho_{ij}: 3:2, 26:2, 43:1, 52:1, 56:1, 71:1$
 $w_i \quad \langle 10t^3 + 20t^2 + 20t + 10, 14t^3 + 28t^2 + 28t + 14 \rangle$

$i = 63, D_i = 16, \rho_{ij}: 3:1, 7:1, 9:1, 10:1, 21:2, 30:1, 38:2, 40:2, 53:1, 60:2, 64:1, 71:2, 73:1$
 $w_i \quad \langle 224t^3 + 858t^2 + 784t + 234, 654t^3 + 1092t^2 + 560t - 52 \rangle$

$i = 64, D_i = 4, \rho_{ij}: 7:2, 10:2, 11:1, 29:2, 31:1, 42:2, 49:1, 54:1, 57:2, 61:1, 63:2$
 $w_i \quad \langle -28t^3 + 140t^2 + 60t - 20, -20t^3 + 100t^2 - 24t + 8 \rangle$

$i = 65, D_i = 2, \rho_{ij}: 3:1, 8:2, 14:2, 16:1, 18:1, 20:1, 29:2, 69:1$
 $w_i \quad \langle -8t^3 - 24t^2 - 12t + 8, 2t^3 - 6t^2 - 18t + 4 \rangle$

$i = 66, D_i = 1, \rho_{ij}: 4:1, 10:1, 17:2, 19:1, 27:1, 29:1, 36:1, 47:1, 53:2, 60:1, 68:1$
 $w_i \quad \langle 52t^3 - 30t^2 + 2t - 6, 2t^3 - 6t^2 - 8t - 18 \rangle$

$i = 67, D_i = 4, \rho_{ij}: 4:1, 11:2, 14:1, 20:2, 35:2, 42:2, 57:2$
 $w_i \quad \langle -8t^3 + 20t^2 - 16t, -16t^3 + 20t^2 - 4t - 20 \rangle$

$i = 68, D_i = 2, \rho_{ij}: 19:2, 47:1, 51:1, 60:2, 66:2$
 $w_i \quad \langle 20t^2 - 2t - 4, \quad \quad \quad \rangle$

$$16t^2 + 14t - 2\rangle$$

$i = 69$, $D_i = 2$, ρ_{ij} : 25:2, 33:2, 34:1, 40:2, 65:2
 w_i $\langle -t^3 - 6t^2 + 2t + 6,$
 $t^3 + 6t^2 - 2t - 6\rangle$

$i = 70$, $D_i = 18$, $e_i = 2$, ρ_{ij} : 22:2, 27:1, 30:2, 39:2
 w_i $\langle 540t^3 - 432t^2 - 756t,$
 $-216t^3 - 1512t^2 - 540t\rangle$

$i = 71$, $D_i = 16$, ρ_{ij} : 13:2, 14:2, 19:2, 21:1, 45:1, 48:2, 61:1, 62:2, 63:1
 W $\langle 154y^2 - 42y + 126,$
 $70y^2 - 77y + 140\rangle$

A $\langle -69632x^{12} + 335872x^{11}y - 823296x^{11} + 397312x^{10}y^2 + 1368064x^{10}y - 3158016x^{10}$
 $-98304x^9y^3 + 3092480x^9y^2 - 811008x^9y - 4278272x^9 + 356352x^8y^4$
 $+116736x^8y^3 + 8402944x^8y^2 - 12795904x^8y + 2398208x^8 + 417792x^7y^5$
 $+1777664x^7y^4 - 51200x^7y^3 + 6680576x^7y^2 - 23298048x^7y + 13584384x^7$
 $+61440x^6y^6 + 4388864x^6y^5 + 1304576x^6y^4 - 800768x^6y^3 - 12648448x^6y^2$
 $-6057984x^6y + 13644800x^6 - 294912x^5y^7 + 2617344x^5y^6 + 11266048x^5y^5$
 $-13684736x^5y^4 + 8048640x^5y^3 - 37025792x^5y^2 + 32894976x^5y - 1664000x^5$
 $-356352x^4y^8 - 980992x^4y^7 + 10454016x^4y^6 + 4870144x^4y^5 - 34753536x^4y^4$
 $+30330880x^4y^3 - 41196544x^4y^2 + 53379072x^4y - 18418688x^4 + 139264x^3y^9$
 $+376832x^3y^8 + 1425408x^3y^7 + 11901952x^3y^6 - 26324992x^3y^5 - 17591296x^3y^4$
 $+33669120x^3y^3 - 13816832x^3y^2 + 32262144x^3y - 23097856x^3 - 315392x^2y^{10}$
 $+878592x^2y^9 + 3283968x^2y^8 + 6217728x^2y^7 + 1019904x^2y^6 - 47808512x^2y^5$
 $+39790592x^2y^4 - 10469888x^2y^3 + 36946432x^2y^2 - 19916800x^2y - 10859008x^2$
 $-376832xy^{11} - 1134592xy^{10} + 2304000xy^9 - 3924992xy^8 + 2007040xy^7 - 6939136xy^6$
 $-27970560xy^5 + 80681472xy^4 - 72322560xy^3 + 77412352xy^2 - 67977216xy + 12261888x$
 $-110592y^{11} - 1063936y^{10} + 3386368y^9 - 4648704y^8 - 2522496y^7 - 6601792y^6$
 $-10256704y^5 + 50461264y^4 - 48189736y^3 + 48761120y^2 - 47147232y + 11962688,$
 $-118784x^{12} - 40960x^{11}y - 790528x^{11} + 524288x^{10}y^2 - 1503232x^{10}y - 1396736x^{10}$
 $+139264x^9y^3 + 2359296x^9y^2 - 7983104x^9y + 1603584x^9 + 147456x^8y^4$
 $+813056x^8y^3 + 1902592x^8y^2 - 14921728x^8y + 8235008x^8 + 712704x^7y^5$
 $-344064x^7y^4 + 833536x^7y^3 - 9246720x^7y^2 - 2217984x^7y + 7825408x^7$
 $+565248x^6y^6 + 2805760x^6y^5 - 6522880x^6y^4 + 2547712x^6y^3 - 26871808x^6y^2$
 $+32716800x^6y - 5577728x^6 + 417792x^5y^7 + 4771840x^5y^6 - 196608x^5y^5$
 $-18663424x^5y^4 + 12732416x^5y^3 - 29626368x^5y^2 + 51894272x^5y - 18570240x^5$
 $-147456x^4y^8 + 1933312x^4y^7 + 9161728x^4y^6 - 17553408x^4y^5 - 8330240x^4y^4$
 $+18128896x^4y^3 - 6160384x^4y^2 + 25484288x^4y - 16686080x^4 + 237568x^3y^9$
 $+335872x^3y^8 + 3198976x^3y^7 - 1260544x^3y^6 - 34240512x^3y^5 + 38673408x^3y^4$
 $-8560640x^3y^3 + 35174400x^3y^2 - 31028224x^3y - 571904x^3 + 229376x^2y^{10}$
 $+2342912x^2y^9 + 2033664x^2y^8 + 2280448x^2y^7 - 18404352x^2y^6 - 18590720x^2y^5$
 $+68760576x^2y^4 - 54772224x^2y^3 + 69664256x^2y^2 - 76911616x^2y + 19270144x^2$
 $-335872xy^{11} + 520192xy^{10} + 1628160xy^9 - 4278272xy^8 + 6071296xy^7 - 18245120xy^6$
 $+20545536xy^5 + 39559680xy^4 - 72111616xy^3 + 62971392xy^2 - 72967680xy + 28207616x$
 $-495616y^{11} + 257024y^{10} + 1671168y^9 - 7153152y^8 + 3063936y^7 - 10151552y^6$
 $+21749664y^5 + 14140448y^4 - 40913432y^3 + 31718928y^2 - 40473792y + 16037568\rangle$

B $\langle 49152x^3y^{12} - 620544x^3y^{11} + 4272128x^3y^{10} + 2113024x^3y^9 - 13802240x^3y^8$
 $+6742912x^3y^7 - 4884416x^3y^6 + 20885312x^3y^5 + 18048432x^3y^4$
 $-47487192x^3y^3 + 21697248x^3y^2 - 25414432x^3y + 18242496x^3 - 907264x^2y^{12}$
 $-1589248x^2y^{11} + 4352512x^2y^{10} - 7369472x^2y^9 + 79104x^2y^8 + 19460928x^2y^7$
 $-20118272x^2y^6 + 34796784x^2y^5 - 36444608x^2y^4 + 4226020x^2y^3$
 $-22342888x^2y^2 + 16896128x^2y + 10801248x^2 + 256000xy^{13} - 735232xy^{12} + 1130496xy^{11}$
 $-1048832xy^{10} - 12639232xy^9 + 25293760xy^8 - 5976192xy^7 - 8869552xy^6 + 24992040xy^5$
 $-82965668xy^4 + 88341964xy^3 - 79990476xy^2 + 70876312xy - 15423584x + 188416y^{14} - 4096y^{13}$
 $-1168384y^{12} + 5210880y^{11} + 2246144y^{10} - 1831616y^9 + 13638176y^8 - 24166640y^7$
 $+15423080y^6 - 4193232y^5 - 41977570y^4 + 76766199y^3 - 61181232y^2 + 56214700y - 17531736,$
 $-69632x^3y^{12} + 552960x^3y^{11} + 3949568x^3y^{10} - 8021504x^3y^9 - 5981184x^3y^8$
 $+11694464x^3y^7 - 3418496x^3y^6 + 26981472x^3y^5 - 40313888x^3y^4$
 $+7485976x^3y^3 - 24505360x^3y^2 + 28521152x^3y + 1032768x^3 - 780288x^2y^{12}$
 $+1164288x^2y^{11} + 2014720x^2y^{10} - 7919104x^2y^9 + 14931328x^2y^8$
 $-3404416x^2y^7 - 9895776x^2y^6 + 24711216x^2y^5 - 73251016x^2y^4$
 $+68298592x^2y^3 - 65645928x^2y^2 + 67734064x^2y - 18700800x^2 - 253952xy^{13}$
 $-1024000xy^{12} + 1333760xy^{11} - 6518272xy^{10} + 1408128xy^9 + 22869696xy^8 - 31445600xy^7$
 $+17509168xy^6 - 17290384xy^5 + 82616414xy^4 + 82616414xy^3 - 66495704xy^2 + 71080360xy$
 $-29424176x + 167936y^{14} - 429056y^{13} + 731136y^{12} + 4399872y^{11} - 4921472y^{10} + 2693312y^9$
 $-2312000y^8 - 18645968y^7 + 25020672y^6 - 26301980y^5 + 16183968y^4 + 29272567y^3$
 $-19691574y^2 + 22226240y - 13802520\rangle$

$i = 72$, $D_i = 2$, ρ_{ij} : 3:1, 4:1, 20:1, 24:1, 45:1, 51:2
 w_i $\langle 8t^2 + 24t - 14,$
 $28t^2 + 6t - 10\rangle$

$i = 73$, $D_i = 2$, ρ_{ij} : 3:1, 7:1, 10:2, 20:1, 21:2, 51:2, 63:2
 w_i $\langle -4t^5 - 14t^4 - 4t^3 + 10t^2 + 2t - 2,$
 $-8t^5 - 4t^4 + 16t^3 + 14t^2 - 2t - 4\rangle$

The third certificate.

$i = 0, D_i = 1, \rho_{ij}: 42:2, 43:2, 46:1, 47:1$

$w_i \quad \langle 4t^2 + 2t, \\ 0 \rangle$

$i = 1, D_i = 14, \rho_{ij}: 13:1, 20:2, 43:2, 46:2, 48:2, 50:1$

$w_i \quad \langle -18t^3 - 34t^2 - 23t + 9, \\ 2t^3 - 32t^2 - 48t - 15 \rangle$

$i = 2, D_i = 2, \rho_{ij}: 4:1, 6:1, 13:1, 15:1, 16:2, 17:2, 42:1, 50:2, 51:1$

$w_i \quad \langle -16t^3 + 2t^2 + 17t + 7, \\ 8t^3 + 20t^2 + 15t - 2 \rangle$

$i = 3, D_i = 4, \rho_{ij}: 13:1, 15:2, 21:1, 29:2, 35:1, 36:1, 39:2, 40:2, 42:1, 49:2, 50:1, 58:1, 60:2$

$w_i \quad \langle -66t^3 + 16t^2 + 65t + 82, \\ -14t^3 + 78t^2 + 66t + 28 \rangle$

$i = 4, D_i = 2, \rho_{ij}: 2:2, 29:2, 30:1, 31:1, 34:1, 39:2, 57:2$

$w_i \quad \langle 4t^2 - 2t + 6, \\ -6t^2 + 4t - 6 \rangle$

$i = 5, D_i = 4, \rho_{ij}: 18:2, 27:2, 33:1, 39:2, 44:1, 45:2, 51:1, 52:2, 56:2, 60:1, 63:2$

$w_i \quad \langle -92t^2 + 4t - 76, \\ -108t^2 + 124t - 4 \rangle$

$i = 6, D_i = 8, \rho_{ij}: 2:2, 21:1, 28:2, 35:1, 38:1, 50:1, 54:2, 59:2, 60:2, 62:2$

$w_i \quad \langle -24t^2 + 72t, \\ -72t^2 - 72t - 72 \rangle$

$i = 7, D_i = 8, \rho_{ij}: 24:1, 25:1, 26:2, 28:2, 31:2, 32:2, 33:1, 35:1, 36:1, 37:1, 38:2, 46:2, 54:2, 61:2$

$w_i \quad \langle 72t^2 + 240t + 648, \\ 216t^2 + 552t + 432 \rangle$

$i = 8, D_i = 4, \rho_{ij}: 22:2, 27:2, 31:1, 36:1, 44:1, 45:2, 49:1, 50:2, 53:1, 61:2, 63:1$

$w_i \quad \langle -48t^2 - 24t + 132, \\ -72t^2 - 36t + 156 \rangle$

$i = 9, D_i = 2, \rho_{ij}: 20:2, 24:1, 28:2, 32:2, 33:1, 36:1, 42:1, 55:2$

$w_i \quad \langle 14t^2 + 2t - 6, \\ 10t^2 + 10t + 6 \rangle$

$i = 10, D_i = 2, \rho_{ij}: 21:1, 23:1, 41:2, 44:2, 45:1, 46:1$

$w_i \quad \langle 2t^2 - 4t + 2, \\ -2t - 18 \rangle$

$i = 11, D_i = 2, \rho_{ij}: 24:2, 29:2, 40:1, 41:2, 51:2, 58:1$

$w_i \quad \langle -28t - 28, \\ -28t^2 + 28 \rangle$

$i = 12, D_i = 2, \rho_{ij}: 19:1, 22:2, 29:1, 31:2, 32:1, 44:2, 52:1, 54:1$

$w_i \quad \langle 8t^2 - 24t, \\ 16t^2 - 48t \rangle$

$i = 13, D_i = 4, \rho_{ij}: 1:2, 2:2, 3:2, 16:1, 22:1, 27:2, 29:1, 32:1, 35:2, 37:1$

$W \quad \langle 8y^2 + 5, \\ 8y^2 + 5 \rangle$

A $\langle 32x^5y^6 + 64x^5y^3 + 32x^5 + 64x^4y^7 + 192x^4y^6 + 128x^4y^4 + 384x^4y^3 \\ + 64x^4y + 192x^4 + 544x^3y^7 + 512x^3y^6 + 1088x^3y^4 + 1024x^3y^3 + 544x^3y \\ + 512x^3 - 160x^2y^9 + 416x^2y^8 + 1664x^2y^7 + 496x^2y^6 + 832x^2y^5 \\ + 3328x^2y^4 + 1472x^2y^3 + 416x^2y^2 + 1664x^2y + 816x^2 - 256xy^{10} - 464xy^9 \\ + 1776xy^8 + 2208xy^7 - 160xy^6 + 3552xy^5 + 5184xy^4 + 1072xy^3 + 1776xy^2 + 2720xy + 768x \\ + 64y^{12} - 192y^{10} - 192y^9 + 960y^8 + 624y^7 - 144y^6 + 1408y^5 + 2336y^4 + 224y^3 \\ + 1088y^2 + 1008y + 312, \\ 64x^5y^6 + 128x^5y^3 + 64x^5 + 256x^4y^7 + 288x^4y^6 + 512x^4y^4 + 576x^4y^3 \\ + 256x^4y + 288x^4 + 448x^3y^8 + 1280x^3y^7 + 640x^3y^6 + 896x^3y^5 + 2560x^3y^4 \\ + 1280x^3y^3 + 448x^3y^2 + 1280x^3y + 640x^3 + 384x^2y^9 + 2400x^2y^8 \\ + 2880x^2y^7 + 1632x^2y^6 + 4800x^2y^5 + 5760x^2y^4 + 2112x^2y^3 + 2400x^2y^2 \\ + 2880x^2y + 864x^2 + 2208xy^9 + 5280xy^8 + 3712xy^7 + 5040xy^6 + 10560xy^5 + 7424xy^4 \\ + 3456xy^3 + 5280xy^2 + 3712xy + 624x + 1008y^9 + 2096y^8 + 1280y^7 + 2176y^6 + 4192y^5 \\ + 2560y^4 + 1328y^3 + 2096y^2 + 1280y + 160 \rangle$

B $\langle -64x^3y^{12} - 128x^3y^{11} - 992x^3y^{10} + 64x^3y^9 + 1888x^3y^8 - 512x^3y^7 \\ + 80x^3y^6 + 4672x^3y^5 + 1440x^3y^4 + 32x^3y^3 + 2016x^3y^2 + 1472x^3y - 120x^3 \\ + 128x^2y^{12} - 832x^2y^{11} - 1472x^2y^{10} + 608x^2y^9 - 96x^2y^8 - 3264x^2y^7 \\ + 320x^2y^6 + 2048x^2y^5 - 1856x^2y^4 - 832x^2y^3 + 1632x^2y^2 - 320x^2y - 572x^2 \\ + 192xy^{13} - 112xy^{12} - 1120xy^{11} - 880xy^{10} - 160xy^9 - 3072xy^8 - 4560xy^7 - 864xy^6 \\ - 2656xy^5 - 5840xy^4 - 1616xy^3 - 864xy^2 - 2224xy - 850x + 128y^{14} + 256y^{13} + 64y^{12} \\ - 704y^{11} - 1008y^{10} - 480y^9 - 3568y^8 - 4800y^7 - 1696y^6 - 4320y^5 - 5744y^4 - 1576y^3 \\ - 1824y^2 - 2016y - 499, \\ -384x^3y^{11} + 512x^3y^{10} + 3584x^3y^9 + 3360x^3y^8 + 2112x^3y^7 + 6912x^3y^6 \\ + 7872x^3y^5 + 2688x^3y^4 + 3072x^3y^3 + 4128x^3y^2 + 1088x^3y - 256x^3 \\ - 448x^2y^{12} - 864x^2y^{11} + 544x^2y^{10} + 1536x^2y^9 - 896x^2y^8 - 448x^2y^7 \\ + 3664x^2y^6 + 800x^2y^5 - 2528x^2y^4 + 928x^2y^3 + 832x^2y^2 - 1536x^2y - 752x^2 \rangle$

$$\begin{aligned}
& -256xy^{13} - 960xy^{12} - 1664xy^{11} - 1728xy^{10} - 3232xy^9 - 7136xy^8 - 6144xy^7 - 4352xy^6 \\
& -9280xy^5 - 8128xy^4 - 2848xy^3 - 3808xy^2 - 3456xy - 768x - 64y^{14} - 256y^{13} - 1328y^{12} \\
& -3232y^{11} - 3568y^{10} - 5856y^9 - 10560y^8 - 8928y^7 - 8048y^6 - 11680y^5 - 8176y^4 \\
& -3840y^3 - 4288y^2 - 2560y - 320)
\end{aligned}$$

$i = 14, D_i = 4, \rho_{ij}: 16:2, 18:1, 19:2, 21:2, 23:2, 24:1, 26:1, 30:1, 32:1, 36:2, 37:2, 40:2, 45:1, 58:1$

$$W \quad (-21y + 42, \\ 42y - 84)$$

$$\begin{aligned}
A \quad & (-8448x^{18} - 14976x^{17}y - 18048x^{17} + 3968x^{16}y^2 + 74880x^{16}y + 19584x^{16} \\
& + 6144x^{15}y^3 + 326272x^{15}y^2 + 430848x^{15}y + 199680x^{15} - 16128x^{14}y^4 \\
& + 263936x^{14}y^3 + 953472x^{14}y^2 + 1044288x^{14}y + 521280x^{14} - 2304x^{13}y^5 \\
& + 17984x^{13}y^4 + 122752x^{13}y^3 + 1123840x^{13}y^2 + 1247424x^{13}y + 738624x^{13} \\
& + 36096x^{12}y^6 + 90560x^{12}y^5 - 572928x^{12}y^4 - 1471872x^{12}y^3 - 699264x^{12}y^2 \\
& + 170496x^{12}y + 451968x^{12} + 41088x^{11}y^7 + 87680x^{11}y^6 + 107200x^{11}y^5 \\
& - 1845056x^{11}y^4 - 4248448x^{11}y^3 - 4474432x^{11}y^2 - 2124192x^{11}y - 492576x^{11} \\
& - 6528x^{10}y^8 - 225792x^{10}y^7 - 149760x^{10}y^6 + 284992x^{10}y^5 - 1704064x^{10}y^4 \\
& - 4738176x^{10}y^3 - 7244576x^{10}y^2 - 4024608x^{10}y - 1607136x^{10} - 12288x^9y^9 \\
& - 569728x^9y^8 - 993440x^9y^7 - 723808x^9y^6 + 1572832x^9y^5 + 2097760x^9y^4 \\
& + 480192x^9y^3 - 5172832x^9y^2 - 3387456x^9y - 1981440x^9 + 32256x^8y^{10} \\
& - 494592x^8y^9 - 1001184x^8y^8 - 1919264x^8y^7 - 196096x^8y^6 + 3381472x^8y^5 \\
& + 7504352x^8y^4 + 9435264x^8y^3 + 2354528x^8y^2 + 185808x^8y - 999792x^8 \\
& + 4608x^7y^{11} - 152128x^7y^{10} - 422784x^7y^9 - 183520x^7y^8 - 574144x^7y^7 \\
& + 1024736x^7y^6 + 3214464x^7y^5 + 9258096x^7y^4 + 14702240x^7y^3 + 10708704x^7y^2 \\
& + 4557360x^7y + 957456x^7 - 46848x^6y^{12} - 112704x^6y^{11} + 178624x^6y^{10} \\
& + 1505664x^6y^9 + 2999712x^6y^8 + 2386416x^6y^7 + 1468464x^6y^6 + 368800x^6y^5 \\
& + 4076912x^6y^4 + 9830176x^6y^3 + 13182528x^6y^2 + 6324672x^6y + 2592480x^6 \\
& - 37248x^5y^{13} - 79744x^5y^{12} + 238784x^5y^{11} + 1637904x^5y^{10} + 4316544x^5y^9 \\
& + 5280560x^5y^8 + 4987408x^5y^7 + 393040x^5y^6 - 4838912x^5y^5 - 5753216x^5y^4 \\
& - 4281184x^5y^3 + 6279984x^5y^2 + 3535704x^5y + 2573112x^5 + 1152x^4y^{14} \\
& + 116352x^4y^{13} + 127360x^4y^{12} + 1178480x^4y^{11} + 1734944x^4y^{10} + 3730640x^4y^9 \\
& + 3137888x^4y^8 + 2537632x^4y^7 - 4010784x^4y^6 - 10764432x^4y^5 - 14991152x^4y^4 \\
& - 19053184x^4y^3 - 6921112x^4y^2 - 2516328x^4y + 577512x^4 + 6144x^3y^{15} \\
& + 229760x^3y^{14} + 320800x^3y^{13} + 503776x^3y^{12} - 357808x^3y^{11} - 1376976x^3y^{10} \\
& - 2716240x^3y^9 - 4270240x^3y^8 - 5176752x^3y^7 - 9719296x^3y^6 - 14049936x^3y^5 \\
& - 17814576x^3y^4 - 23691216x^3y^3 - 17791240x^3y^2 - 7805712x^3y - 2238720x^3 \\
& - 16128x^2y^{16} + 238848x^2y^{15} + 430240x^2y^{14} + 201216x^2y^{13} - 1001440x^2y^{12} \\
& - 4077104x^2y^{11} - 6392208x^2y^{10} - 9836752x^2y^9 - 10368032x^2y^8 \\
& - 9559296x^2y^7 - 8267680x^2y^6 - 8570832x^2y^5 - 9697968x^2y^4 - 12314560x^2y^3 \\
& - 17624136x^2y^2 - 8121324x^2y - 3879036x^2 - 2304xy^{17} + 139712xy^{16} + 255360xy^{15} \\
& + 72096xy^{14} - 1393960xy^{13} - 3441032xy^{12} - 6241104xy^{11} - 8420032xy^{10} - 10691736xy^9 \\
& - 7024520xy^8 - 3229984xy^7 + 5428400xy^6 + 8899168xy^5 + 8647380xy^4 + 10855512xy^3 \\
& - 4101760xy^2 - 2277540xy + 2861868x + 2816y^{18} + 6720y^{17} - 64576y^{16} - 205696y^{15} \\
& - 218216y^{14} + 525720y^{13} + 1829544y^{12} + 4109776y^{11} + 5928520y^{10} + 7894344y^9 \\
& + 6292968y^8 + 4234792y^7 + 667464y^6 + 138668y^5 + 1516288y^4 + 1662080y^3 + 8840280y^2 \\
& + 3262884y + 3061644, \\
& -9984x^{18} - 39936x^{17}y - 41856x^{17} - 50048x^{16}y^2 - 120192x^{16}y - 79488x^{16} \\
& - 33792x^{15}y^3 + 40064x^{15}y^2 - 55296x^{15}y - 25728x^{15} - 43008x^{14}y^4 \\
& + 222976x^{14}y^3 + 767616x^{14}y^2 + 545664x^{14}y + 251712x^{14} - 43776x^{13}y^5 \\
& + 37504x^{13}y^4 + 961856x^{13}y^3 + 2060288x^{13}y^2 + 1761600x^{13}y + 749760x^{13} \\
& + 8448x^{12}y^6 - 4160x^{12}y^5 - 141696x^{12}y^4 + 935808x^{12}y^3 + 2331264x^{12}y^2 \\
& + 2693760x^{12}y + 1183296x^{12} + 84480x^{11}y^7 + 185728x^{11}y^6 - 114496x^{11}y^5 \\
& - 1551040x^{11}y^4 - 1527104x^{11}y^3 - 721664x^{11}y^2 + 1821120x^{11}y + 1093344x^{11} \\
& + 101760x^{10}y^8 + 188160x^{10}y^7 + 296832x^{10}y^6 - 556096x^{10}y^5 - 3685952x^{10}y^4 \\
& - 5673216x^{10}y^3 - 6496288x^{10}y^2 - 1476384x^{10}y + 266208x^{10} + 67584x^9y^9 \\
& - 93440x^9y^8 - 390208x^9y^7 - 19808x^9y^6 - 881920x^9y^5 - 3293920x^9y^4 \\
& - 6785280x^9y^3 - 10275104x^9y^2 - 5495808x^9y - 942048x^9 + 86016x^8y^{10} \\
& - 327936x^8y^9 - 1062624x^8y^8 - 2142496x^8y^7 - 360896x^8y^6 + 972896x^8y^5 \\
& + 1798240x^8y^4 - 1391808x^8y^3 - 6691232x^8y^2 - 6966624x^8y - 1708848x^8 \\
& + 87552x^7y^{11} - 146432x^7y^{10} - 1061568x^7y^9 - 2379872x^7y^8 - 2745728x^7y^7 \\
& + 109888x^7y^6 + 3798912x^7y^5 + 8970624x^7y^4 + 7991536x^7y^3 + 4166304x^7y^2 \\
& - 3717552x^7y + 1339344x^7 + 13056x^6y^{12} - 33984x^6y^{11} - 488512x^6y^{10} \\
& - 585408x^6y^9 - 1297440x^6y^8 - 505632x^6y^7 + 796368x^6y^6 + 5243984x^6y^5 \\
& + 12552400x^6y^4 + 14045600x^6y^3 + 15192768x^6y^2 + 3006240x^6y + 121488x^6 \\
& - 49152x^5y^{13} - 135296x^5y^{12} - 896x^5y^{11} + 537120x^5y^{10} + 2627280x^5y^9 \\
& + 2952352x^5y^8 + 4143680x^5y^7 + 2334128x^5y^6 + 3663200x^5y^5 + 8800064x^5y^4 \\
& + 11156176x^5y^3 + 17315808x^5y^2 + 8762928x^5y + 1771512x^5 - 53376x^4y^{14} \\
& - 84864x^4y^{13} + 5120x^4y^{12} + 1055344x^4y^{11} + 3386752x^4y^{10} + 5977936x^4y^9 \\
& + 7181248x^4y^8 + 6658304x^4y^7 + 1733328x^4y^6 - 2035824x^4y^5 - 2210080x^4y^4 \\
& - 259568x^4y^3 + 6767464x^4y^2 + 8931768x^4y + 2378520x^4 - 33792x^3y^{15} \\
& + 25216x^3y^{14} + 81920x^3y^{13} + 1051232x^3y^{12} + 1844464x^3y^{11} + 3943056x^3y^{10} \\
& + 4452928x^3y^9 + 5117200x^3y^8 + 1455888x^3y^7 - 4671200x^3y^6 - 11700336x^3y^5 \\
& - 16457568x^3y^4 - 14041728x^3y^3 - 11024024x^3y^2 + 2292384x^3y + 1280904x^3 \\
& - 43008x^2y^{16} + 97536x^2y^{15} + 359648x^2y^{14} + 756096x^2y^{13} + 853312x^2y^{12}
\end{aligned}$$

$$\begin{aligned}
& -137488x^2y^{11} - 596784x^2y^{10} - 2991488x^2y^9 - 3857344x^2y^8 - 9203856x^2y^7 \\
& -12817376x^2y^6 - 19740720x^2y^5 - 25734240x^2y^4 - 21533024x^2y^3 \\
& -24070008x^2y^2 - 7370424x^2y - 1007244x^2 - 43776xy^{17} + 111232xy^{16} + 401856xy^{15} \\
& + 659520xy^{14} + 163312xy^{13} - 2074744xy^{12} - 4612464xy^{11} - 8061008xy^{10} - 11660784xy^9 \\
& -12384136xy^8 - 15134864xy^7 - 13365152xy^6 - 16029184xy^5 - 20481720xy^4 - 16178028xy^3 \\
& -21636752xy^2 - 13228812xy - 2960772x + 3328y^{18} + 33600y^{17} - 25856y^{16} - 214016y^{15} \\
& -443272y^{14} - 341304y^{13} + 598536y^{12} + 1978160y^{11} + 3495752y^{10} + 5673816y^9 \\
& + 6051816y^8 + 9098624y^7 + 10179288y^6 + 14255140y^5 + 18544952y^4 + 14663152y^3 \\
& + 17226816y^2 + 6594024y + 1891992) \\
B \quad & (16384x^3y^{18} + 16128x^3y^{17} + 172736x^3y^{16} - 95104x^3y^{15} - 1193808x^3y^{14} \\
& - 2573040x^3y^{13} - 5002384x^3y^{12} - 7909632x^3y^{11} - 8820224x^3y^{10} \\
& - 8716080x^3y^9 + 66400x^3y^8 + 8267504x^3y^7 + 24674096x^3y^6 + 30385976x^3y^5 \\
& + 26996040x^3y^4 + 30242600x^3y^3 + 6687696x^3y^2 + 3156300x^3y - 2519796x^3 \\
& + 11136x^2y^{19} + 18304x^2y^{18} + 28352x^2y^{17} - 472688x^2y^{16} - 967872x^2y^{15} \\
& - 1485200x^2y^{14} - 1052248x^2y^{13} - 317096x^2y^{12} + 2544064x^2y^{11} \\
& + 5624256x^2y^{10} + 11586936x^2y^9 + 21912744x^2y^8 + 30284680x^2y^7 \\
& + 46908904x^2y^6 + 54283008x^2y^5 + 48912440x^2y^4 + 49992012x^2y^3 \\
& + 31066516x^2y^2 + 13444524x^2y + 2918340x^2 + 1408xy^{20} + 40128xy^{19} - 47424xy^{18} \\
& - 465008xy^{17} - 707304xy^{16} - 616624xy^{15} + 1207120xy^{14} + 5203144xy^{13} + 10727936xy^{12} \\
& + 18715568xy^{11} + 23580512xy^{10} + 30074280xy^9 + 33396592xy^8 + 35585044xy^7 + 40038388xy^6 \\
& + 46250148xy^5 + 42703228xy^4 + 37882990xy^3 + 37147792xy^2 + 15982356xy + 6769140x + 14400y^{20} \\
& - 45056y^{19} - 261312y^{18} - 409960y^{17} + 175536y^{16} + 1946176y^{15} + 5492648y^{14} \\
& + 11066952y^{13} + 17478624y^{12} + 22406352y^{11} + 22387616y^{10} + 20691456y^9 + 13397444y^8 \\
& + 10134524y^7 + 4513296y^6 + 10779696y^5 + 15135740y^4 + 9824647y^3 + 19686450y^2 + 8637276y \\
& + 4715616, \\
& -14848x^3y^{18} + 5376x^3y^{17} + 305920x^3y^{16} + 654784x^3y^{15} - 140784x^3y^{14} \\
& - 2047680x^3y^{13} - 6258080x^3y^{12} - 10908192x^3y^{11} - 16378096x^3y^{10} \\
& - 18760752x^3y^9 - 16939744x^3y^8 - 15498560x^3y^7 - 9102896x^3y^6 \\
& - 9554744x^3y^5 - 15971832x^3y^4 - 12984488x^3y^3 - 19671528x^3y^2 - 16671984x^3y \\
& - 4910964x^3 + 4608x^2y^{19} - 16000x^2y^{18} + 187456x^2y^{17} + 183104x^2y^{16} \\
& - 502608x^2y^{15} - 2019088x^2y^{14} - 4756064x^2y^{13} - 8956984x^2y^{12} \\
& - 11772064x^2y^{11} - 14718720x^2y^{10} - 10307760x^2y^9 - 3341688x^2y^8 \\
& + 9359744x^2y^7 + 21505064x^2y^6 + 29280504x^2y^5 + 25684240x^2y^4 \\
& + 16529352x^2y^3 + 12582812x^2y^2 - 6943164x^2y - 3232764x^2 + 1664xy^{20} + 19200xy^{19} \\
& + 77376xy^{18} - 148432xy^{17} - 588624xy^{16} - 1540712xy^{15} - 2310304xy^{14} - 3021064xy^{13} \\
& - 2521544xy^{12} + 67864xy^{11} + 4077184xy^{10} + 14970456xy^9 + 24852944xy^8 + 44288864xy^7 \\
& + 53894180xy^6 + 65182488xy^5 + 64537436xy^4 + 42075548xy^3 + 41683412xy^2 + 10513248xy \\
& + 871248x + 28992y^{20} + 12608y^{19} - 232704y^{18} - 592904y^{17} - 839616y^{16} - 164656y^{15} \\
& + 1708840y^{14} + 4784904y^{13} + 10366128y^{12} + 15684768y^{11} + 22145272y^{10} + 30622728y^9 \\
& + 36527308y^8 + 50998876y^7 + 53566680y^6 + 60400248y^5 + 60910744y^4 + 37979018y^3 \\
& + 35345436y^2 + 15188424y + 2450400) \\
i = 15, D_i = 16, \rho_{ij}: 2:2, 3:1, 20:2, 21:1, 33:1, 35:1, 36:1, 58:2 \\
W \quad & (8y^2 - 16y + 34, \\
& -16y^2 + 18y + 9) \\
A \quad & (20480x^6y^3 + 45056x^6y^2 - 8192x^6y - 32768x^6 + 28672x^5y^4 + 180224x^5y^3 \\
& + 270336x^5y^2 - 20480x^5y - 139264x^5 + 49152x^4y^5 + 212992x^4y^4 + 712704x^4y^3 \\
& + 745472x^4y^2 - 77824x^4y - 274432x^4 + 81920x^3y^6 + 282624x^3y^5 + 823296x^3y^4 \\
& + 1648640x^3y^3 + 1226752x^3y^2 - 114688x^3y - 315392x^3 + 94208x^2y^7 \\
& + 423936x^2y^6 + 880640x^2y^5 + 1761280x^2y^4 + 2306048x^2y^3 + 1224704x^2y^2 \\
& - 10240x^2y - 139264x^2 + 73728xy^8 + 497664xy^7 + 1099776xy^6 + 1658880xy^5 + 2148352xy^4 \\
& + 1652736xy^3 + 532480xy^2 + 227328xy + 182272x + 116736y^8 + 416768y^7 + 770560y^6 + 745472y^5 \\
& + 866816y^4 + 340928y^3 + 93920y^2 + 214320y + 129544, \\
& -12288x^6y^3 + 53248x^6y^2 + 45056x^6y - 20480x^6 - 57344x^5y^4 + 12288x^5y^3 \\
& + 319488x^5y^2 + 212992x^5y - 36864x^5 - 69632x^4y^5 - 167936x^4y^4 + 294912x^4y^3 \\
& + 917504x^4y^2 + 528384x^4y + 4096x^4 - 49152x^3y^6 - 249856x^3y^5 - 12288x^3y^4 \\
& + 1017856x^3y^3 + 1632256x^3y^2 + 931840x^3y + 129024x^3 - 16384x^2y^7 \\
& - 174080x^2y^6 - 126976x^2y^5 + 749568x^2y^4 + 1767424x^2y^3 + 1894400x^2y^2 \\
& + 1110016x^2y + 264192x^2 - 4096xy^8 - 77824xy^7 + 22528xy^6 + 811008xy^5 + 1681408xy^4 \\
& + 1697792xy^3 + 1185792xy^2 + 706560xy + 251904x + 10240y^8 + 40960y^7 + 290304y^6 + 628992y^5 \\
& + 1009024y^4 + 403200y^3 + 298016y^2 + 153648y + 148536) \\
B \quad & (20480x^3y^9 + 235520x^3y^8 + 697344x^3y^7 + 880128x^3y^6 + 825344x^3y^5 \\
& + 1536x^3y^4 - 726976x^3y^3 - 688864x^3y^2 + 170704x^3y + 298488x^3 + 120832x^2y^9 \\
& + 454656x^2y^8 + 670720x^2y^7 + 597760x^2y^6 - 209024x^2y^5 - 1320128x^2y^4 \\
& - 1832384x^2y^3 - 1124656x^2y^2 - 86248x^2y + 51692x^2 - 6144xy^{10} - 1024xy^9 \\
& - 42496xy^8 - 222464xy^7 - 933120xy^6 - 1815872xy^5 - 2330464xy^4 - 2090896xy^3 \\
& - 985104xy^2 - 537792xy - 168796x - 36864y^{11} - 173056y^{10} - 402432y^9 - 590592y^8 - 927104y^7 \\
& - 1170112y^6 - 1070016y^5 - 1331808y^4 - 732728y^3 - 349300y^2 - 219850y - 203627, \\
& -12288x^3y^9 - 40960x^3y^8 + 53248x^3y^7 + 666112x^3y^6 + 1318656x^3y^5 \\
& + 637568x^3y^4 - 11008x^3y^3 - 763936x^3y^2 - 414768x^3y - 94264x^3 - 12288x^2y^9 \\
& - 41984x^2y^8 + 325120x^2y^7 + 1108992x^2y^6 + 820352x^2y^5 - 170048x^2y^4 \\
& - 1239712x^2y^3 - 1385088x^2y^2 - 1106768x^2y - 330464x^2 - 16384xy^{10} - 69632xy^9 \\
& - 4608xy^8 + 364288xy^7 + 352896xy^6 - 309760xy^5 - 1270144xy^4 - 1419216xy^3 - 1286696xy^2)
\end{aligned}$$

	$-846308xy - 250758x + 2048y^{11} + 23552y^{10} + 5632y^9 - 119808y^8 - 260352y^7 - 223680y^6$ $-455904y^5 - 770640y^4 - 451536y^3 - 312564y^2 - 91334y - 116395)$
$i = 16, D_i = 2, \rho_{ij}: 2:1, 13:2, 14:1, 19:2, 20:2, 27:2, 30:1, 34:1, 54:1$	
w_i	$\langle -344t^3 + 136t^2 - 44t + 16,$ $-80t^3 + 100t^2 + 8t + 104 \rangle$
$i = 17, D_i = 2, \rho_{ij}: 2:1, 18:1, 26:2, 30:2, 33:2, 38:2, 57:1$	
w_i	$\langle 40t^3 - 20t^2 - 4t - 4,$ $32t^3 - 16t^2 + 8t + 8 \rangle$
$i = 18, D_i = 64, \rho_{ij}: 5:1, 14:2, 17:2, 30:1, 36:1, 37:2, 39:2, 40:2, 46:2, 48:2, 50:2, 55:1, 56:2, 58:1, 60:1, 63:2$	
w_i	$\langle 10673t^3 + 5357t^2 - 3339t - 4974,$ $8114t^3 + 10275t^2 + 4816t + 240 \rangle$
$i = 19, D_i = 160, \rho_{ij}: 12:2, 14:1, 16:1, 28:1, 30:2, 33:1, 36:2, 46:1, 48:1, 51:2, 52:1, 53:1, 62:1, 63:2$	
w_i	$\langle -9506t^3 - 6650t^2 + 6650t + 10724,$ $3626t^3 + 8162t^2 + 5278t - 1652 \rangle$
$i = 20, D_i = 4, \rho_{ij}: 1:1, 9:1, 15:1, 16:1, 53:1, 54:1, 57:1, 58:2, 59:2$	
w_i	$\langle -33t^3 - 21t^2 + 9t + 18,$ $3t^3 + 3t + 6 \rangle$
$i = 21, D_i = 16, \rho_{ij}: 3:2, 6:2, 10:2, 14:1, 15:2, 26:2, 37:1, 49:2, 51:2, 54:1, 58:1, 63:1$	
w_i	$\langle 448t^3 + 289t^2 - 218t - 308,$ $91t^3 + 129t^2 + 51t - 98 \rangle$
$i = 22, D_i = 2, \rho_{ij}: 8:1, 12:1, 13:2, 23:2, 28:2, 48:2, 54:1, 56:2$	
w_i	$\langle 60t^3 + 20t^2 - 4t - 24,$ $36t^3 - 4t^2 - 12t - 24 \rangle$
$i = 23, D_i = 16, \rho_{ij}: 10:2, 14:1, 22:1, 30:2, 39:2, 41:1, 44:2, 47:1, 53:1, 54:2, 56:2, 59:2$	
w_i	$\langle 515t^3 + 496t^2 - 424t - 588,$ $117t^3 + 1054t^2 + 1313t + 412 \rangle$
$i = 24, D_i = 4, \rho_{ij}: 7:2, 9:2, 11:1, 14:2, 31:2, 32:1, 36:1, 48:1, 49:2, 51:2, 55:1$	
w_i	$\langle -124t^3 + 86t^2 + 63t + 5,$ $-169t^3 - 85t^2 + 98t + 1 \rangle$
$i = 25, D_i = 2, \rho_{ij}: 7:2, 41:2, 58:1, 63:2$	
w_i	$\langle 4t^3 + 2t + 6,$ $-4t^3 - 4t + 4 \rangle$
$i = 26, D_i = 8, \rho_{ij}: 7:1, 14:2, 17:1, 21:1, 28:1, 30:1, 35:1, 37:2, 40:2, 41:2, 47:2, 48:2$	
w_i	$\langle 179t^3 + 103t^2 - 4t - 26,$ $97t^3 + 228t^2 + 127t + 73 \rangle$
$i = 27, D_i = 2, \rho_{ij}: 5:1, 8:1, 13:1, 16:1, 36:1, 53:1, 55:2$	
w_i	$\langle -4t^3 + 4,$ $4t^3 - 4 \rangle$
$i = 28, D_i = 8, \rho_{ij}: 6:1, 7:1, 9:1, 19:2, 22:1, 26:2, 37:2, 45:2, 50:2, 58:2$	
w_i	$\langle 376t^3 + 475t^2 + 232t - 53,$ $303t^3 + 485t^2 + 344t + 33 \rangle$
$i = 29, D_i = 1, \rho_{ij}: 3:1, 4:1, 11:1, 12:2, 13:2, 32:2, 58:1$	
w_i	$\langle 0,$ $8t^3 - 4t^2 - 16t - 16 \rangle$
$i = 30, D_i = 128, \rho_{ij}: 4:2, 14:2, 16:2, 17:1, 18:2, 19:1, 23:1, 26:2, 31:2, 39:1, 46:1, 50:2, 52:2, 57:2, 58:2, 63:2$	
w_i	$\langle -37588t^3 - 44661t^2 - 23872t + 5587,$ $-20941t^3 + 17768t^2 + 48710t + 37700 \rangle$
$i = 31, D_i = 2, \rho_{ij}: 4:2, 7:1, 8:2, 12:1, 24:1, 30:1, 44:1, 55:2, 63:1$	
w_i	$\langle -11t^3 - 22t^2 - 12t + 2,$ $-7t^3 - 9t^2 + 4t + 3 \rangle$
$i = 32, D_i = 4, \rho_{ij}: 7:1, 9:1, 12:2, 13:2, 14:2, 24:2, 29:1, 47:1, 49:1, 59:1, 60:2$	
w_i	$\langle -48t^3 + 48t^2 + 32t - 16,$ $-28t^3 + 8t^2 + 48t - 48 \rangle$
$i = 33, D_i = 8, \rho_{ij}: 5:2, 7:2, 9:2, 15:2, 17:1, 19:2, 34:1, 35:2, 43:2, 52:2, 54:1$	
w_i	$\langle 104t^3 + 205t^2 + 200t + 93,$ $-17t^3 + 209t^2 + 278t + 181 \rangle$
$i = 34, D_i = 2, \rho_{ij}: 4:2, 16:2, 33:2, 39:1, 46:1, 55:1$	
w_i	$\langle 8t^3 - 6t^2 - 2t,$ $12t^3 + 4t^2 - 2t \rangle$
$i = 35, D_i = 16, \rho_{ij}: 3:2, 6:2, 7:2, 13:1, 15:2, 26:2, 33:1, 40:1, 53:1, 55:2, 56:1, 58:2, 62:2$	
w_i	$\langle -244t^3 - 1216t^2 - 1213t - 168,$ $1027t^3 - 104t^2 - 581t - 666 \rangle$
$i = 36, D_i = 8, \rho_{ij}: 3:2, 7:2, 8:2, 9:2, 14:1, 15:2, 18:2, 19:1, 24:2, 27:2, 40:1, 51:2$	
w_i	$\langle -858t^3 - 1051t^2 + 315t + 475,$ $-719t^3 + 172t^2 + 1239t + 585 \rangle$
$i = 37, D_i = 4, \rho_{ij}: 7:2, 13:2, 14:1, 18:1, 21:2, 26:1, 28:1, 48:1, 50:2, 51:2, 55:1, 60:1$	
w_i	$\langle 92t^3 - 30t^2 - 23t + 23,$ $75t^3 + 25t^2 + 95t - 4 \rangle$

$i = 38, D_i = 1, \rho_{ij}: 6:2, 7:1, 17:1, 52:1$
 $w_i \quad \langle -2t - 2, 0 \rangle$

$i = 39, D_i = 4, \rho_{ij}: 3:1, 4:1, 5:1, 18:1, 23:1, 30:2, 34:2, 58:2, 59:2$
 $w_i \quad \langle -20t^3 - 272t^2 - 532t - 256, 32t^3 - 280t^2 - 380t - 104 \rangle$

$i = 40, D_i = 16, \rho_{ij}: 3:1, 11:2, 14:1, 18:1, 26:1, 35:2, 36:2, 52:1, 54:2, 57:1, 58:1, 59:1, 61:1$
 $w_i \quad \langle 6162t^3 - 9996t^2 + 11883t - 3630, 6495t^3 - 11130t^2 + 8622t - 1104 \rangle$

$i = 41, D_i = 2, \rho_{ij}: 10:1, 11:1, 23:2, 25:1, 26:1, 52:1, 59:2$
 $w_i \quad \langle -2t^3 + 24t^2 - 66t - 20, -10t^3 + 36t^2 - 36t - 16 \rangle$

$i = 42, D_i = 4, \rho_{ij}: 0:1, 2:2, 3:2, 9:2, 58:1, 63:1$
 $w_i \quad \langle -144t^3 + 136t^2 + 56t + 40, -40t^3 + 72t^2 + 8 \rangle$

$i = 43, D_i = 4, \rho_{ij}: 0:1, 1:1, 33:1, 44:1, 46:2, 47:2, 56:1$
 $w_i \quad \langle -8t^3 - 64t^2 - 52t - 24, -16t^3 - 20t^2 - 12t - 16 \rangle$

$i = 44, D_i = 4, \rho_{ij}: 5:2, 8:2, 10:1, 12:1, 23:1, 31:2, 43:2, 46:2, 50:1, 56:1, 59:1$
 $w_i \quad \langle -8t^3 + 604t^2 - 380t + 208, -368t^3 + 864t^2 - 188t + 260 \rangle$

$i = 45, D_i = 4, \rho_{ij}: 5:1, 8:1, 10:2, 14:2, 28:1, 49:2$
 $w_i \quad \langle -24t^3 + 88t^2 - 76t + 40, -8t^3 + 24t^2 - 60t + 4 \rangle$

$i = 46, D_i = 4, \rho_{ij}: 0:2, 1:1, 7:1, 10:2, 18:1, 19:2, 30:2, 34:2, 43:1, 44:1, 51:2, 54:2$
 $w_i \quad \langle 304t^5 - 436t^4 + 240t^3 - 652t^2 - 144t - 124, -8t^5 + 284t^4 + 360t^3 - 608t^2 + 36t - 176 \rangle$

$i = 47, D_i = 4, \rho_{ij}: 0:2, 23:2, 26:1, 32:2, 43:1, 50:1, 58:2, 61:1, 62:2, 63:2$
 $w_i \quad \langle 124t^5 + 16t^4 - 128t^3 + 144t^2 + 128t + 228, 8t^5 + 104t^4 - 160t^3 + 180t^2 + 76t + 96 \rangle$

$i = 48, D_i = 16, \rho_{ij}: 1:1, 18:1, 19:2, 22:1, 24:2, 26:1, 37:2$
 $w_i \quad \langle 48t^4 + 16t^3 - 32t^2 - 32t, 32t^5 + 96t^4 + 80t^3 + 16t^2 \rangle$

$i = 49, D_i = 32, \rho_{ij}: 3:1, 8:2, 21:1, 24:1, 32:2, 45:1, 50:1$
 $w_i \quad \langle 32t^5 - 32t^4 - 32t^3 - 64t^2, 64t^5 - 64t^4 + 32t^3 - 32t^2 + 96t \rangle$

$i = 50, D_i = 32768, \rho_{ij}: 1:2, 2:1, 3:2, 6:2, 8:1, 18:1, 28:1, 30:1, 37:1, 44:2, 47:2, 49:2, 54:1, 61:1, 62:2$
 $w_i \quad \langle 6684672t^5 - 55640064t^4 + 26935296t^3 - 10321920t^2 - 1867776t - 36765696, -28606464t^5 + 46497792t^4 - 24379392t^3 - 3244032t^2 + 15532032t - 5505024 \rangle$

$i = 51, D_i = 32, \rho_{ij}: 2:2, 5:2, 11:1, 19:1, 21:1, 24:1, 36:1, 37:1, 46:1$
 $w_i \quad \langle 96t^4 + 32t^3 - 64t^2 - 64t, -64t^5 - 96t^4 - 128t^3 - 96t^2 - 64t \rangle$

$i = 52, D_i = 8, \rho_{ij}: 5:1, 12:2, 19:2, 30:1, 33:1, 38:2, 40:2, 41:2, 56:2, 59:2, 60:2$
 $w_i \quad \langle -96t^5 + 56t^4 + 96t^3 - 32t^2 - 40t + 80, -40t^5 + 152t^4 + 64t^3 + 24t^2 - 56t - 112 \rangle$

$i = 53, D_i = 64, \rho_{ij}: 8:2, 19:2, 20:2, 23:2, 27:2, 35:2, 57:2$
 $w_i \quad \langle 384t^5 + 2304t^4 + 2688t^3 + 3072t^2 + 2304t + 1536, 384t^5 + 2304t^4 + 2688t^3 + 3072t^2 + 2304t + 1536 \rangle$

$i = 54, D_i = 8192, \rho_{ij}: 6:1, 7:1, 12:2, 16:2, 20:2, 21:2, 22:2, 23:1, 33:2, 40:1, 46:1, 50:2, 63:2$
 $w_i \quad \langle -8085504t^5 + 16244736t^4 - 18718720t^3 + 13754368t^2 - 9322496t + 5439488, -11059200t^5 + 35790848t^4 - 40034304t^3 + 36700160t^2 - 19087360t + 17072128 \rangle$

$i = 55, D_i = 64, \rho_{ij}: 9:1, 18:2, 24:2, 27:1, 31:1, 34:2, 35:1, 37:2, 59:1$
 $w_i \quad \langle -768t^5 - 448t^4 + 768t^3 - 320t^2 + 320t - 448, -576t^5 - 384t^4 - 256t^3 - 128t^2 + 320t - 768 \rangle$

$i = 56, D_i = 128, \rho_{ij}: 5:1, 18:1, 22:1, 23:1, 35:2, 43:2, 44:2, 52:1, 58:2$
 $w_i \quad \langle 1408t^5 - 2176t^4 - 2688t^3 + 1024t^2 - 512t - 1280, 128t^5 - 2688t^4 + 768t^3 + 1024t^2 - 2048t - 1152 \rangle$

$i = 57, D_i = 32, \rho_{ij}: 4:1, 17:2, 20:2, 30:1, 40:2, 53:1$
 $w_i \quad \langle 64t^5 + 160t^4 + 32t^3 + 32t^2 - 32t, 96t^5 + 128t^4 + 160t^3 + 160t^2 + 64t \rangle$

$i = 58, D_i = 262144, \rho_{ij}: 3:2, 11:2, 14:2, 15:1, 18:2, 20:1, 21:2, 25:2, 28:1, 29:2, 30:1, 35:1, 39:1, 40:2, 42:2, 47:1, 56:1, 60:2, 63:1$
 $w_i \quad \langle 1932001280t^5 + 1109917696t^4 + 3083337728t^3 + 2095579136t^2 + 652214272t - 1580204032, -1714946048t^5 - 1124073472t^4 - 2915041280t^3 - 1919418368t^2 - 516947968t + 1429209088 \rangle$

$i = 59, D_i = 256, \rho_{ij}: 6:1, 20:1, 23:1, 32:2, 39:1, 40:2, 41:1, 44:2, 52:1, 55:2$
 $w_i \quad \langle 8960t^5 + 3072t^4 - 14080t^3 + 9728t^2 + 7168t + 6144, 13824t^5 - 1536t^4 - 256t^3 + 8192t^2 + 4096t + 10240 \rangle$

$i = 60, D_i = 4, \rho_{ij}: 3:1, 5:2, 6:1, 18:2, 32:1, 37:2, 52:1, 58:1$

w_i	$\langle 8t^4 - 12t^2 + 4t + 4,$ $-4t^5 + 8t^3 - 8t^2 - 4t + 4 \rangle$
$i = 61, D_i = 18, e_i = 0, \rho_{ij}: 7:1, 8:1, 40:2, 47:2, 50:2$	
w_i	$\langle 54t^4 - 108t^2 + 162t + 216,$ $108t^4 - 540t^2 + 324t + 270 \rangle$
$i = 62, D_i = 72, e_i = 1, \rho_{ij}: 6:1, 19:2, 35:1, 47:1, 50:1$	
w_i	$\langle -432t^3 - 432t^2,$ $-216t^4 - 648t^3 - 432t^2 \rangle$
$i = 63, D_i = 2, \rho_{ij}: 5:1, 8:2, 18:1, 19:1, 21:2, 25:1, 30:1, 31:2, 42:2, 47:1, 54:1, 58:2$	
w_i	$\langle 22t^7 - 2t^6 + 73t^5 + 37t^4 + 43t^3 + 77t^2 - 8t + 32,$ $24t^7 - 20t^6 + 100t^5 - 64t^4 + 108t^3 - 21t^2 + 25t + 12 \rangle$

Table 5: The edges (i, j) with a linear endpoint, the linear endpoint k used, and the norm of $R_{ij} = \text{Res}_t(f_k, g_{kl})$.

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
<i>The first certificate.</i>					
(0, 6)	0	3	(16, 64)	64	$7^3 \cdot 13$
(0, 15)	0	$2^2 \cdot 7^2$	(17, 30)	17	$2^2 \cdot 3 \cdot 7^2$
(0, 27)	0	7	(17, 50)	17	$2^2 \cdot 3 \cdot 7$
(0, 47)	0	$2^2 \cdot 3^2 \cdot 7^2$	(17, 54)	17	$2^4 \cdot 3 \cdot 7^2$
(0, 48)	0	13	(17, 55)	17	$3^5 \cdot 7$
(0, 57)	0	7^2	(18, 20)	18	2^{10}
(0, 63)	0	$3 \cdot 7 \cdot 13$	(18, 26)	18	7^2
(0, 66)	0	2^2	(18, 32)	18	$2^2 \cdot 3 \cdot 7^2$
(1, 3)	1	1	(18, 34)	18	$3 \cdot 7$
(1, 6)	1	3	(18, 50)	18	$2^4 \cdot 3 \cdot 7^2$
(1, 16)	1	$2^2 \cdot 7^2$	(18, 58)	18	$2^4 \cdot 3 \cdot 7^2$
(1, 17)	1	7^2	(19, 22)	19	$2^4 \cdot 7 \cdot 13$
(1, 18)	1	7^2	(19, 26)	19	$7 \cdot 13^3$
(1, 22)	1	13	(19, 45)	19	$7 \cdot 13^4$
(1, 37)	1	3	(19, 55)	19	$7 \cdot 13^2$
(1, 49)	1	2^2	(19, 56)	19	$7^3 \cdot 13^2$
(1, 50)	1	7^2	(19, 60)	19	7^3
(1, 51)	1	$2^2 \cdot 7^2$	(20, 33)	20	7
(1, 52)	1	7^2	(20, 36)	20	13
(1, 55)	1	13	(20, 38)	20	$2^2 \cdot 13^2$
(1, 65)	1	$3 \cdot 7^4$	(20, 41)	20	13
(2, 15)	2	$2^2 \cdot 7^2$	(20, 42)	20	$2^2 \cdot 13^2$
(2, 17)	2	7^2	(20, 49)	20	3^2
(2, 18)	2	7^2	(20, 50)	20	$2^6 \cdot 7 \cdot 13$
(2, 26)	2	3	(20, 62)	20	2^4
(2, 28)	2	1	(21, 27)	21	$2^2 \cdot 7^2$
(2, 35)	2	3	(21, 30)	21	$2^2 \cdot 7^2$
(2, 52)	2	1	(21, 50)	21	$2^4 \cdot 7^2$
(2, 63)	2	7^2	(21, 56)	21	$2^2 \cdot 7^2$
(3, 19)	3	7	(21, 57)	21	3^4
(3, 25)	3	1	(21, 60)	21	$3^2 \cdot 7^2$
(3, 44)	3	1	(21, 61)	21	$2^4 \cdot 3^2 \cdot 7^2$
(3, 59)	3	2^4	(21, 66)	21	$2^6 \cdot 7^2$
(3, 61)	3	7	(22, 24)	22	2^2
(3, 66)	3	2^4	(22, 25)	22	$2^2 \cdot 13^2$
(4, 21)	4	2^2	(22, 26)	22	$7 \cdot 13$
(4, 30)	4	7	(22, 28)	22	$7 \cdot 13$
(4, 36)	4	2^2	(22, 43)	22	$2^2 \cdot 13^2$
(4, 47)	4	$2^2 \cdot 7 \cdot 13$	(22, 44)	22	13
(4, 51)	4	$2^4 \cdot 7$	(22, 49)	22	3^2
(4, 52)	4	13	(22, 52)	22	$2^2 \cdot 7 \cdot 13$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(4, 62)	4	$2^6 \cdot 7$	(22, 56)	22	13
(4, 64)	4	7	(23, 31)	23	$2^4 \cdot 13$
(5, 17)	5	3^4	(23, 32)	23	2^2
(5, 18)	5	3^4	(23, 50)	23	2^4
(5, 30)	5	3^2	(23, 56)	23	2^{12}
(5, 33)	5	3^3	(24, 37)	24	3
(5, 37)	5	3	(24, 40)	24	3
(5, 39)	5	1	(24, 54)	24	$3^2 \cdot 7^2$
(5, 40)	5	$3^2 \cdot 5^2$	(24, 57)	24	$2^4 \cdot 3^4$
(5, 41)	5	3^3	(24, 58)	24	$2^4 \cdot 7^2$
(5, 42)	5	$3^3 \cdot 13$	(24, 61)	24	$2^8 \cdot 3^2$
(5, 45)	5	3	(25, 52)	25	$7 \cdot 13^2$
(5, 53)	5	13	(26, 41)	26	$2^8 \cdot 7$
(5, 56)	5	3^3	(26, 43)	26	$3^3 \cdot 7^3$
(5, 58)	5	$3^2 \cdot 13^2$	(27, 34)	27	7^3
(5, 60)	5	3^2	(27, 37)	27	7^2
(5, 63)	5	3^3	(27, 43)	27	$2^2 \cdot 7^2 \cdot 13$
(5, 66)	5	5^2	(27, 53)	27	$2^2 \cdot 13$
(6, 18)	6	$3 \cdot 7$	(27, 55)	27	7
(6, 23)	6	2^2	(27, 56)	27	$2^4 \cdot 3^2 \cdot 7^2$
(6, 25)	6	7	(27, 57)	27	$2^6 \cdot 7^2$
(6, 26)	6	3	(28, 38)	28	$2^2 \cdot 7 \cdot 13$
(6, 32)	6	$3 \cdot 7$	(28, 41)	28	$3 \cdot 13$
(6, 38)	6	$7 \cdot 13$	(28, 48)	28	$2^4 \cdot 7^2$
(6, 39)	6	7	(28, 51)	28	$2^2 \cdot 7^2 \cdot 13$
(6, 46)	6	$2^2 \cdot 3$	(28, 53)	28	7
(6, 48)	6	2^2	(28, 56)	28	$3 \cdot 13^2$
(6, 50)	6	$2^2 \cdot 3 \cdot 7$	(28, 62)	28	$2^6 \cdot 13$
(6, 53)	6	7	(29, 57)	29	2^8
(6, 60)	6	3	(29, 64)	29	2^{12}
(6, 62)	6	$2^2 \cdot 3 \cdot 13$	(30, 50)	30	$2^2 \cdot 3^2$
(6, 64)	6	$3^2 \cdot 7$	(30, 51)	30	7^3
(6, 66)	6	7	(31, 33)	31	$3 \cdot 13$
(7, 41)	7	7	(31, 37)	31	$7 \cdot 13$
(7, 42)	7	$2^6 \cdot 7$	(31, 43)	31	$2^2 \cdot 7 \cdot 13$
(7, 45)	7	2^2	(31, 46)	31	$2^4 \cdot 3^3$
(7, 47)	7	$7 \cdot 13$	(31, 47)	31	3^4
(7, 51)	7	7	(31, 50)	31	$2^2 \cdot 3^3 \cdot 7$
(7, 52)	7	$7 \cdot 13$	(31, 52)	31	$3 \cdot 13^2$
(7, 56)	7	2^2	(31, 56)	31	$2^4 \cdot 3 \cdot 7$
(8, 27)	8	7	(32, 44)	32	$2^2 \cdot 7^2$
(8, 28)	8	$2^2 \cdot 3 \cdot 7$	(32, 48)	32	$2^4 \cdot 7$
(8, 30)	8	$3 \cdot 7$	(32, 55)	32	$3 \cdot 7^2$
(8, 37)	8	3	(32, 58)	32	$3^3 \cdot 7^2$
(8, 38)	8	7	(32, 59)	32	$2^8 \cdot 3 \cdot 7$
(8, 39)	8	7	(32, 63)	32	$2^4 \cdot 3$
(8, 40)	8	3	(33, 42)	33	$7^3 \cdot 13$
(8, 42)	8	$3^2 \cdot 7$	(33, 55)	33	$3 \cdot 13^2$
(8, 47)	8	$2^2 \cdot 3$	(33, 58)	33	$3 \cdot 7^2 \cdot 13$
(8, 58)	8	3	(33, 64)	33	13^3
(8, 59)	8	$2^2 \cdot 3 \cdot 7$	(34, 41)	34	$2^4 \cdot 3^2 \cdot 7$
(8, 63)	8	$2^2 \cdot 3$	(34, 65)	34	7^2
(8, 64)	8	$3^2 \cdot 7$	(35, 60)	35	$3^2 \cdot 7$
(8, 65)	8	3	(36, 42)	36	$2^2 \cdot 7^2 \cdot 13$
(8, 66)	8	7	(36, 49)	36	$2^2 \cdot 7$
(9, 22)	9	$7 \cdot 13$	(36, 51)	36	$2^2 \cdot 7$
(9, 25)	9	13^2	(36, 57)	36	$2^6 \cdot 7^2$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(9, 43)	9	$2^4 \cdot 13$	(36, 62)	36	2^6
(9, 46)	9	7	(37, 38)	37	7
(9, 48)	9	13^2	(37, 42)	37	$3^3 \cdot 7^3$
(9, 49)	9	$2^4 \cdot 7$	(37, 48)	37	$7 \cdot 13$
(9, 58)	9	$7^2 \cdot 13$	(37, 51)	37	7^3
(10, 20)	10	$7 \cdot 13$	(37, 52)	37	13^2
(10, 26)	10	$7 \cdot 13$	(37, 64)	37	$3^2 \cdot 7^2$
(10, 33)	10	$7^2 \cdot 13$	(38, 50)	38	13^2
(10, 42)	10	$2^4 \cdot 13$	(38, 63)	38	7^2
(10, 56)	10	$2^4 \cdot 7$	(39, 56)	39	$2^2 \cdot 3^4 \cdot 7^2$
(11, 20)	11	$2^4 \cdot 7 \cdot 13$	(40, 43)	40	$2^4 \cdot 3^4$
(11, 22)	11	$2^4 \cdot 7 \cdot 13$	(40, 59)	40	2^8
(11, 25)	11	$7 \cdot 13^2$	(40, 64)	40	$2^6 \cdot 3^2$
(11, 26)	11	$7 \cdot 13^2$	(40, 66)	40	$2^6 \cdot 5^2$
(11, 29)	11	2^{10}	(41, 43)	41	$7^3 \cdot 13$
(11, 39)	11	$2^2 \cdot 7^2$	(41, 64)	41	13^3
(11, 42)	11	$2^4 \cdot 7^2$	(41, 65)	41	$7^2 \cdot 13$
(11, 53)	11	$2^4 \cdot 7^2$	(42, 45)	42	$2^4 \cdot 3^2 \cdot 7$
(11, 59)	11	$2^4 \cdot 7^2$	(42, 58)	42	$2^6 \cdot 13^2$
(12, 21)	12	$2^8 \cdot 3^2$	(42, 60)	42	$3^4 \cdot 7^3$
(12, 28)	12	3^4	(42, 61)	42	$2^8 \cdot 3^2 \cdot 7$
(12, 46)	12	3^4	(43, 53)	43	$2^6 \cdot 13^2$
(12, 52)	12	$2^8 \cdot 3^2$	(43, 54)	43	$2^4 \cdot 3^2 \cdot 7^2$
(12, 58)	12	$2^6 \cdot 3^4$	(43, 59)	43	$2^8 \cdot 7^2$
(13, 23)	23	2^{10}	(44, 45)	44	$2^2 \cdot 13^2$
(13, 35)	35	2^{10}	(44, 48)	44	$7^3 \cdot 13$
(13, 36)	36	2^{10}	(44, 50)	44	7^5
(14, 25)	25	$2^6 \cdot 7$	(44, 56)	44	$3^2 \cdot 7 \cdot 13^2$
(14, 26)	26	$2^4 \cdot 3 \cdot 7 \cdot 13$	(45, 53)	45	13^2
(14, 28)	28	$2^4 \cdot 3^2 \cdot 13$	(45, 55)	45	13^2
(14, 31)	31	$2^4 \cdot 3^2 \cdot 13$	(45, 60)	45	$3 \cdot 7$
(14, 32)	32	$2^4 \cdot 3^2 \cdot 7^2$	(45, 62)	45	$2^4 \cdot 3^2 \cdot 7^2$
(14, 33)	33	$2^4 \cdot 3^2 \cdot 7$	(45, 63)	45	13^2
(14, 35)	35	$2^4 \cdot 3^3 \cdot 7$	(46, 60)	46	7^2
(14, 37)	37	$2^4 \cdot 3 \cdot 7 \cdot 13$	(46, 62)	46	$2^4 \cdot 7^2$
(14, 38)	38	$2^6 \cdot 7$	(47, 52)	47	$3 \cdot 13^2$
(14, 42)	42	$2^4 \cdot 3^3 \cdot 7$	(48, 51)	48	7^4
(14, 55)	55	$3^2 \cdot 7^2$	(48, 56)	48	$2^2 \cdot 7^2$
(15, 22)	22	$2^2 \cdot 7 \cdot 13$	(51, 59)	51	$2^2 \cdot 7^2$
(15, 31)	31	$2^2 \cdot 7 \cdot 13$	(51, 61)	51	$2^2 \cdot 7^2$
(15, 32)	32	$2^2 \cdot 7^2$	(51, 65)	51	$7^2 \cdot 13$
(15, 36)	36	$2^2 \cdot 7^3$	(52, 61)	52	7^3
(15, 37)	37	$2^2 \cdot 7^3$	(53, 58)	53	$2^6 \cdot 13^2$
(15, 55)	55	$7^3 \cdot 13^3$	(54, 55)	54	3^6
(15, 64)	64	$7^3 \cdot 13$	(54, 57)	54	$2^4 \cdot 3^4$
(16, 20)	20	$2^2 \cdot 7 \cdot 13$	(54, 63)	54	$2^4 \cdot 3^7$
(16, 26)	26	$2^2 \cdot 7^3$	(58, 64)	58	$2^{12} \cdot 13^2$
(16, 27)	27	$2^2 \cdot 7^3$	(62, 63)	62	2^{12}
(16, 44)	44	$2^2 \cdot 7^2$			
<i>The second certificate.</i>					
(0, 4)	0	3	(19, 60)	19	$2^2 \cdot 7^2$
(0, 24)	0	$2^2 \cdot 3^2$	(19, 66)	19	$3 \cdot 7$
(0, 38)	0	3	(19, 68)	19	$2^2 \cdot 3 \cdot 7^2$
(0, 43)	0	3	(19, 71)	19	$2^2 \cdot 7^2$
(0, 48)	0	2^2	(20, 29)	20	$3 \cdot 7$
(1, 3)	1	7	(20, 32)	20	$3 \cdot 7^2$
(1, 5)	1	3	(20, 59)	20	$2^4 \cdot 3^2 \cdot 7$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(1, 8)	1	2^{10}	(20, 65)	20	1
(1, 10)	1	$2^2 \cdot 3^2 \cdot 7^2$	(20, 67)	20	$2^8 \cdot 13$
(1, 11)	1	$2^2 \cdot 7^2$	(20, 72)	20	$7^3 \cdot 13$
(1, 23)	1	$2^4 \cdot 3^4$	(20, 73)	20	$3^2 \cdot 7^2$
(1, 31)	1	7^3	(21, 38)	21	$3 \cdot 7^2$
(1, 33)	1	2^4	(21, 57)	21	3^5
(1, 36)	1	$2^6 \cdot 7^2$	(21, 60)	21	$2^2 \cdot 3^3 \cdot 7$
(1, 41)	1	$2^2 \cdot 3$	(21, 63)	21	$7 \cdot 13$
(1, 43)	1	$2^2 \cdot 3 \cdot 7 \cdot 13$	(21, 71)	21	$2^2 \cdot 7^3$
(1, 47)	1	$2^2 \cdot 3 \cdot 7 \cdot 13$	(21, 73)	21	7^2
(1, 48)	1	$2^2 \cdot 7^4$	(22, 27)	22	$2^2 \cdot 7^2 \cdot 13^2$
(1, 58)	1	3^2	(22, 36)	22	$2^6 \cdot 7 \cdot 13$
(1, 59)	1	7	(22, 37)	22	$2^6 \cdot 13^2$
(2, 14)	2	13	(22, 42)	22	$2^2 \cdot 3^2 \cdot 13$
(2, 16)	2	2^2	(22, 61)	22	$2^8 \cdot 13$
(2, 19)	2	2^8	(22, 70)	22	$2^4 \cdot 3^4 \cdot 7$
(2, 35)	2	7^2	(23, 26)	23	$2^4 \cdot 3^2 \cdot 7$
(2, 38)	2	7^3	(23, 27)	23	$3^3 \cdot 13$
(2, 39)	2	13^2	(23, 34)	23	$2^8 \cdot 7^2$
(2, 50)	2	$2^4 \cdot 7^2 \cdot 13$	(23, 36)	23	$2^6 \cdot 13^2$
(2, 52)	2	$2^2 \cdot 3^4 \cdot 7 \cdot 13$	(23, 40)	23	$2^2 \cdot 3^4 \cdot 7^2 \cdot 13$
(2, 60)	2	2^2	(23, 46)	23	$2^4 \cdot 3^3 \cdot 7$
(3, 14)	3	13	(23, 49)	23	$2^2 \cdot 3^2 \cdot 7 \cdot 13$
(3, 20)	3	13	(24, 26)	24	$2^2 \cdot 3^2$
(3, 27)	3	$2^4 \cdot 7$	(24, 30)	24	$3^4 \cdot 7^2$
(3, 37)	3	$2^6 \cdot 7$	(24, 31)	24	$2^2 \cdot 7 \cdot 13$
(3, 41)	3	13^2	(24, 39)	24	$3^4 \cdot 7 \cdot 13$
(3, 43)	3	7	(24, 43)	24	$2^2 \cdot 3^2 \cdot 7^5$
(3, 45)	3	2^2	(24, 45)	24	$2^4 \cdot 3^2$
(3, 50)	3	$2^6 \cdot 7$	(24, 48)	24	7^4
(3, 62)	3	$2^4 \cdot 7$	(24, 72)	24	$2^2 \cdot 3^2 \cdot 7^2$
(3, 63)	3	13	(25, 29)	25	2^{20}
(3, 65)	3	7	(25, 30)	25	$3^2 \cdot 7 \cdot 13^2$
(3, 72)	3	$7^2 \cdot 13$	(25, 34)	25	$2^2 \cdot 13^2$
(3, 73)	3	7	(25, 35)	25	$3^2 \cdot 7 \cdot 13^2$
(4, 13)	4	13	(25, 39)	25	$3^2 \cdot 13$
(4, 14)	4	13	(25, 42)	25	$2^2 \cdot 3^3 \cdot 13$
(4, 29)	4	$3^2 \cdot 13^2$	(25, 49)	25	$2^2 \cdot 3 \cdot 13$
(4, 39)	4	3^2	(25, 69)	25	$2^2 \cdot 7^2$
(4, 47)	4	3	(26, 30)	26	$3^4 \cdot 7 \cdot 13$
(4, 48)	4	3^2	(26, 33)	26	13^2
(4, 59)	4	3^3	(26, 37)	26	$2^4 \cdot 3^2 \cdot 7^2$
(4, 66)	4	3^4	(26, 42)	26	$2^8 \cdot 3 \cdot 13$
(4, 67)	4	13	(26, 44)	26	7
(4, 72)	4	$3^3 \cdot 13$	(26, 45)	26	$2^2 \cdot 3^4$
(5, 12)	5	7	(26, 55)	26	$2^2 \cdot 3^4 \cdot 13^2$
(5, 14)	5	13	(26, 56)	26	$2^4 \cdot 13$
(5, 24)	5	$3^2 \cdot 7$	(26, 57)	26	$2^4 \cdot 3^4 \cdot 13$
(5, 28)	5	$2^2 \cdot 3$	(26, 62)	26	3^5
(5, 33)	5	7	(27, 38)	27	$3^2 \cdot 7^2 \cdot 13^2$
(5, 36)	5	3	(27, 45)	27	$2^2 \cdot 3^5 \cdot 13$
(5, 37)	5	$2^2 \cdot 3 \cdot 13$	(27, 66)	27	7^2
(5, 48)	5	$2^2 \cdot 3 \cdot 7$	(27, 70)	27	$3^4 \cdot 7 \cdot 13^4$
(5, 55)	5	$3^2 \cdot 13$	(28, 31)	28	$3 \cdot 13^2$
(5, 56)	5	2^2	(28, 41)	28	13^2
(5, 58)	5	3	(29, 32)	29	$3^3 \cdot 7^2 \cdot 13$
(5, 59)	5	$3 \cdot 13$	(29, 50)	29	$2^2 \cdot 7^2 \cdot 13$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(6, 13)	6	13	(29, 54)	29	$7 \cdot 13$
(6, 14)	6	7	(29, 61)	29	$2^2 \cdot 7^2 \cdot 13$
(6, 20)	6	$7 \cdot 13$	(29, 64)	29	$7 \cdot 13^2$
(6, 25)	6	13	(29, 65)	29	$7 \cdot 13^2$
(6, 38)	6	13	(29, 66)	29	$3 \cdot 7$
(6, 39)	6	$2^4 \cdot 7$	(30, 33)	30	13
(6, 43)	6	7^2	(30, 34)	30	$2^2 \cdot 13$
(6, 53)	6	2^2	(30, 57)	30	3^2
(7, 15)	7	$3^2 \cdot 7$	(30, 63)	30	$2^4 \cdot 7$
(7, 21)	7	3	(30, 70)	30	$2^4 \cdot 3^6$
(7, 36)	7	3	(31, 64)	31	$7 \cdot 13^2$
(7, 42)	7	3	(32, 56)	32	2^4
(7, 45)	7	$2^2 \cdot 3 \cdot 13$	(32, 57)	32	7^3
(7, 56)	7	2^2	(33, 36)	33	$2^6 \cdot 13^2$
(7, 63)	7	$3 \cdot 13$	(33, 57)	33	$2^2 \cdot 7 \cdot 13$
(7, 64)	7	7	(33, 59)	33	13^2
(7, 73)	7	3	(33, 69)	33	$2^2 \cdot 7$
(8, 14)	8	$2^4 \cdot 7 \cdot 13$	(34, 35)	34	$2^2 \cdot 13$
(8, 32)	8	$2^4 \cdot 7 \cdot 13$	(34, 40)	34	2^2
(8, 45)	8	2^{10}	(34, 45)	34	2^8
(8, 50)	8	$2^4 \cdot 7^2$	(34, 50)	34	$2^8 \cdot 7^2$
(8, 65)	8	$7 \cdot 13^2$	(34, 60)	34	2^4
(9, 34)	9	2^{12}	(34, 69)	34	2^2
(9, 36)	9	$2^6 \cdot 3^4$	(35, 38)	35	3^2
(9, 50)	9	$2^4 \cdot 3^4$	(35, 41)	35	$2^4 \cdot 3^4$
(9, 51)	9	$2^8 \cdot 3^5$	(35, 43)	35	$2^4 \cdot 3^2 \cdot 7^3$
(9, 56)	9	2^{10}	(35, 53)	35	$2^2 \cdot 13$
(9, 60)	9	$2^8 \cdot 3^2$	(35, 55)	35	$3^4 \cdot 7 \cdot 13$
(9, 63)	9	3^4	(35, 67)	35	1
(10, 13)	13	$2^4 \cdot 7 \cdot 13$	(36, 43)	36	$2^4 \cdot 7^3 \cdot 13^2$
(10, 14)	14	$2^4 \cdot 7 \cdot 13$	(36, 44)	36	2^4
(10, 16)	16	$2^4 \cdot 3^2 \cdot 13$	(36, 54)	36	$7^2 \cdot 13$
(10, 17)	17	$2^4 \cdot 3^2 \cdot 7^2$	(36, 57)	36	$2^4 \cdot 7^2 \cdot 13$
(10, 18)	18	$2^6 \cdot 3^2$	(36, 66)	36	$2^4 \cdot 3 \cdot 7^2$
(10, 21)	21	$2^4 \cdot 3 \cdot 7 \cdot 13$	(37, 40)	37	$2^8 \cdot 3^2 \cdot 7$
(10, 25)	25	$2^2 \cdot 3^4 \cdot 13^2$	(37, 44)	37	7
(10, 40)	40	$3^2 \cdot 7^2$	(37, 47)	37	$2^2 \cdot 3$
(10, 47)	47	$2^4 \cdot 3 \cdot 7^2 \cdot 13$	(37, 56)	37	2^{12}
(10, 53)	53	$2^2 \cdot 3^2 \cdot 13^2$	(37, 61)	37	2^6
(10, 57)	57	$2^6 \cdot 3 \cdot 7^2$	(38, 46)	38	3^3
(10, 59)	59	$2^4 \cdot 3^2 \cdot 7$	(38, 56)	38	3^6
(10, 60)	60	$2^8 \cdot 3^2$	(38, 63)	38	7
(10, 63)	63	$2^4 \cdot 3^2 \cdot 13$	(39, 43)	39	3^4
(10, 64)	64	$2^6 \cdot 7$	(39, 49)	39	$3^2 \cdot 7^2 \cdot 13$
(10, 66)	66	$2^8 \cdot 3^2$	(39, 54)	39	$2^6 \cdot 13^2$
(10, 73)	73	$3^2 \cdot 7^2$	(39, 56)	39	2^{12}
(11, 15)	15	$2^2 \cdot 7^3$	(39, 59)	39	7
(11, 16)	16	$2^2 \cdot 7 \cdot 13$	(39, 70)	39	$2^8 \cdot 3^6$
(11, 17)	17	$2^2 \cdot 7^2$	(40, 45)	40	$3^2 \cdot 13^2$
(11, 48)	48	7^4	(40, 56)	40	13^2
(11, 57)	57	$2^2 \cdot 7^2 \cdot 13$	(40, 57)	40	$2^2 \cdot 3^2 \cdot 7^2 \cdot 13$
(11, 64)	64	$2^2 \cdot 7 \cdot 13$	(40, 59)	40	13^3
(11, 67)	67	$2^2 \cdot 7^2 \cdot 13$	(40, 63)	40	$7^2 \cdot 13$
(12, 23)	12	$2^{10} \cdot 7^2$	(40, 69)	40	$2^4 \cdot 7^2$
(12, 41)	12	$3^2 \cdot 7^2$	(41, 55)	41	$3^3 \cdot 13$
(12, 42)	12	3^2	(41, 56)	41	3^6
(12, 51)	12	$3^4 \cdot 7$	(42, 49)	42	$2^4 \cdot 3 \cdot 13$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(13, 17)	13	7	(42, 56)	42	$2^4 \cdot 3^4$
(13, 23)	13	$2^2 \cdot 13^2$	(42, 61)	42	2^4
(13, 24)	13	$2^2 \cdot 7^4$	(42, 64)	42	13^3
(13, 30)	13	7^2	(42, 67)	42	13
(13, 31)	13	$2^2 \cdot 7 \cdot 13$	(43, 44)	43	2^2
(13, 36)	13	$2^4 \cdot 7 \cdot 13$	(43, 60)	43	$3^2 \cdot 7^2$
(13, 37)	13	2^4	(43, 62)	43	$2^2 \cdot 3 \cdot 7^2 \cdot 13$
(13, 39)	13	13	(44, 45)	44	2^2
(13, 42)	13	$3^2 \cdot 13$	(44, 48)	44	2^4
(13, 43)	13	7^2	(44, 61)	44	2^2
(13, 47)	13	$2^2 \cdot 7 \cdot 13$	(45, 59)	45	$2^2 \cdot 13^2$
(13, 52)	13	$3^4 \cdot 7 \cdot 13$	(45, 71)	45	$2^2 \cdot 13^2$
(13, 54)	13	$7 \cdot 13$	(45, 72)	45	$2^2 \cdot 3^5 \cdot 13$
(13, 71)	13	$2^2 \cdot 7 \cdot 13$	(47, 49)	47	$2^4 \cdot 3 \cdot 13$
(14, 15)	14	$7 \cdot 13$	(47, 50)	47	$2^2 \cdot 3^4 \cdot 7$
(14, 35)	14	7^2	(47, 52)	47	$3^4 \cdot 13^2$
(14, 40)	14	$2^4 \cdot 7 \cdot 13$	(47, 54)	47	$2^2 \cdot 7 \cdot 13$
(14, 41)	14	$3^2 \cdot 13^2$	(47, 66)	47	13^2
(14, 43)	14	$7 \cdot 13$	(47, 68)	47	$2^6 \cdot 13$
(14, 45)	14	$2^2 \cdot 13^2$	(48, 50)	48	$2^8 \cdot 7^3$
(14, 48)	14	$3^2 \cdot 7^2$	(48, 51)	48	$2^6 \cdot 3^7 \cdot 7^2$
(14, 65)	14	$2^2 \cdot 13^2$	(48, 56)	48	$2^4 \cdot 3^4$
(14, 67)	14	13	(48, 71)	48	7^4
(14, 71)	14	$2^2 \cdot 7 \cdot 13$	(49, 55)	49	$3 \cdot 13^2$
(15, 24)	15	$3^2 \cdot 7$	(49, 57)	49	$2^2 \cdot 3 \cdot 7 \cdot 13$
(15, 30)	15	$3^4 \cdot 13$	(49, 61)	49	$2^6 \cdot 7$
(15, 50)	15	$3^3 \cdot 7^3$	(49, 64)	49	$7 \cdot 13$
(15, 60)	15	$2^2 \cdot 3^3 \cdot 7$	(50, 55)	50	$2^4 \cdot 3^2 \cdot 7$
(16, 32)	16	$2^2 \cdot 3^3 \cdot 7$	(50, 56)	50	$2^4 \cdot 13$
(16, 33)	16	7	(50, 58)	50	$2^4 \cdot 3^3 \cdot 7$
(16, 35)	16	$2^4 \cdot 7$	(50, 59)	50	$7^3 \cdot 13$
(16, 38)	16	13	(51, 68)	51	3^{10}
(16, 40)	16	$7^2 \cdot 13$	(51, 72)	51	$2^4 \cdot 3^7 \cdot 7$
(16, 41)	16	$2^4 \cdot 7^2$	(51, 73)	51	$2^4 \cdot 3^{13} \cdot 7$
(16, 43)	16	$2^4 \cdot 7$	(52, 53)	52	$3^4 \cdot 7 \cdot 13^2$
(16, 59)	16	$3 \cdot 13$	(52, 62)	52	$2^4 \cdot 3^{10} \cdot 13$
(16, 65)	16	1	(53, 54)	53	$2^4 \cdot 13$
(17, 34)	17	$2^4 \cdot 7^2$	(53, 60)	53	$2^2 \cdot 7^2$
(17, 35)	17	$2^2 \cdot 7$	(53, 63)	53	$2^8 \cdot 3 \cdot 13$
(17, 38)	17	7^3	(53, 66)	53	$3^5 \cdot 7$
(17, 40)	17	1	(54, 57)	54	$2^4 \cdot 7$
(17, 45)	17	2^8	(54, 58)	54	7
(17, 47)	17	$2^2 \cdot 7^4$	(54, 64)	54	13^2
(17, 53)	17	$2^2 \cdot 3$	(56, 62)	56	$2^4 \cdot 13$
(17, 66)	17	$2^2 \cdot 3 \cdot 7^2$	(57, 59)	57	$7^2 \cdot 13$
(18, 19)	18	3^2	(57, 64)	57	$2^4 \cdot 7 \cdot 13$
(18, 29)	18	$2^2 \cdot 3^4$	(57, 67)	57	7
(18, 31)	18	$3 \cdot 7 \cdot 13$	(58, 61)	58	2^{10}
(18, 32)	18	$3^3 \cdot 7$	(60, 63)	60	2^2
(18, 35)	18	7	(60, 66)	60	$2^2 \cdot 7^4$
(18, 41)	18	$2^2 \cdot 13$	(60, 68)	60	7^2
(18, 65)	18	7^3	(61, 64)	61	$3^4 \cdot 7$
(19, 30)	19	$2^2 \cdot 7$	(61, 71)	61	$2^2 \cdot 7^3$
(19, 32)	19	$2^2 \cdot 3^2$	(62, 71)	62	$2^2 \cdot 7^2 \cdot 13$
(19, 34)	19	$2^4 \cdot 7^2$	(63, 64)	63	1
(19, 36)	19	$3^3 \cdot 7^2$	(63, 71)	63	$2^2 \cdot 7 \cdot 13$
(19, 40)	19	1	(63, 73)	63	$2^2 \cdot 3 \cdot 7$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(19, 43)	19	$2^4 \cdot 7$	(65, 69)	65	$3^4 \cdot 7$
(19, 52)	19	$2^2 \cdot 3^4 \cdot 7^2$	(66, 68)	66	$3^2 \cdot 7^2 \cdot 13$
<i>The third certificate.</i>					
(0, 42)	0	2^{10}	(18, 39)	18	$2^2 \cdot 7 \cdot 13$
(0, 43)	0	$2^2 \cdot 13^2$	(18, 40)	18	$2^2 \cdot 13^2$
(0, 46)	0	13	(18, 46)	18	7^2
(0, 47)	0	13	(18, 48)	18	$2^6 \cdot 7 \cdot 13$
(1, 13)	1	$2^2 \cdot 7^2$	(18, 50)	18	$2^2 \cdot 7 \cdot 13$
(1, 20)	1	3	(18, 55)	18	13
(1, 43)	1	$2^2 \cdot 3^2$	(18, 56)	18	2^4
(1, 46)	1	1	(18, 58)	18	$2^4 \cdot 7 \cdot 13$
(1, 48)	1	7^2	(18, 60)	18	2^4
(1, 50)	1	1	(18, 63)	18	$7 \cdot 13$
(2, 4)	2	1	(19, 28)	19	2^2
(2, 6)	2	3	(19, 30)	19	$2^2 \cdot 7^2$
(2, 13)	2	$2^2 \cdot 7^2$	(19, 33)	19	$2^2 \cdot 7^2$
(2, 15)	2	$2^2 \cdot 7^2$	(19, 36)	19	$2^2 \cdot 3^3 \cdot 7$
(2, 16)	2	7^2	(19, 46)	19	$2^2 \cdot 7^2$
(2, 17)	2	7^2	(19, 48)	19	$2^4 \cdot 7^2$
(2, 42)	2	$2^2 \cdot 3^2$	(19, 51)	19	$2^4 \cdot 7^2$
(2, 50)	2	7^2	(19, 52)	19	2^4
(2, 51)	2	7^2	(19, 53)	19	3^4
(3, 13)	3	$2^2 \cdot 13^2$	(19, 62)	19	$2^4 \cdot 3^7 \cdot 7$
(3, 15)	3	$2^2 \cdot 7^2$	(19, 63)	19	$3^2 \cdot 7^2$
(3, 21)	3	7	(20, 53)	20	$2^4 \cdot 3^4$
(3, 29)	3	2^{10}	(20, 54)	20	$2^4 \cdot 7^2$
(3, 35)	3	2^2	(20, 57)	20	$3^2 \cdot 7^2$
(3, 36)	3	3	(20, 58)	20	$2^4 \cdot 3^4$
(3, 39)	3	$2^2 \cdot 3 \cdot 7 \cdot 13$	(20, 59)	20	3^2
(3, 40)	3	$2^4 \cdot 3^4$	(21, 26)	21	$7 \cdot 13$
(3, 42)	3	$2^2 \cdot 3^2 \cdot 7^2$	(21, 37)	21	7
(3, 49)	3	7^3	(21, 49)	21	$2^2 \cdot 7 \cdot 13$
(3, 50)	3	7^3	(21, 51)	21	$2^6 \cdot 7 \cdot 13$
(3, 58)	3	$2^2 \cdot 3^2$	(21, 54)	21	$2^4 \cdot 7 \cdot 13$
(3, 60)	3	13	(21, 58)	21	$2^4 \cdot 7 \cdot 13$
(4, 29)	4	2^4	(21, 63)	21	7^2
(4, 30)	4	7	(22, 23)	22	$2^4 \cdot 13$
(4, 31)	4	1	(22, 28)	22	$2^4 \cdot 13$
(4, 34)	4	1	(22, 48)	22	2^4
(4, 39)	4	$2^2 \cdot 7$	(22, 54)	22	13^2
(4, 57)	4	1	(22, 56)	22	2^{12}
(5, 18)	5	7	(23, 30)	23	2^8
(5, 27)	5	7	(23, 39)	23	$2^8 \cdot 3$
(5, 33)	5	2^2	(23, 41)	23	$2^2 \cdot 3^2 \cdot 7 \cdot 13$
(5, 39)	5	$7 \cdot 13$	(23, 44)	23	$2^4 \cdot 7$
(5, 44)	5	$7 \cdot 13$	(23, 47)	23	3^2
(5, 45)	5	$2^2 \cdot 7 \cdot 13$	(23, 53)	23	$2^4 \cdot 7$
(5, 51)	5	2^2	(23, 54)	23	$2^4 \cdot 7^2 \cdot 13$
(5, 52)	5	2^2	(23, 56)	23	$2^4 \cdot 3 \cdot 7$
(5, 56)	5	$2^6 \cdot 7$	(23, 59)	23	$2^8 \cdot 7$
(5, 60)	5	13	(24, 31)	24	$2^2 \cdot 7 \cdot 13$
(5, 63)	5	7^2	(24, 32)	24	7^3
(6, 21)	6	13	(24, 36)	24	$7 \cdot 13^2$
(6, 28)	6	3^2	(24, 48)	24	7^2
(6, 35)	6	13	(24, 49)	24	$7 \cdot 13^2$
(6, 38)	6	3^3	(24, 51)	24	13^2
(6, 50)	6	3^4	(24, 55)	24	$3^2 \cdot 13^2$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(6, 54)	6	$3^2 \cdot 13^2$	(25, 41)	25	$2^4 \cdot 3^3 \cdot 7$
(6, 59)	6	3^3	(25, 58)	25	$3^2 \cdot 7$
(6, 60)	6	1	(25, 63)	25	$3 \cdot 7$
(6, 62)	6	3^8	(26, 28)	26	$7 \cdot 13$
(7, 24)	7	7	(26, 30)	26	7
(7, 25)	7	3	(26, 35)	26	7^2
(7, 26)	7	3	(26, 37)	26	$7 \cdot 13^2$
(7, 28)	7	$3 \cdot 13$	(26, 40)	26	$3^2 \cdot 7$
(7, 31)	7	$2^2 \cdot 3 \cdot 7$	(26, 41)	26	$3^3 \cdot 7^3$
(7, 32)	7	$3 \cdot 7$	(26, 47)	26	$3^4 \cdot 13$
(7, 33)	7	$3 \cdot 7$	(26, 48)	26	$7 \cdot 13$
(7, 35)	7	7	(27, 36)	27	7^2
(7, 36)	7	$3^2 \cdot 7$	(27, 53)	27	$2^6 \cdot 7^2$
(7, 37)	7	$7 \cdot 13$	(27, 55)	27	$2^8 \cdot 3^2 \cdot 7$
(7, 38)	7	$2^2 \cdot 3$	(28, 37)	28	$2^2 \cdot 7 \cdot 13$
(7, 46)	7	2^2	(28, 45)	28	$2^4 \cdot 3^3$
(7, 54)	7	3	(28, 50)	28	$3 \cdot 13^2$
(7, 61)	7	3^7	(28, 58)	28	$7^2 \cdot 13$
(8, 22)	8	2^2	(29, 32)	29	$2^8 \cdot 3^2$
(8, 27)	8	2^2	(29, 58)	29	2^{12}
(8, 31)	8	2^2	(30, 31)	30	$2^8 \cdot 3^2$
(8, 36)	8	7	(30, 39)	30	$2^2 \cdot 7^4$
(8, 44)	8	$2^2 \cdot 7 \cdot 13$	(30, 46)	30	$2^4 \cdot 7^3$
(8, 45)	8	$7 \cdot 13$	(30, 50)	30	$2^4 \cdot 3 \cdot 7$
(8, 49)	8	$7 \cdot 13$	(30, 52)	30	$2^4 \cdot 7^2$
(8, 50)	8	13	(30, 57)	30	$2^4 \cdot 3$
(8, 53)	8	7	(30, 58)	30	1
(8, 61)	8	$3^6 \cdot 7$	(30, 63)	30	$2^4 \cdot 7$
(8, 63)	8	7	(31, 44)	31	$2^4 \cdot 3^3$
(9, 20)	9	3	(31, 55)	31	$2^2 \cdot 3 \cdot 13$
(9, 24)	9	$7 \cdot 13$	(31, 63)	31	$2^4 \cdot 7$
(9, 28)	9	$2^2 \cdot 3 \cdot 7$	(32, 47)	32	7
(9, 32)	9	$3 \cdot 7$	(32, 49)	32	$3 \cdot 7 \cdot 13$
(9, 33)	9	$2^2 \cdot 3$	(32, 59)	32	$2^2 \cdot 3 \cdot 7$
(9, 36)	9	3	(32, 60)	32	13
(9, 42)	9	$3^2 \cdot 7$	(33, 34)	33	$2^2 \cdot 3 \cdot 7^2$
(9, 55)	9	$3 \cdot 13$	(33, 35)	33	$2^2 \cdot 7^2$
(10, 21)	10	$7 \cdot 13$	(33, 43)	33	7
(10, 23)	10	$7^2 \cdot 13$	(33, 52)	33	$2^4 \cdot 7^2$
(10, 41)	10	$2^4 \cdot 13$	(33, 54)	33	$3^3 \cdot 7^2$
(10, 44)	10	7	(34, 39)	34	$2^6 \cdot 13$
(10, 45)	10	$2^4 \cdot 7$	(34, 46)	34	7^2
(10, 46)	10	13^2	(34, 55)	34	$3^3 \cdot 7^2$
(11, 24)	11	$7 \cdot 13^2$	(35, 40)	35	$2^2 \cdot 7^2 \cdot 13$
(11, 29)	11	2^{10}	(35, 53)	35	$2^6 \cdot 7^2$
(11, 40)	11	$2^4 \cdot 7^2$	(35, 55)	35	$3^2 \cdot 7$
(11, 41)	11	$2^4 \cdot 7^2$	(35, 56)	35	2^6
(11, 51)	11	$2^4 \cdot 7 \cdot 13$	(35, 58)	35	$2^2 \cdot 7$
(11, 58)	11	$2^2 \cdot 7^2$	(35, 62)	35	$3^4 \cdot 7$
(12, 19)	12	$2^8 \cdot 3^2$	(36, 40)	36	$3^3 \cdot 7^3$
(12, 22)	12	2^{10}	(36, 51)	36	$7 \cdot 13$
(12, 29)	12	$2^6 \cdot 3^4$	(37, 48)	37	13^2
(12, 31)	12	3^4	(37, 50)	37	$7 \cdot 13^2$
(12, 32)	12	$2^2 \cdot 3^4$	(37, 51)	37	7^2
(12, 44)	12	3^4	(37, 55)	37	$3^2 \cdot 13^2$
(12, 52)	12	2^{12}	(37, 60)	37	$7^2 \cdot 13$
(12, 54)	12	$2^6 \cdot 3^4$	(38, 52)	38	2^4

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(13, 16)	16	$2^4 \cdot 7 \cdot 13$	(39, 58)	39	$2^2 \cdot 3^4 \cdot 13$
(13, 22)	22	2^{10}	(39, 59)	39	$2^6 \cdot 3^2 \cdot 13$
(13, 27)	27	2^{10}	(40, 52)	40	$2^8 \cdot 7^2$
(13, 29)	29	2^{10}	(40, 54)	40	$2^6 \cdot 13^2$
(13, 32)	32	$2^4 \cdot 7^2$	(40, 57)	40	$2^4 \cdot 3^2 \cdot 7^2$
(13, 35)	35	2^{10}	(40, 58)	40	$2^2 \cdot 3^4 \cdot 7^2 \cdot 13$
(13, 37)	37	$2^4 \cdot 7 \cdot 13$	(40, 59)	40	$2^6 \cdot 3^6$
(14, 16)	16	$2^8 \cdot 3^2$	(40, 61)	40	$3^8 \cdot 7^3$
(14, 18)	18	$2^4 \cdot 7 \cdot 13$	(41, 52)	41	$2^8 \cdot 7^2$
(14, 19)	19	$2^8 \cdot 3^2$	(41, 59)	41	$2^2 \cdot 3^2 \cdot 7^4$
(14, 21)	21	$2^4 \cdot 7 \cdot 13$	(42, 58)	42	$3^4 \cdot 7^2$
(14, 23)	23	$2^6 \cdot 3 \cdot 7^2$	(42, 63)	42	$2^2 \cdot 3^2 \cdot 7^5$
(14, 24)	24	$2^6 \cdot 7$	(43, 44)	43	$2^{16} \cdot 7$
(14, 26)	26	$2^4 \cdot 3 \cdot 7 \cdot 13$	(43, 46)	43	$2^2 \cdot 13$
(14, 30)	30	$2^4 \cdot 3^2 \cdot 7^2$	(43, 47)	43	$3^4 \cdot 7 \cdot 13$
(14, 32)	32	$2^6 \cdot 3^2$	(43, 56)	43	$2^4 \cdot 3^2 \cdot 7^2$
(14, 36)	36	$2^4 \cdot 3 \cdot 7 \cdot 13$	(44, 46)	44	$2^2 \cdot 13$
(14, 37)	37	$2^6 \cdot 7$	(44, 50)	44	$3 \cdot 13^2$
(14, 40)	40	$2^4 \cdot 3^3 \cdot 7$	(44, 56)	44	$2^4 \cdot 7^2$
(14, 45)	45	$2^2 \cdot 3^2 \cdot 7^2$	(44, 59)	44	$2^2 \cdot 3 \cdot 7 \cdot 13$
(14, 58)	58	$3^2 \cdot 7^2$	(45, 49)	45	$3 \cdot 13^2$
(15, 20)	20	2^2	(46, 51)	46	$7 \cdot 13$
(15, 21)	21	$2^2 \cdot 7 \cdot 13$	(46, 54)	46	$7^3 \cdot 13$
(15, 33)	33	$2^2 \cdot 7^2$	(47, 50)	47	$2^4 \cdot 13$
(15, 35)	35	$2^2 \cdot 7^3$	(47, 58)	47	$3^4 \cdot 7$
(15, 36)	36	$2^2 \cdot 7^3$	(47, 61)	47	$3^8 \cdot 7 \cdot 13$
(15, 58)	58	$7^3 \cdot 13$	(47, 62)	47	3^8
(16, 19)	16	$2^2 \cdot 7^4$	(47, 63)	47	$2^4 \cdot 3^2 \cdot 7^3$
(16, 20)	16	2^{10}	(49, 50)	49	$2^4 \cdot 3^6$
(16, 27)	16	7^2	(50, 54)	50	$3 \cdot 7 \cdot 13$
(16, 30)	16	$2^2 \cdot 3 \cdot 7^2$	(50, 61)	50	$3^7 \cdot 13^2$
(16, 34)	16	$3^2 \cdot 7^2 \cdot 13$	(50, 62)	50	$2^{10} \cdot 3^8$
(16, 54)	16	$2^4 \cdot 3 \cdot 7^2$	(52, 56)	52	2^{14}
(17, 18)	17	2^{10}	(52, 59)	52	2^{14}
(17, 26)	17	7^2	(52, 60)	52	1
(17, 30)	17	$3 \cdot 7$	(53, 57)	53	$2^4 \cdot 3^4$
(17, 33)	17	$2^2 \cdot 3 \cdot 7^2$	(54, 63)	54	$2^4 \cdot 7^3 \cdot 13^2$
(17, 38)	17	$2^8 \cdot 3^2$	(55, 59)	55	$2^2 \cdot 3^4$
(17, 57)	17	$2^4 \cdot 3 \cdot 7^2$	(56, 58)	56	$2^8 \cdot 3^2 \cdot 7$
(18, 30)	18	7	(58, 60)	58	$2^4 \cdot 7^2$
(18, 36)	18	$7 \cdot 13$	(58, 63)	58	$2^2 \cdot 3^6 \cdot 7^2$
(18, 37)	18	$2^2 \cdot 13^2$			

Table 6: The terminal residue classes $x \equiv a$, $y \equiv b \pmod{p^k}$ of Step 4, grouped by p , k and the value of $(B_p^{(g)})_g$.

p	k	value	classes (a, b)
2	3	(1, 1, 2)	(0, 5), (2, 6), (4, 1), (6, 0), (6, 4)
2	4	(1, 1, 2)	(2, 11), (6, 7), (10, 3), (14, 15)
2	5	(1, 1, 2)	(18, 2), (18, 10), (18, 18), (18, 26)
3	4	(0, 0, 1)	(8, 23), (17, 59), (26, 14), (35, 50), (44, 5), (53, 41), (62, 77), (71, 32), (80, 68)
3	4	(0, 0, 2)	(20, 6), (20, 33), (20, 60), (47, 15), (47, 42), (47, 69), (74, 24), (74, 51), (74, 78)

p	k	value	classes (a, b)
3	4	(0, 1, 0)	(2, 44), (11, 26), (20, 8), (29, 71), (38, 53), (47, 35), (56, 17), (65, 0), (65, 9), (65, 18), (65, 27), (65, 36), (65, 45), (65, 54), (65, 63), (65, 72), (65, 80), (74, 62)
3	4	(0, 2, 1)	(20, 12), (20, 39), (20, 66), (47, 3), (47, 30), (47, 57), (74, 21), (74, 48), (74, 75)
3	4	(0, 2, 2)	(5, 29), (14, 38), (23, 47), (32, 56), (41, 65), (50, 74), (59, 2), (68, 11), (77, 20)
5	1	(0, 0, 0)	(1, 0), (1, 4), (4, 1), (4, 2)
7	2	(0, 0, 1)	(0, 44), (4, 20), (7, 16), (19, 19), (21, 9), (31, 31), (35, 2)
7	3	(0, 0, 1)	(1, 127), (3, 227), (5, 117), (8, 330), (10, 31), (11, 300), (12, 19), (14, 86), (15, 288), (18, 41), (22, 1), (24, 80), (25, 272), (26, 117), (28, 128), (29, 155), (32, 307), (33, 313), (36, 64), (38, 31), (39, 146), (40, 264), (42, 170), (43, 71), (45, 227), (46, 132), (47, 313), (50, 176), (52, 227), (54, 117), (57, 36), (59, 31), (60, 202), (61, 19), (63, 233), (64, 337), (66, 325), (67, 286), (71, 50), (73, 80), (74, 174), (75, 117), (77, 275), (78, 204), (81, 209), (82, 313), (85, 113), (87, 31), (88, 7), (88, 42), (88, 48), (88, 56), (88, 91), (88, 105), (88, 140), (88, 154), (88, 189), (88, 203), (88, 238), (88, 252), (88, 287), (88, 301), (88, 336), (89, 264), (91, 317), (92, 120), (94, 227), (95, 34), (96, 313), (99, 225), (101, 227), (103, 117), (106, 85), (108, 31), (109, 104), (110, 19), (112, 37), (113, 43), (115, 325), (116, 188), (120, 99), (122, 80), (123, 76), (124, 117), (126, 79), (127, 253), (130, 111), (131, 313), (134, 162), (136, 31), (137, 293), (138, 264), (141, 169), (143, 227), (144, 279), (145, 313), (148, 274), (150, 227), (152, 117), (155, 134), (157, 31), (158, 6), (159, 19), (161, 184), (162, 92), (164, 325), (165, 90), (169, 148), (171, 80), (172, 321), (173, 117), (175, 226), (176, 302), (179, 13), (180, 313), (183, 211), (185, 31), (186, 14), (186, 35), (186, 63), (186, 84), (186, 112), (186, 133), (186, 161), (186, 182), (186, 195), (186, 210), (186, 231), (186, 259), (186, 280), (186, 308), (186, 329), (187, 264), (189, 268), (190, 218), (192, 227), (193, 181), (194, 313), (197, 323), (199, 227), (201, 117), (204, 183), (206, 31), (207, 251), (208, 19), (210, 331), (211, 141), (213, 325), (214, 335), (218, 197), (220, 80), (222, 117), (224, 30), (225, 8), (228, 258), (229, 313), (234, 31), (235, 21), (235, 28), (235, 70), (235, 97), (235, 119), (235, 126), (235, 168), (235, 175), (235, 217), (235, 224), (235, 266), (235, 273), (235, 315), (235, 322), (236, 264), (238, 72), (239, 267), (241, 227), (243, 313), (246, 29), (248, 227), (250, 117), (253, 232), (255, 31), (256, 153), (257, 19), (259, 135), (260, 190), (262, 325), (263, 237), (267, 246), (269, 80), (270, 125), (271, 117), (273, 177), (274, 57), (277, 160), (278, 313), (281, 309), (283, 31), (284, 0), (284, 49), (284, 98), (284, 147), (284, 196), (284, 245), (284, 294), (285, 264), (287, 219), (288, 316), (290, 227), (291, 328), (292, 313), (295, 78), (297, 227), (299, 117), (302, 281), (304, 31), (305, 55), (308, 282), (309, 239), (311, 325), (312, 139), (316, 295), (318, 80), (319, 27), (320, 117), (322, 324), (323, 106), (326, 62), (327, 313), (330, 15), (332, 31), (334, 264), (336, 23), (337, 22), (339, 227), (340, 230), (341, 313)

p	k	value	classes (a, b)
7	4	(0, 0, 1)	(17, 325), (140, 121), (221, 223), (232, 1289), (242, 2141), (284, 342), (306, 2077), (333, 930), (360, 325), (483, 1150), (564, 1938), (575, 1632), (578, 77), (578, 420), (578, 763), (578, 1106), (578, 1449), (578, 1792), (578, 2135), (585, 1455), (627, 2057), (649, 2077), (676, 244), (703, 325), (826, 2179), (907, 1252), (918, 1975), (928, 769), (970, 1371), (992, 2077), (1019, 1959), (1046, 325), (1169, 807), (1250, 566), (1261, 2318), (1271, 83), (1313, 685), (1335, 2077), (1362, 1273), (1389, 325), (1512, 1836), (1593, 2281), (1604, 260), (1614, 1798), (1656, 2400), (1678, 2077), (1705, 587), (1732, 325), (1855, 464), (1936, 1595), (1947, 603), (1957, 1112), (1999, 1714), (2021, 2077), (2048, 2302), (2075, 325), (2198, 1493), (2279, 909), (2290, 946), (2300, 426), (2342, 1028), (2364, 2077), (2391, 1616)
13	1	(1, 2, 0)	(10, 10)
13	2	(0, 0, 0)	(3, 69), (4, 66), (16, 82), (17, 118), (29, 95), (30, 1), (42, 108), (43, 53), (55, 121), (56, 105), (68, 134), (69, 157), (81, 147), (82, 40), (94, 160), (95, 92), (107, 4), (108, 144), (120, 17), (121, 27), (133, 30), (134, 79), (147, 131), (159, 56), (160, 14)
13	2	(0, 1, 0)	(4, 108), (5, 139), (17, 4), (18, 139), (25, 120), (30, 69), (31, 139), (43, 134), (44, 139), (51, 146), (56, 30), (57, 139), (64, 159), (69, 95), (70, 139), (77, 3), (82, 160), (83, 139), (90, 16), (95, 56), (96, 139), (108, 121), (109, 139), (116, 42), (121, 17), (122, 139), (129, 55), (134, 82), (135, 139), (142, 68), (147, 147), (148, 139), (155, 81), (160, 43), (168, 94)
13	2	(0, 2, 0)	(4, 163), (17, 46), (30, 98), (43, 150), (56, 33), (69, 85), (95, 20), (108, 72), (134, 7), (147, 59), (160, 111)
13	2	(1, 0, 0)	(3, 98), (9, 29), (16, 7), (22, 133), (29, 85), (35, 68), (42, 163), (48, 3), (55, 72), (61, 107), (68, 150), (74, 42), (81, 59), (87, 146), (94, 137), (100, 81), (107, 46), (113, 16), (120, 124), (126, 120), (133, 33), (139, 55), (146, 111), (152, 159), (159, 20), (165, 94)
13	2	(1, 1, 0)	(6, 103), (19, 103), (32, 103), (58, 103), (71, 103), (84, 103), (97, 103), (110, 103), (123, 103), (136, 103), (149, 103), (162, 103)
13	2	(1, 2, 0)	(1, 62), (27, 75), (40, 166), (53, 88), (66, 10), (92, 23), (105, 114), (118, 36), (131, 127), (144, 49)
13	2	(2, 0, 0)	(3, 1), (16, 79), (29, 157), (42, 66), (55, 144), (68, 53), (81, 131), (94, 40), (107, 118), (120, 27), (133, 105), (146, 14), (159, 92)
13	2	(2, 2, 0)	(7, 28), (20, 132), (33, 67), (46, 2), (59, 106), (72, 41), (111, 15), (124, 119), (150, 158), (163, 93)

p	k	value	classes (a, b)
13	3	(0, 0, 0)	(11, 1279), (24, 1396), (37, 330), (50, 278), (63, 1240), (76, 1019), (89, 1812), (102, 1422), (115, 2046), (128, 1487), (141, 1942), (146, 888), (154, 1214), (167, 1500), (180, 603), (193, 720), (206, 1851), (219, 1799), (232, 564), (245, 343), (258, 1136), (271, 746), (284, 1370), (297, 811), (310, 1266), (315, 1057), (323, 538), (336, 824), (349, 2124), (362, 44), (375, 1175), (388, 1123), (401, 2085), (414, 1864), (427, 460), (440, 70), (453, 694), (466, 135), (479, 590), (484, 1226), (492, 2059), (505, 148), (518, 1448), (531, 1565), (544, 499), (557, 447), (570, 1409), (583, 1188), (596, 1981), (609, 1591), (622, 18), (635, 1656), (648, 2111), (653, 1395), (661, 1383), (674, 1669), (687, 772), (700, 889), (713, 2020), (726, 1968), (739, 733), (752, 512), (765, 1305), (778, 915), (791, 1539), (804, 980), (817, 1435), (822, 1564), (830, 707), (843, 993), (856, 96), (869, 213), (882, 1344), (895, 1292), (908, 57), (921, 2033), (934, 629), (947, 239), (960, 863), (973, 304), (986, 759), (991, 1733), (999, 31), (1012, 317), (1025, 1617), (1038, 1734), (1051, 668), (1064, 616), (1077, 1578), (1090, 1357), (1103, 2150), (1116, 1760), (1129, 187), (1142, 1825), (1155, 83), (1160, 1902), (1168, 1552), (1181, 1838), (1194, 941), (1207, 1058), (1220, 2189), (1233, 2137), (1246, 902), (1259, 681), (1272, 1474), (1285, 1084), (1298, 1708), (1311, 1149), (1324, 1604), (1329, 2071), (1337, 876), (1350, 1162), (1363, 265), (1376, 382), (1389, 1513), (1402, 1461), (1415, 226), (1428, 5), (1441, 798), (1454, 408), (1467, 1032), (1480, 473), (1493, 928), (1498, 43), (1506, 200), (1519, 486), (1532, 1786), (1545, 1903), (1558, 837), (1571, 785), (1584, 1747), (1597, 1526), (1610, 122), (1623, 1929), (1636, 356), (1649, 1994), (1662, 252), (1667, 212), (1675, 1721), (1688, 2007), (1701, 1110), (1714, 1227), (1727, 161), (1740, 109), (1753, 1071), (1766, 850), (1779, 1643), (1792, 1253), (1805, 1877), (1818, 1318), (1831, 1773), (1836, 381), (1844, 1045), (1857, 1331), (1870, 434), (1883, 551), (1896, 1682), (1909, 1630), (1922, 395), (1935, 174), (1948, 967), (1961, 577), (1974, 1201), (1987, 642), (2000, 1097), (2005, 550), (2013, 369), (2026, 655), (2039, 1955), (2052, 2072), (2065, 1006), (2078, 954), (2091, 1916), (2104, 1695), (2117, 291), (2130, 2098), (2143, 525), (2156, 2163), (2169, 421), (2174, 719), (2182, 1890), (2195, 2176)
13	3	(0, 1, 0)	(12, 1459), (38, 1992), (103, 1212), (161, 308), (181, 1628), (207, 2161), (272, 1381), (330, 308), (350, 1797), (376, 133), (441, 1550), (499, 308), (519, 1966), (545, 302), (610, 1719), (668, 308), (688, 2135), (714, 471), (779, 1888), (837, 308), (857, 107), (883, 640), (948, 2057), (1006, 308), (1026, 276), (1052, 809), (1117, 29), (1175, 308), (1195, 445), (1221, 978), (1286, 198), (1344, 308), (1364, 614), (1390, 1147), (1455, 367), (1513, 308), (1533, 783), (1559, 1316), (1624, 536), (1682, 308), (1702, 952), (1728, 1485), (1793, 705), (1851, 308), (1871, 1121), (1897, 1654), (1962, 874), (2020, 308), (2040, 1290), (2066, 1823), (2131, 1043), (2189, 308)
13	3	(0, 2, 0)	(82, 982), (121, 1138), (251, 1658), (290, 1814), (420, 137), (459, 293), (589, 813), (628, 969), (758, 1489), (797, 1645), (927, 2165), (966, 124), (1096, 644), (1135, 800), (1265, 1320), (1304, 1476), (1434, 1996), (1473, 2152), (1603, 475), (1642, 631), (1772, 1151), (1811, 1307), (1941, 1827), (1980, 1983), (2110, 306), (2149, 462)
13	3	(1, 1, 0)	(45, 1962), (214, 1962), (383, 1962), (552, 1962), (721, 1962), (890, 1962), (1059, 1962), (1228, 1962), (1397, 1962), (1566, 1962), (1735, 1962), (1904, 1962), (2073, 1962)

p	k	value	classes (a, b)
13	3	(1, 2, 0)	(14, 1843), (79, 608), (157, 1661), (183, 829), (248, 1791), (326, 647), (352, 2012), (417, 777), (495, 1830), (521, 998), (586, 1960), (664, 816), (690, 2181), (755, 946), (833, 1999), (859, 1167), (924, 2129), (1002, 985), (1028, 153), (1093, 1115), (1171, 2168), (1197, 1336), (1262, 101), (1340, 1154), (1366, 322), (1431, 1284), (1509, 140), (1535, 1505), (1600, 270), (1678, 1323), (1704, 491), (1769, 1453), (1847, 309), (1873, 1674), (1938, 439), (2016, 1492), (2042, 660), (2107, 1622), (2185, 478)
13	3	(2, 2, 0)	(85, 1497), (98, 1432), (137, 1744), (254, 652), (267, 587), (306, 899), (423, 2004), (436, 1939), (475, 54), (592, 1159), (605, 1094), (644, 1406), (761, 314), (774, 249), (813, 561), (930, 1666), (943, 1601), (982, 1913), (1099, 821), (1112, 756), (1151, 1068), (1268, 2173), (1281, 2108), (1320, 223), (1437, 1328), (1450, 1263), (1489, 1575), (1606, 483), (1619, 418), (1658, 730), (1775, 1835), (1788, 1770), (1827, 2082), (1944, 990), (1957, 925), (2113, 145), (2126, 80), (2165, 392)
13	4	(2, 2, 0)	(1996, 18813), (4193, 7828), (6390, 25404), (8587, 14419), (10784, 3434), (12981, 21010), (15178, 10025), (17375, 27601), (19572, 16616), (21769, 5631), (23966, 23207), (26163, 12222), (28360, 1237)
19	1	(0, 0, 0)	(0, 3), (0, 5), (0, 10), (1, 14), (2, 1), (3, 4), (4, 14), (7, 17), (8, 8), (9, 17), (10, 14), (12, 12), (12, 13), (13, 17), (16, 12), (16, 13), (17, 8)